



Child and Adolescent Development & Facilitating Learning

MR. ENGLEVERT REYES, M. ED. - SPED

CHILD AND ADOLESCENT DEVELOPMENT

TOPIC OUTLINE

A. Basic Concepts on Human Development

- ▶ Human Development: Meaning, Concepts, and Approaches
- ▶ The Stages of Development and Developmental Tasks
- ▶ Issues on Human Development

B. Developmental Theories

- ▶ Freud's Psychoanalytic Theory
- ▶ Piaget's Theory of Cognitive Development
- ▶ Erikson's Psychosocial Theory of Development
- ▶ Kohlberg's Theory of Moral Development
- ▶ Vygotsky's Socio-Cultural Theory
- ▶ Bronfenbrenner's Ecological Theory

HUMAN DEVELOPMENT: Meaning, Concepts and Approaches



William Shakespeare



All the world's a stage, and all
the men and women merely
players: they have their exits
and their entrances; and one
man in his time plays many
parts, his acts being seven ages.



Definitions of Growth and Development

Growth

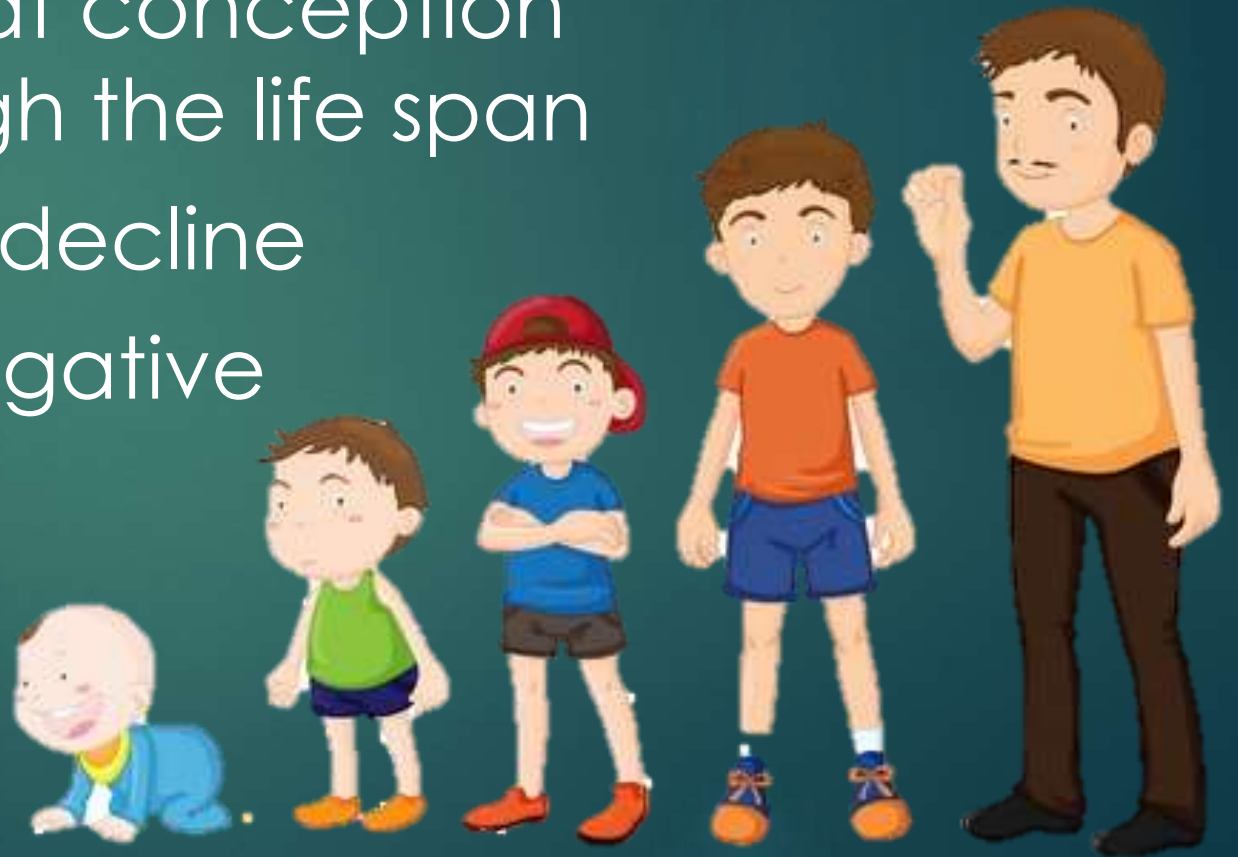
- ▶ Increase in physical size of a whole or any of its parts, or an increase in number and size of cells: Growth can be measured.

Development

- ▶ A continuous, orderly series of conditions that leads to activities, new motives for activities, and patterns of behavior

Human Development

- the pattern of movement or change that begins at conception and continues through the life span
- includes growth and decline
- can be positive or negative

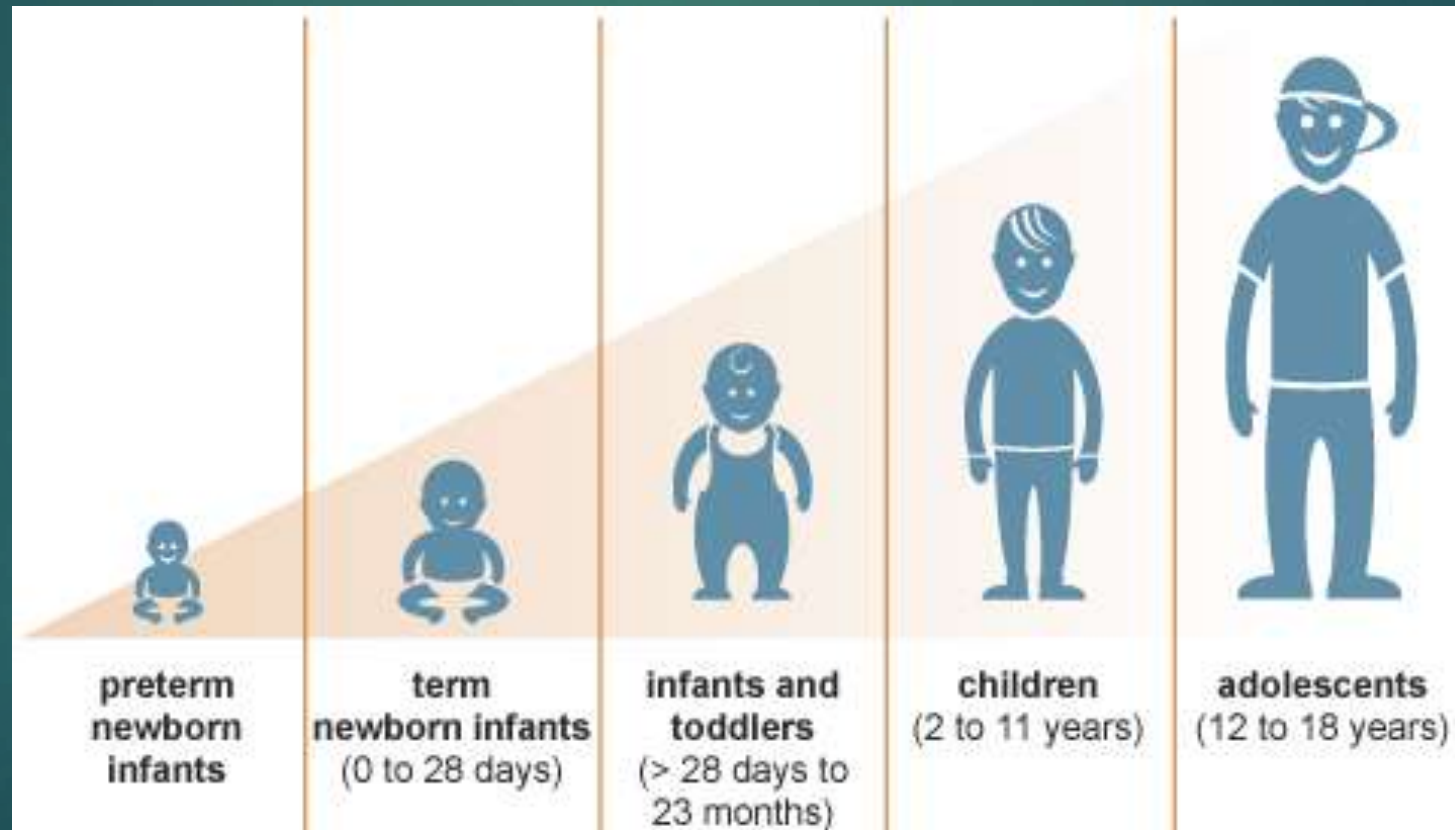


SOME PRINCIPLES OF HUMAN DEVELOPMENT



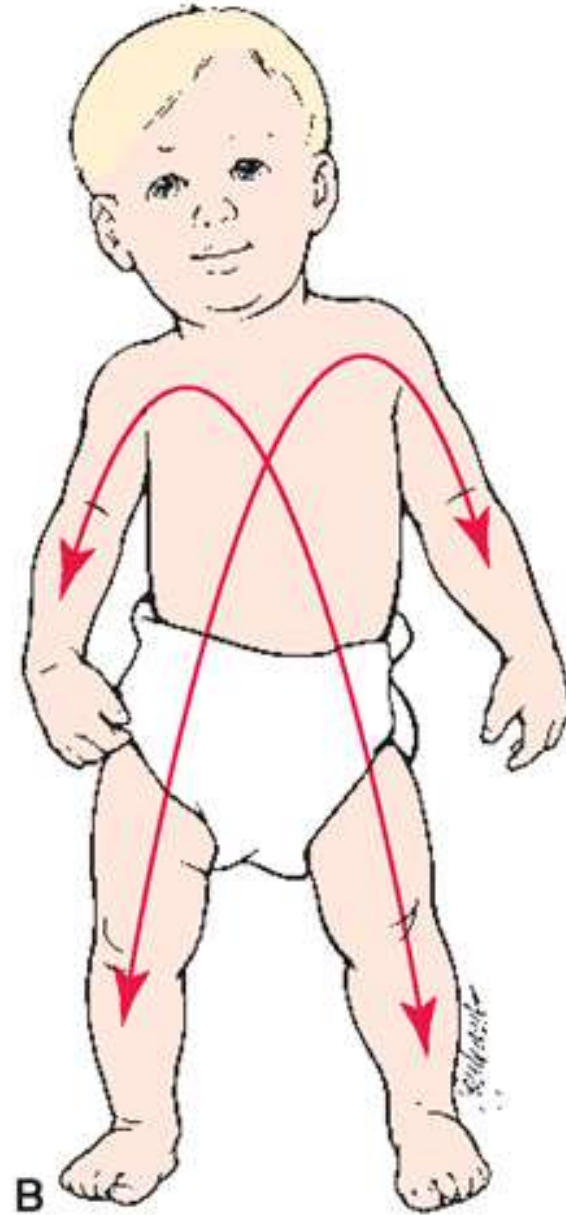
Principle 1

Development is relatively orderly.

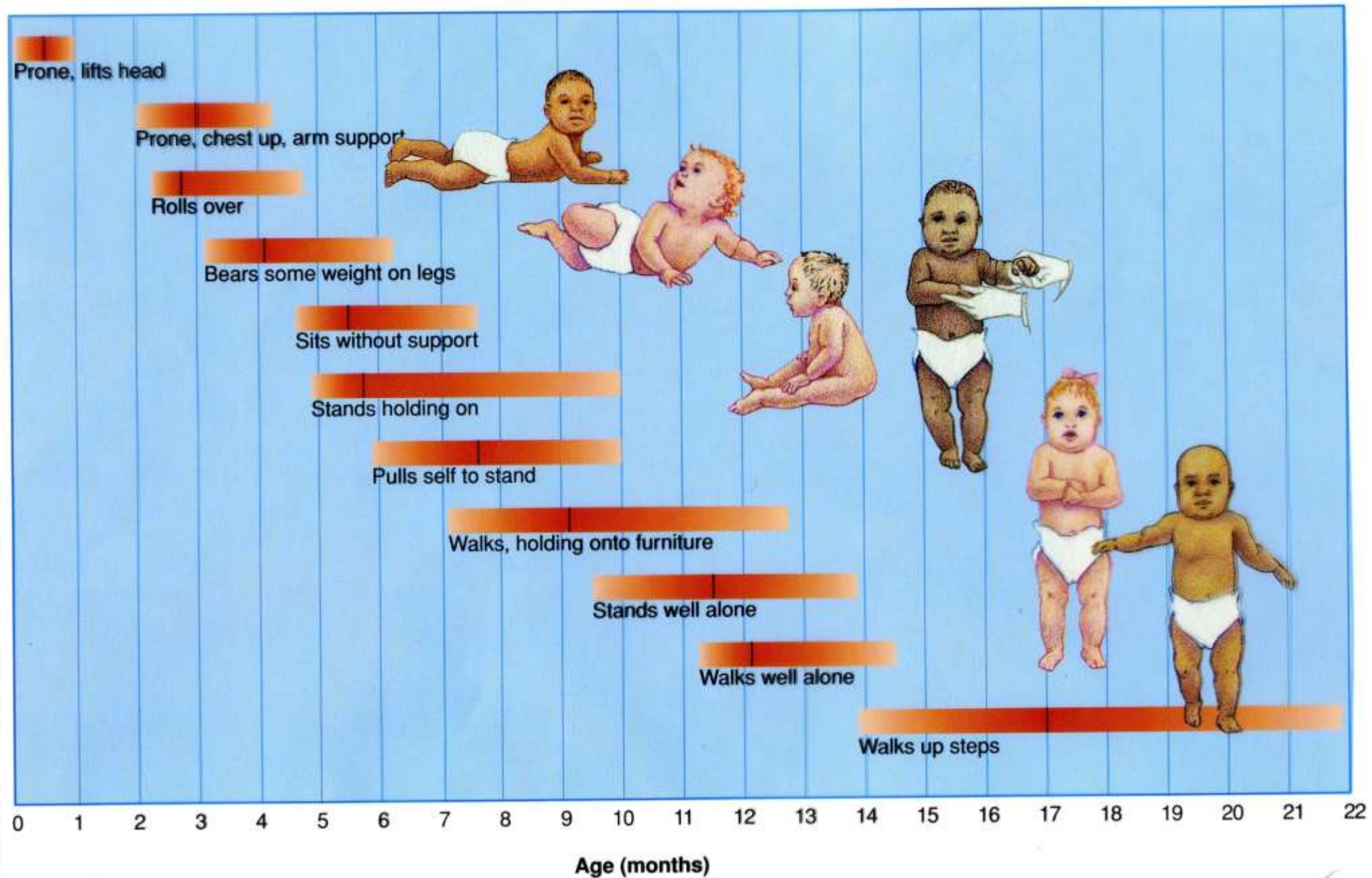


Proximodistal Pattern

Development proceeds from the center of the body outward.

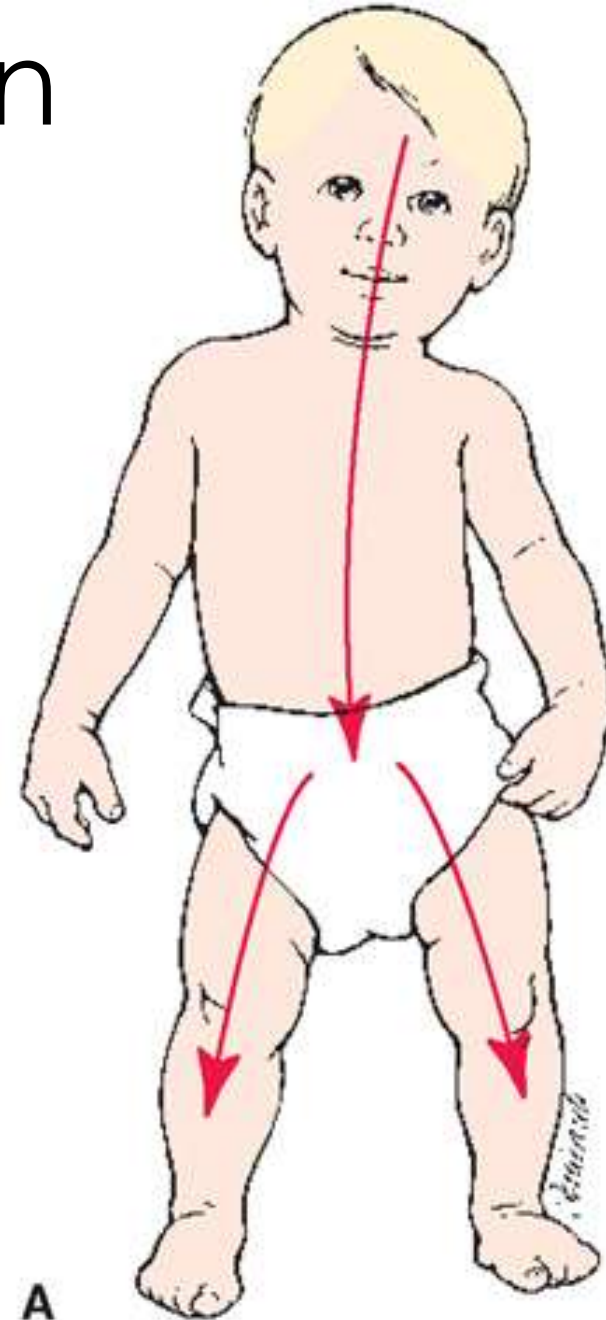


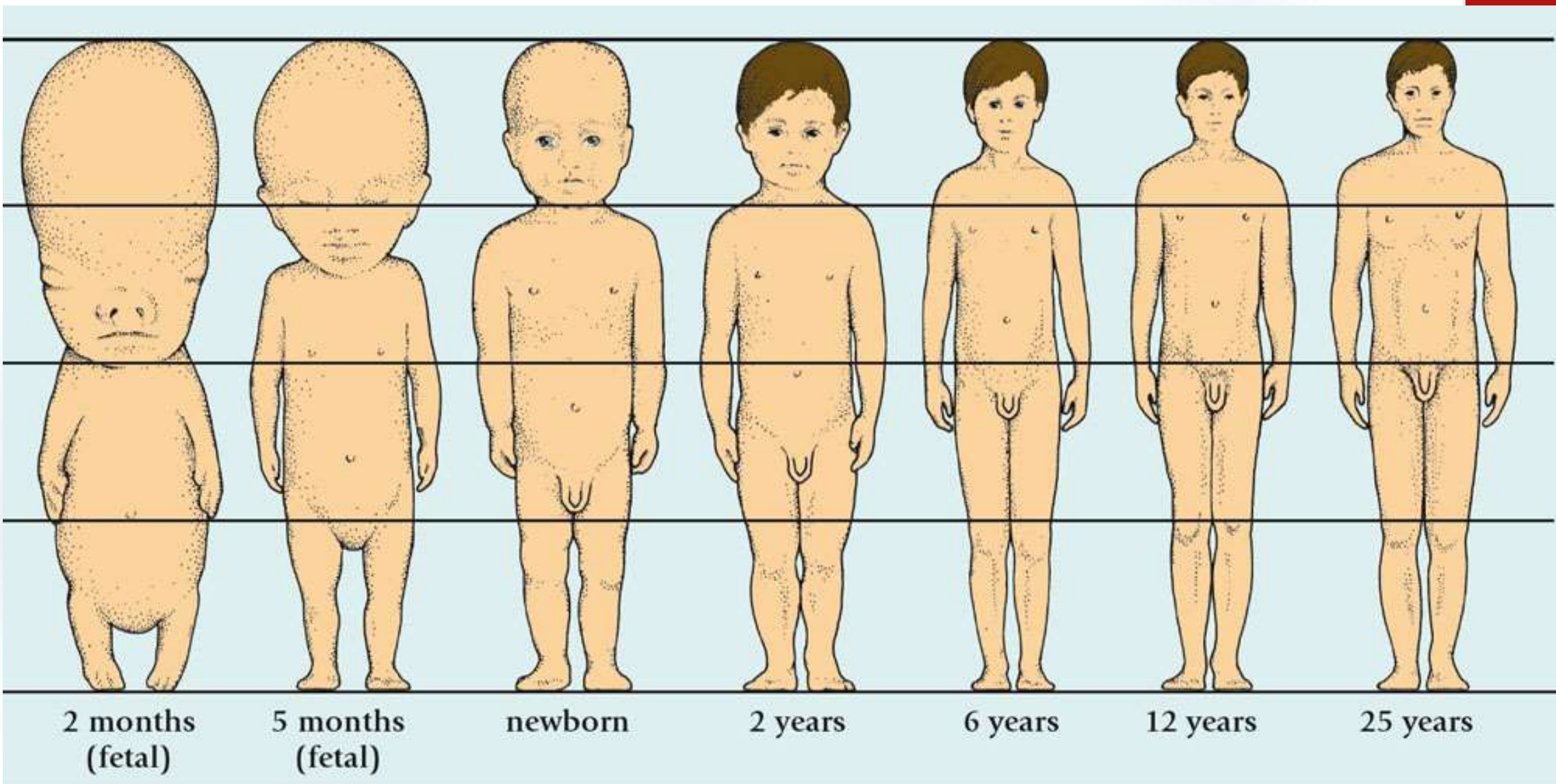
Landmarks in motor development



Cephalocaudal Pattern

Development proceeds from the head downward.





Principle 2

While the pattern of development is likely to be similar, the outcome of developmental process and the rate of development are likely to vary among individuals.



Principle 3

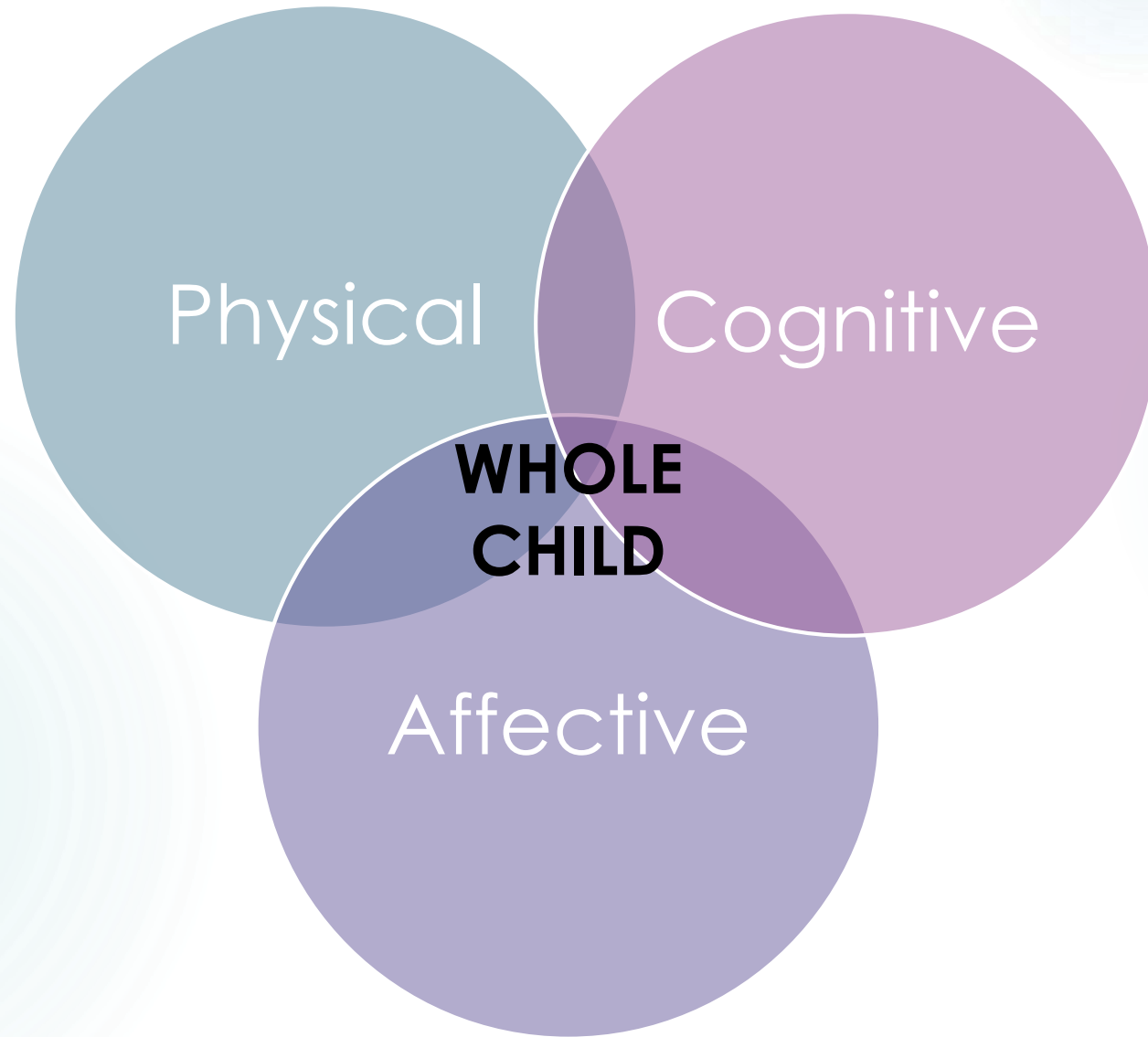


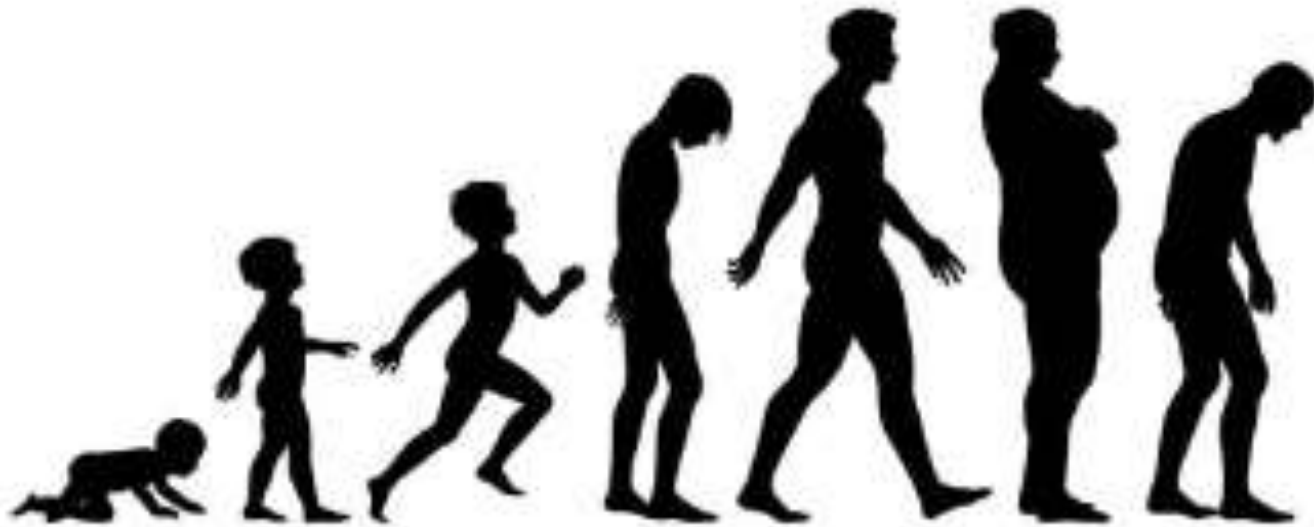
Development takes place gradually.

Principle 4

Development as a process complex because it is the product of biological, cognitive and socio-emotional process.







APPROACHES TO HUMAN DEVELOPMENT

Traditional

Human development shows extensive change from birth to adolescence, little or no change in adulthood and decline in the late old age.



Life-span

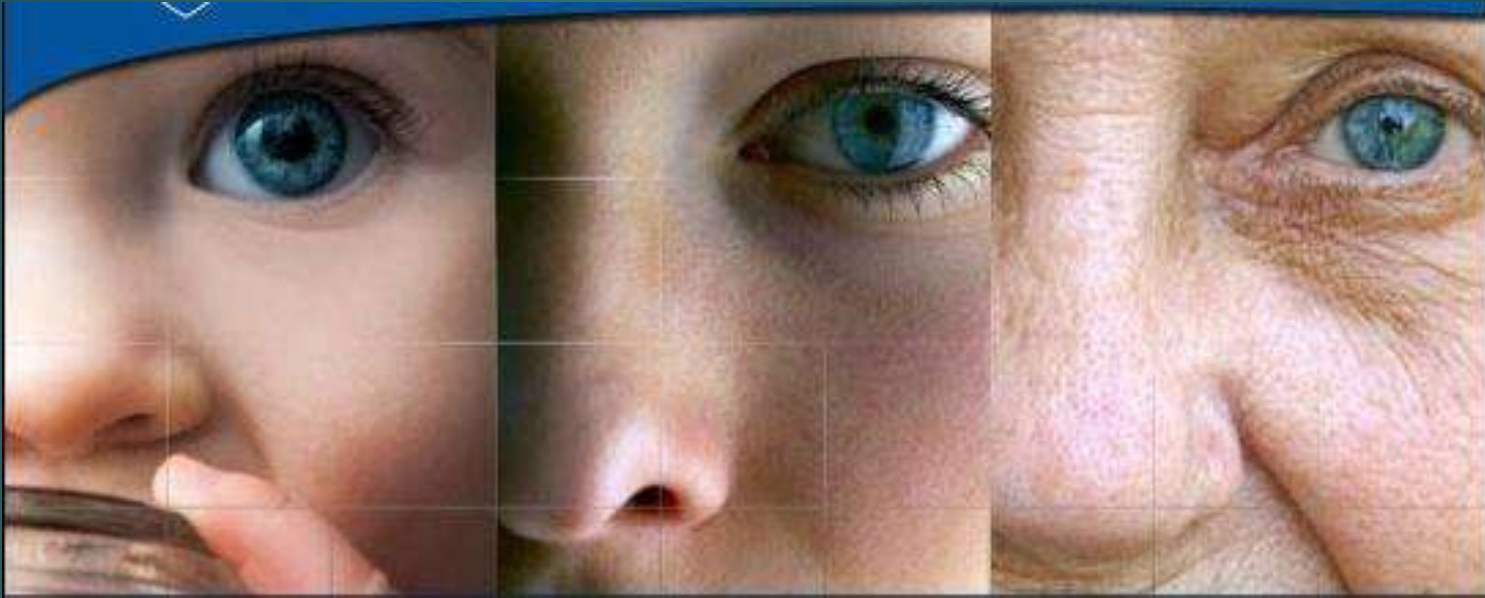
Human development occurs all throughout the life-span of an individual.



Characteristics of Life-span Development



Life-long



It does not end by adulthood and
no development stage dominates
the development.

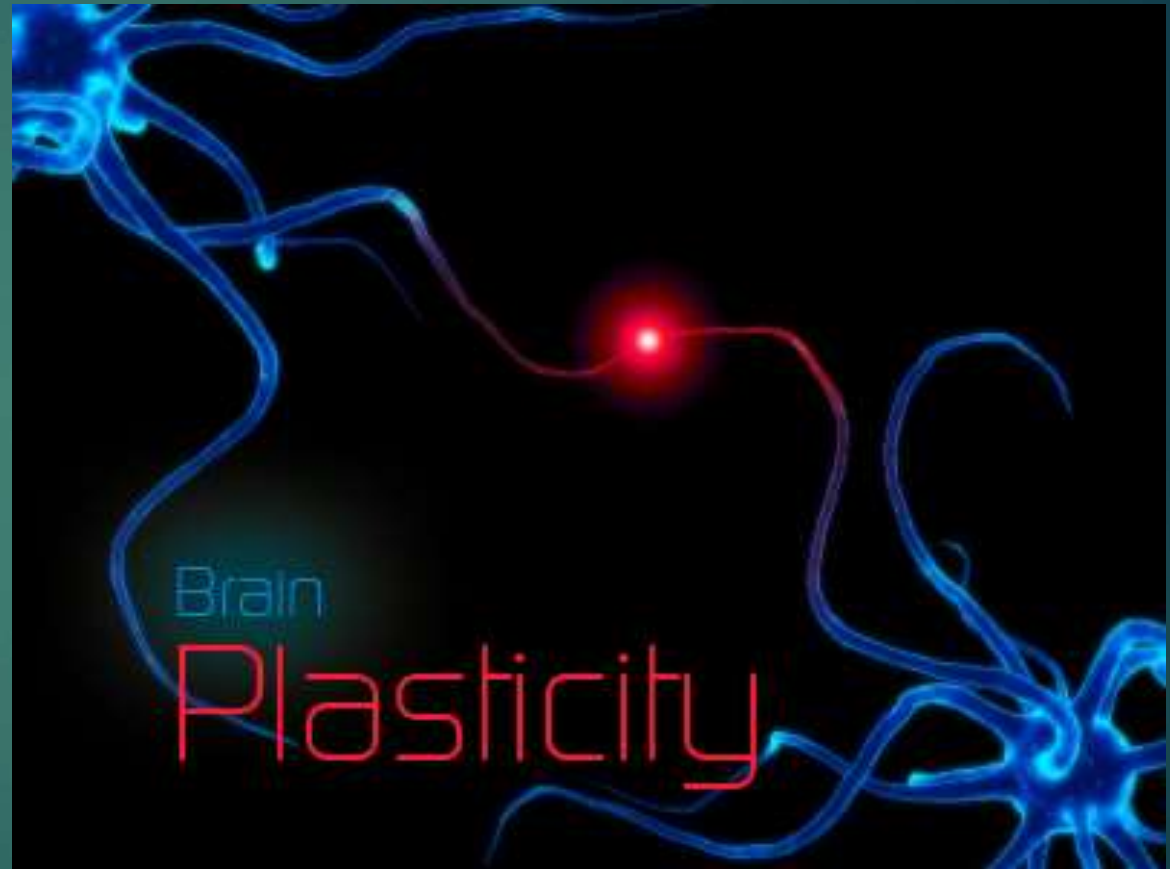


Multi-dimensional

Development consists of
biological, cognitive,
and socio-emotional
dimensions.

Plastic

Development is possible throughout the life-span.



Contextual

Individuals are changing beings in a changing world.



Growth, maintenance and regulation

These are considered goals in that varies in every developmental change.



TEST YOUR UNDERSTANDING

1. Development is a pattern of change.
2. Development is either growth or decline.
3. From both traditional and life-span perspectives development is lifelong.
4. In the development process, there are things that hold true to all people.
5. Individuals develop uniformly.

TEST YOUR UNDERSTANDING

- 6. Development is predictable because it follows an orderly process.
- 7. Development is unidimensional.
- 8. Development takes place in a vacuum.
- 9. The effect of biological process on development is isolated from the effect of cognitive and socio-dimensional processes.
- 10. The rate of development of children is uniform.

Stages of Human Development

DEVELOPMENTAL TASKS



Stages of Human Development

- 1. Pre-natal Period**
- 2. Infancy (birth – 2 years)**
- 3. Early Childhood (3 – 5 years)**
- 4. Middle and Late Childhood (6 to 12 years)**
- 5. Adolescence (13 – 18 years)**
- 6. Early Adulthood (19 – 29 years)**
- 7. Middle Adulthood (30 – 60 years)**
- 8. Late Adulthood (61 years and above)**

Progress Before Birth: Prenatal Development

3 PHASES

germinal stage = first 2 weeks

conception, implantation, formation of placenta

embryonic stage = 2 weeks – 2 months

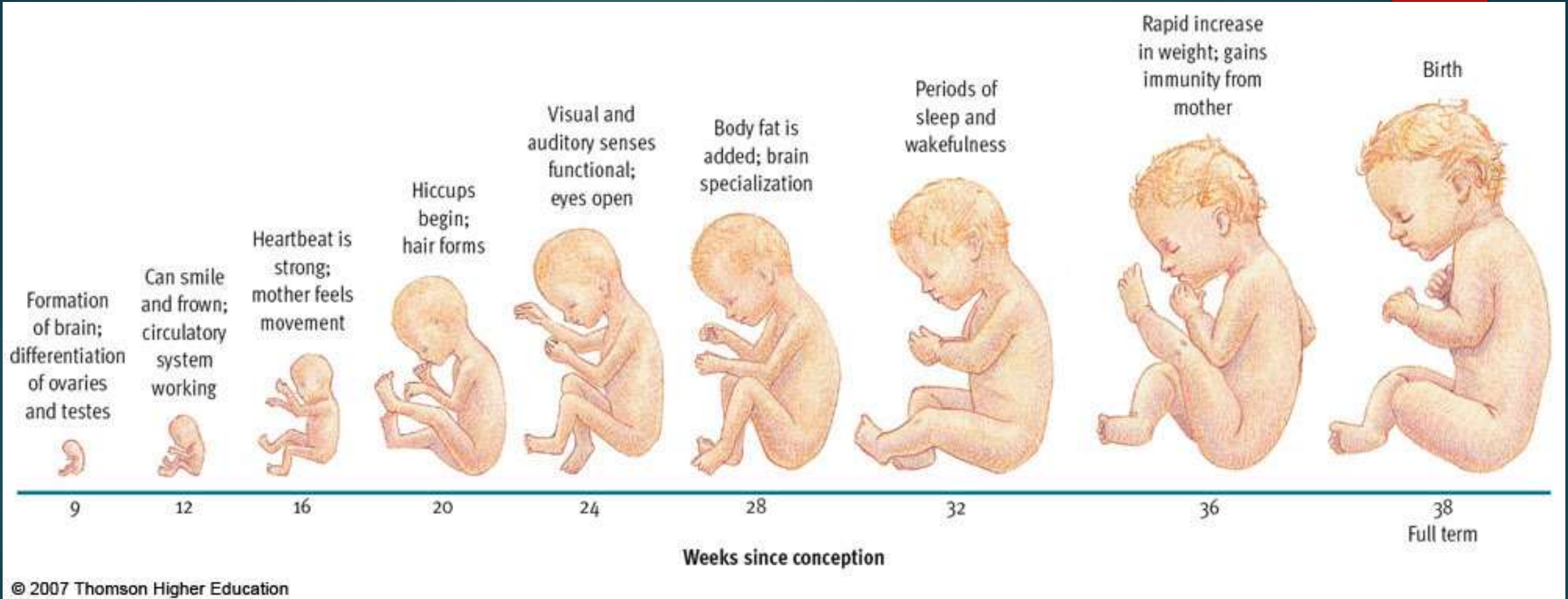
formation of vital organs and systems

fetal stage = 2 months – birth

bodily growth continues, movement capability begins, brain cells multiply age of viability

Prenatal

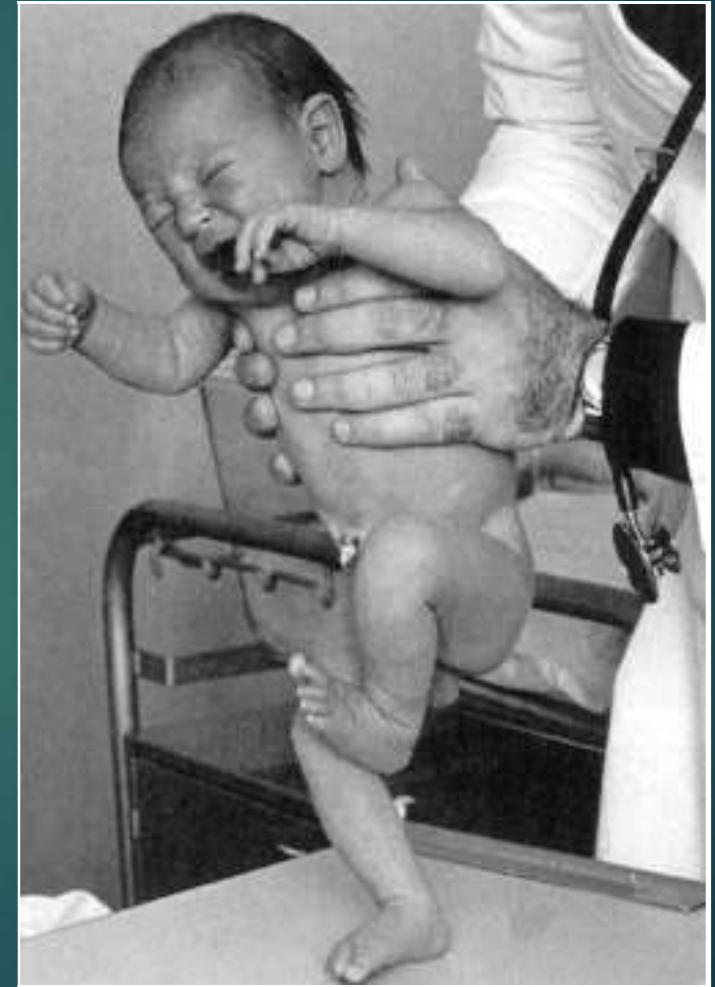
Overview of fetal development



It involves tremendous growth – from a single cell to an organism complete with brain and behavioral capabilities

Infancy (Birth – 2 years)

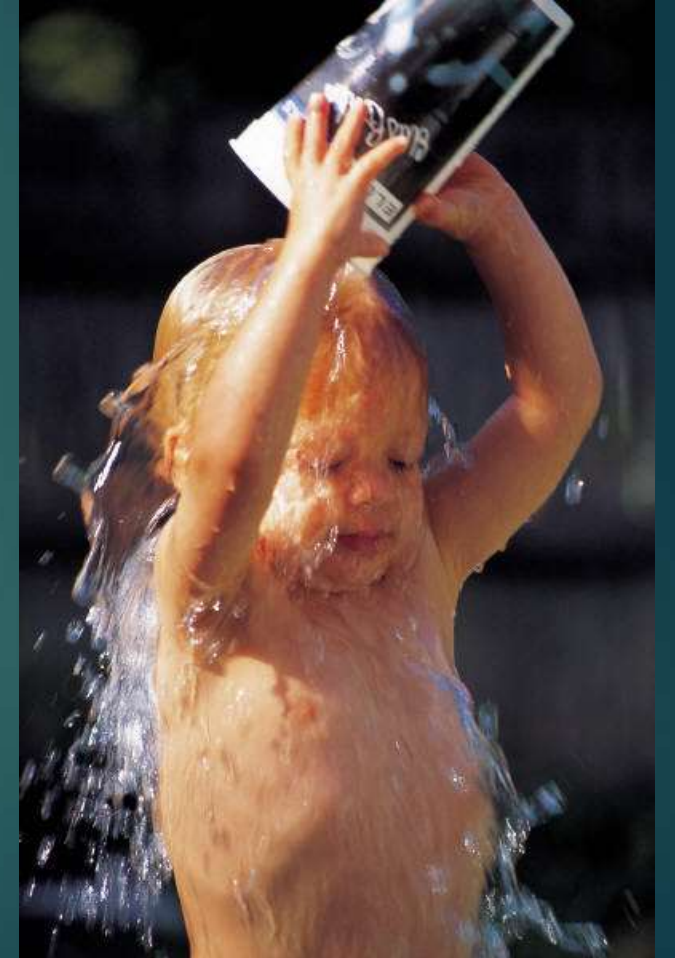
- extreme dependence on adults
- beginning psychological activities:
 - language, symbolic thought, sensorimotor coordination and social learning
- language of newborn is the cry
- usually eats every 2 to 3 hours
- uncoordinated movements
- toothless
- poor vision (focusing range 8 to 12 inches)
- usually doubles weight by 9 months
- responds to human voice & touch



Infancy

1 year old

- change from plump baby to leaner more muscular toddler
- begins to walk & talk
- ability for passive language (better understanding of what's being said)
- tentative sense of independence
- determined explorer



Infancy

2 years old

- begins to communicate verbally (name, etc.)
- can usually speak in 3 to 4 word sentences
- famous for negative behavior

“NO!” to everything!

temper tantrums

- will play side by side other children, but does not actively play with them
- great imitators



Early Childhood (3 – 5 years old)



Early Childhood

3 years old

- wants to be just like parents
- vocabulary and pronunciation continue to expand
- climbs stairs with alternating feet
- can briefly stand on one foot



Early Childhood

4 years old

- sentences are more complex; speaks well enough for strangers to understand
- imagination is vivid; line between what is real & imaginary is often indistinct
- develops fears (common fears: fear of dark, fear of animals, & fear of death)



Early Childhood

5 years old

- can hop on one foot & skip
- can accurately copy figures
- may begin to read
- socialize with other children their age



Middle and Late Childhood (6 – 12 years old)



Middle and Late Childhood

6 - 12 years old

- both large & small muscles well-developed
- developed complex motor skills
- from independent activities to same sex group activities
- acceptance by peers very important
- parental approval still important



Adolescence

13 - 18 years old

- traumatic life stage for child & parent
- puberty occurs
- extremely concerned with appearance
- trying to establish self-identity
- confrontations with authority



Early Adulthood/Young Adult

19 - 29 years old

- physical development complete
- emotional maturation continues to develop
- usually learned to accept responsibility for actions & accept criticism
- usually knows how to profit from errors
- socially progress from age-related peer groups to people with similar interests



Middle Adulthood

30 - 60 years old

- physical changes begin to occur:
 - hair begins to thin & gray
 - wrinkles appear
 - hearing & vision decrease
 - muscles lose tone
- main concerns: children, health, job security, aging parents, & fear of aging
- love & acceptance still take a major role



Late Adulthood

61 years and above

- fastest growing age bracket of society
- physical deterioration (brittle bones, poor coordination)
- some memory problems
- coping with retirement & forms of entertainment
- very concerned with health & finances
- significant number become depressed; suicide rate is high



“Who are you?” asked the caterpillar. Alice replied rather shyly, “I-I hardly know, Sir, just at present – at least I know who I was when I got up this morning, but I must have hanged several times since then.”

- LEWIS CARROLL



Developmental Tasks

- ▶ One that “arises at a certain period in our life, the **successful achievement** of which leads to happiness and success with later tasks while **failure** leads to unhappiness, social disapproval, and difficulty with later tasks.” by Robert Havighurst , 1972

Infancy and Early Childhood (0-5)	Middle Childhood (6-12)
Learning to w__k	Learning physical sk__ls necessary for ordinary games
Learning to take solid f__d	Building a wholesome att__de toward oneself
Learning to t__k	Learning to get along with age__
Learning to control the elimination of body w__te	Learning an appropriate s_x role
Learning s_x differences and sexual modesty	Developing fundamental skills in reading, writing, and cal____ing
Acquiring concepts and language to describe so__l and physical reality	Developing concepts necessary for everyday living
Readiness for r__ding	Developing conscience, mo__lity, and a scale or val__s
Learning to distinguish right from ____ and developing conscience	Achieving personal in____dence Developing acceptable attitudes toward society

Adolescence (13-18)	Early Adulthood (19-29)
Achieving mature relations with both sexes	Selecting a mate
Achieving a masculine or feminine social role	Learning to live with a partner
Accepting one's physique	Starting a family
Achieving emotional independence of adults	Rearing children
Preparing for marriage and family life	Managing a home
Preparing for an economic career	Starting an occupation
Acquiring values and an ethical system to guide behavior	Assuming civic responsibility
Desiring and achieving socially responsible behavior	

Middle Adulthood (30-60)	Later Maturity (61 and over)
Helping teenage children to become happy and responsible adults	Adjusting to decreasing strength and health
Achieving adult social and civic responsibility	Adjusting to retirement and reduced income
Satisfactory career achievement	Adjusting to death of spouse
Developing adult leisure time activities	Establishing relations with one's own age group
Relating to one's spouse as a person	Meeting social and civic obligation
Accepting the physiological changes of middle age	Establishing satisfactory living quarters
Adjusting to aging parent	

TEST YOUR UNDERSTANDING



1. Development tasks are only for the first 3 stages of human development.
2. Failure of achieving developmental tasks in an earlier stage also means failure for the learner to master the developmental tasks in the next stage.
3. Preschool age corresponds to early childhood stage.
4. Adolescence is middle and late childhood stage.

TEST YOUR UNDERSTANDING

- 5. Teenage is middle childhood.
- 6. Mastery of fundamental skills is a major concern during early childhood.
- 7. Play is a great need of children in middle childhood.
- 8. Preparing children for school readiness is the major concern of middle childhood.

ISSUES ON HUMAN DEVELOPMENT

THE

NATURE

VERSUS

NURTURE

DEBATE

► Behavior Genetics

- The study of the influences of heredity and the environment on behavior

► “Nature”

vs.

“Nurture”



Our genetic influences
– inborn traits



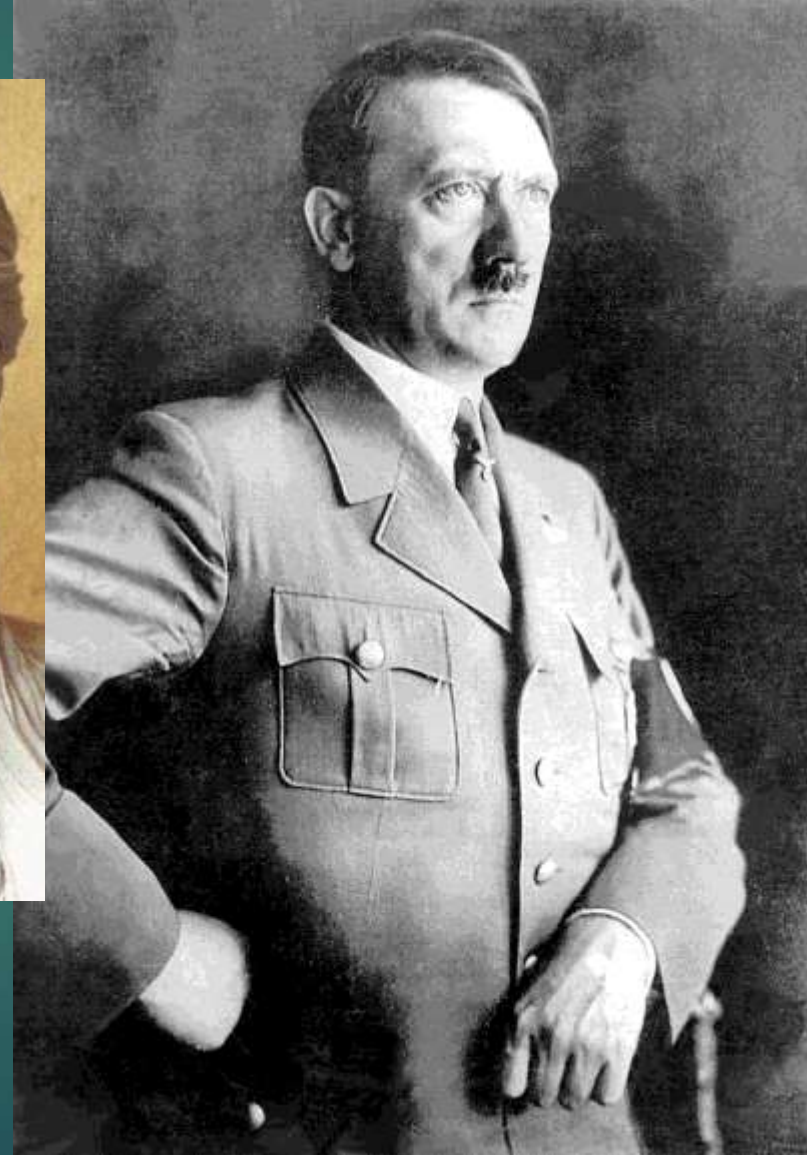
Environmental influences
(everything non-genetic)

What makes someone...



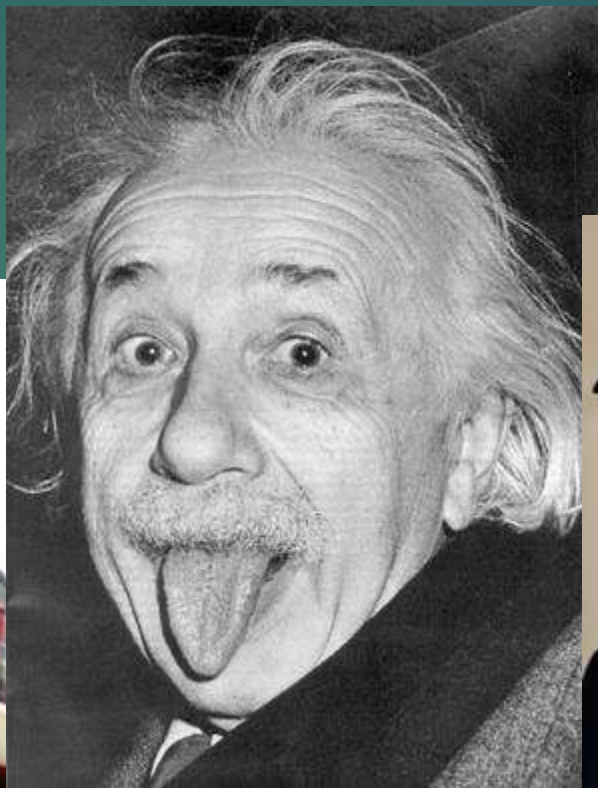
A superior athlete?

What makes someone...



Mean, cruel, or evil?

What makes someone...



Intelligent?

What makes someone...



Brave?

What makes someone...



Shy?
or
outgoing?



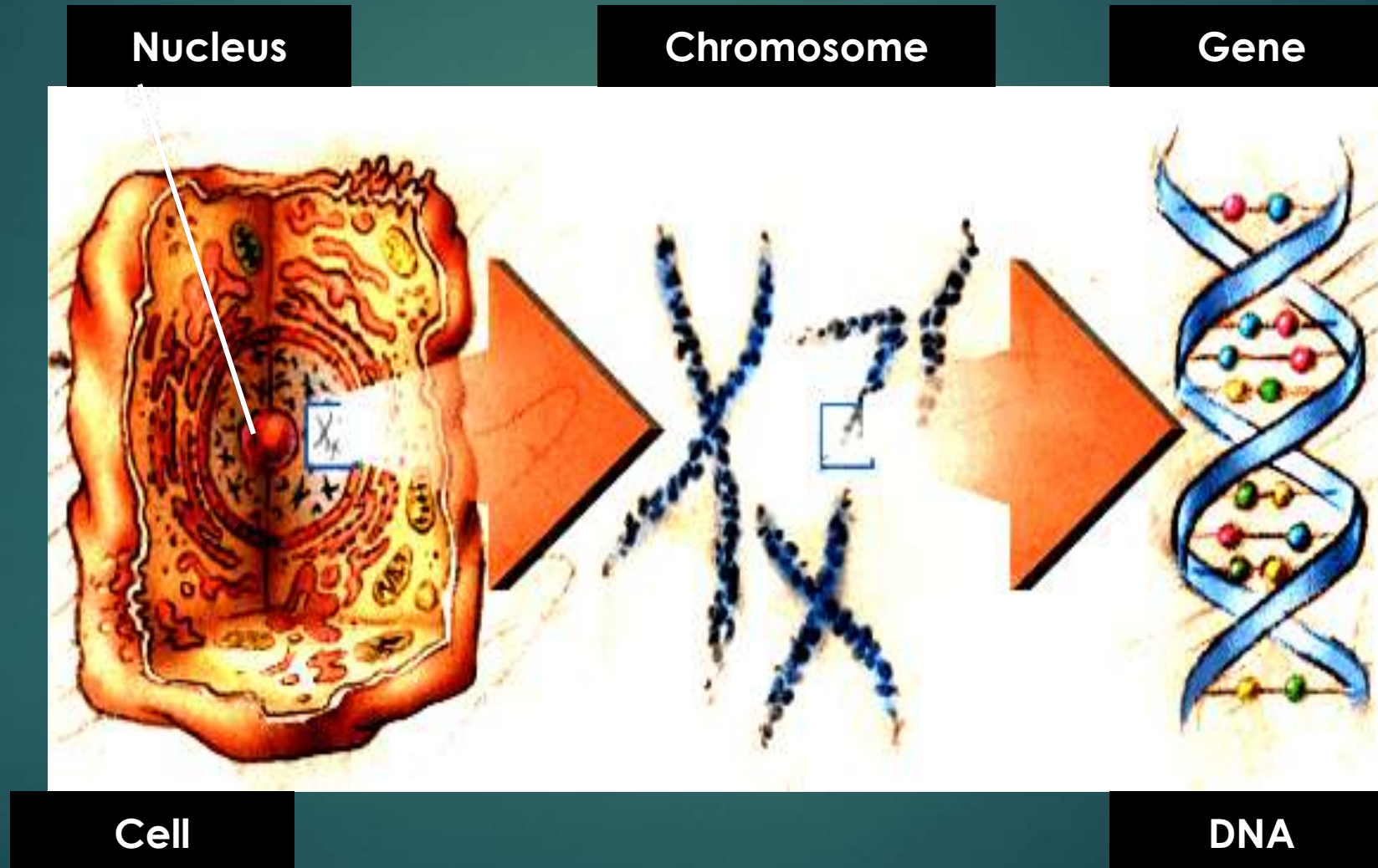
Is it
“nature”
or
“nurture”?

Nature

- ▶ People behave the way they do because they are animals who act in accordance with their animal instincts and are *determined by their biology.*



Genetics and Behavior





Nurture

- ▶ People behave the way they do because they are **determined** by the things other people teach them, the things they observe around them, and because of the different situations they are put in.



Is it your or genes/dna



It seems as if it is easy to prove that our physical traits – how we look, what our facial features are like, what our body type is like, is the result of natural inheritance or our genetic code.



See the
resemblance



Genes /
Natural
inheritance



Adoption Studies



?

=



?

=



Is the child more like the biological parents (“nature”) or adoptive parents (“nurture”)?

Twin Studies

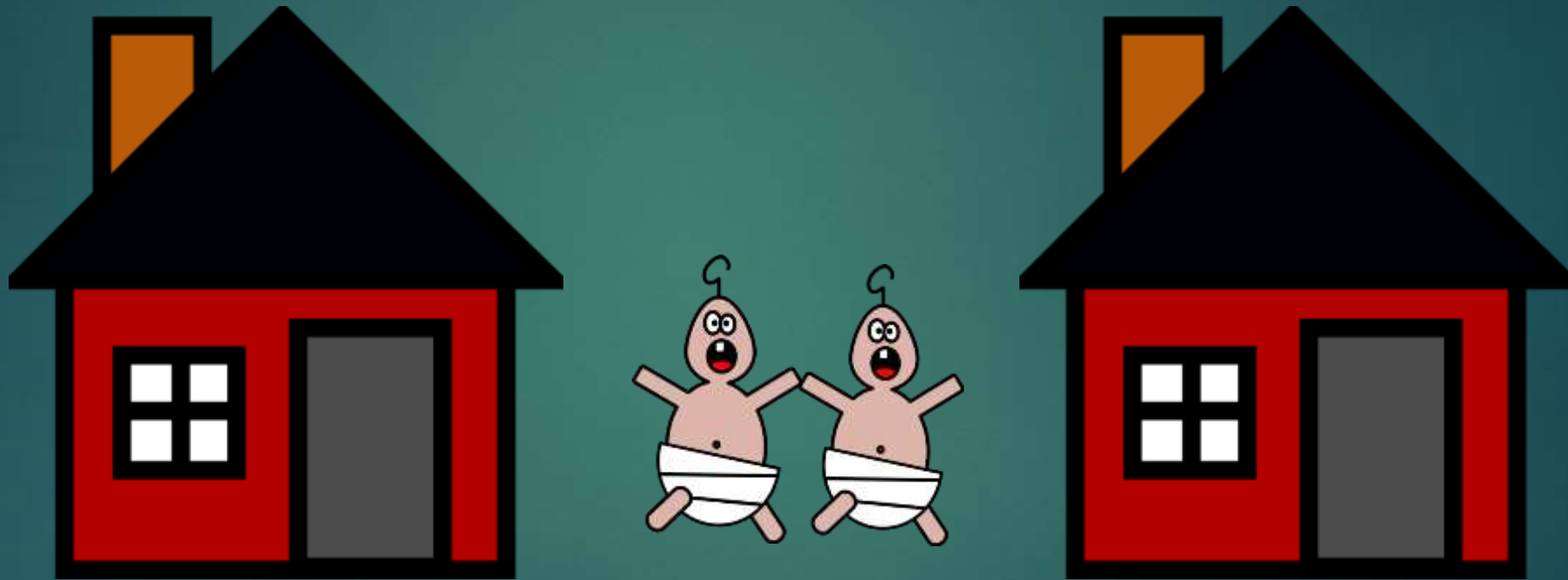


Can compare identical twins with fraternal twins

Monozygotic = identical twins (one egg, exact same genetics)

Dizygotic = fraternal twins (two eggs, no more genetically similar than other siblings)

Twin Studies

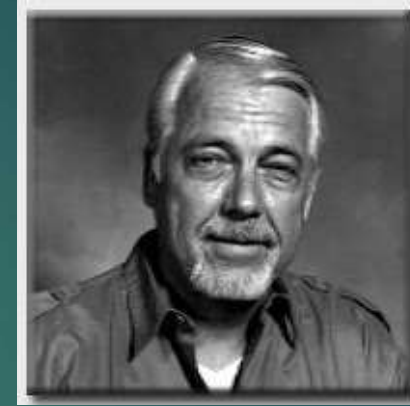


Can study identical twins separated at birth

Same “nature” ...

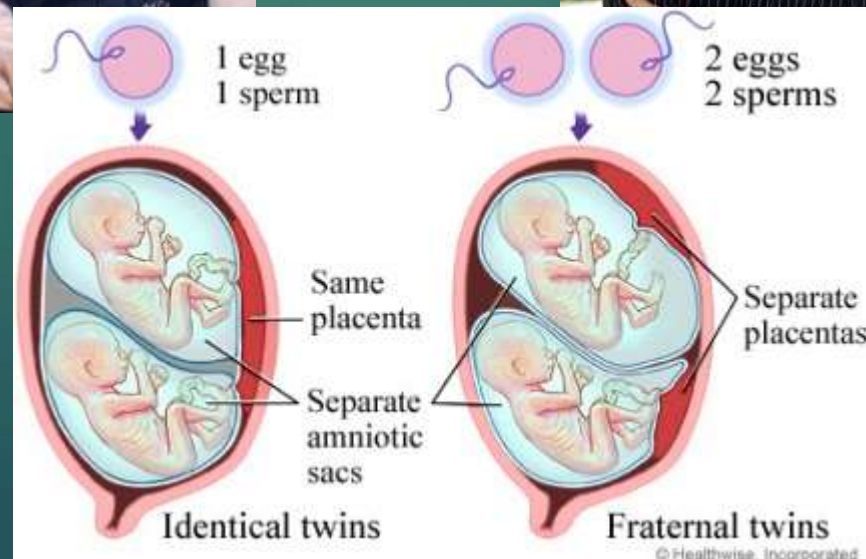
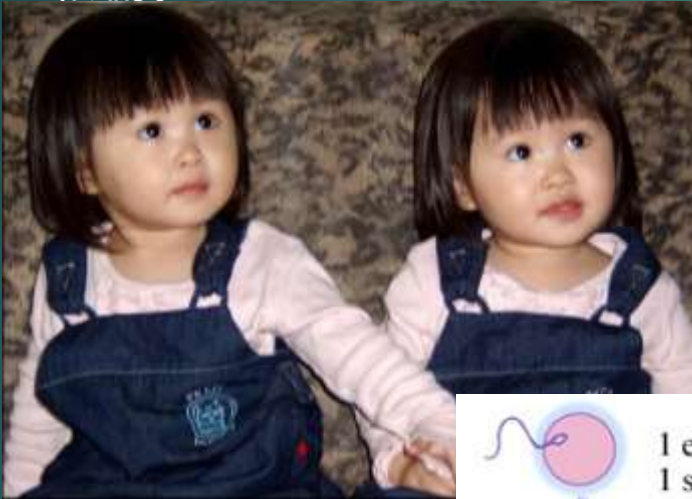
different “nurture”

Minnesota Twin Studies



- ▶ Dr. Thomas Bouchard & Dr. David Lykken
 - ▶ Conducted famous longitudinal studies on twins at U of M to attempt to determine which characteristics we inherit (nature) and which we acquire (nurture)
 - ▶ Included 8,000 pairs of twins born b/w 1936-1955
 - ▶ Hypothesis: Even if identical twins were reared apart, they should exhibit the same traits in many areas due to the fact that they share 100% of genetic blueprint. Those traits that differed were due to environmental influences.

- The Nature versus Nurture Debate has been going on for centuries.
- Most of the studies done on the Nature versus Nurture debate have been conducted on the study of twin behavior.
- Biologically there are two types of twins – Identical Twins and Fraternal



TASK: Listen/read along to the summary of the Bouchard-twin study in Minnesota

Jim (real name
James)
Lewis



Jim (real name
James)
Springer

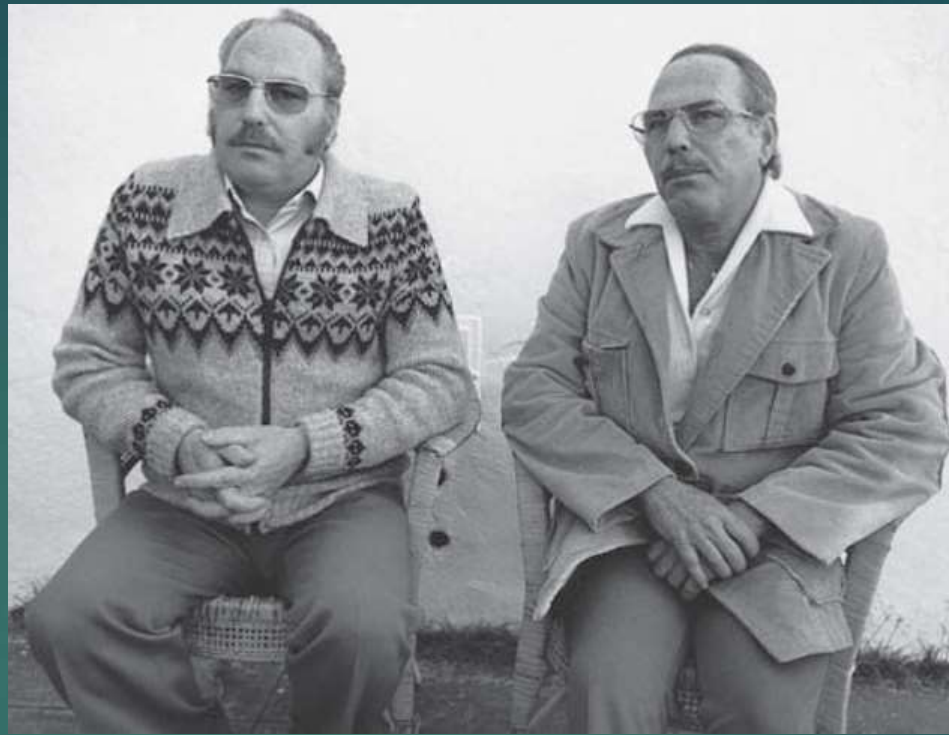
“Thomas Bouchard of the University of Minnesota did the most famous research on genetic influences in humans. He studied *identical twins separated since birth*. Identical twins come from a single egg, fertilized by a single sperm, which splits after the egg starts to develop. Therefore identical twins are closer to being genetically identical than any other humans. By studying identical twins who were separated at birth and raised by different families, Bouchard could see which similarities might emerge despite a different family environment. These similarities might be those that are heavily influenced by a person's genetic heritage.

Bouchard's data set was unique, probably a one-time event in history, because modern adoption agencies no longer break up sets of identical twins. Bouchard's project started when he read news reports of two identical twins reunited after a lifetime apart: James Lewis and James Springer were separated weeks after birth. When they were reunited, an extraordinary collection of coincidences emerged.”

http://www.intropsych.com/ch11_personality/bouchards_twin_research.html

"When the two first met, Lewis described it as "like looking into a mirror." For starters, both had the same first name. They were physically identical. But when they got talking, the similarities were astounding. Both had childhood dogs named Toy. Both had been nail biters and fretful sleepers. Both had migraines. Both had married first wives names Linda, second wives named Betty. Lewis named his first son James Allen, Springer named his James Alan. For years, they both had taken holidays on the same Florida beach. They both drank Miller Lite, smoked Salem cigarettes, loved stock car racing, disliked baseball, left regular love notes to their wives, made doll furniture in their basements, and had added circular white benches around the trees in their backyards. Their IQs, habits, facial expressions, brain waves, heartbeats, and handwriting were nearly identical. The Jim twins lived apart but died on the same day, from the same illness."

<http://blog.lib.umn.edu/reife014/myblog2/2012/04/twin-studies.html>



Oskar Stohr and Jack Yufe

In one case, identical twin babies (Oskar and Jack) were raised in extremely different cultures. The two were born in Trinidad and separated shortly after birth. After that, their childhoods were very different.

The mother took Oskar back to Germany, where his grandmother raised him as a Catholic and a Nazi youth. Jack was raised in the Caribbean as a Jew, by his father, and spent part of his youth on an Israeli kibbutz.

But similarities started cropping up as soon as Oskar arrived at the airport. Both were wearing wire-rimmed glasses and mustaches, both sported two-pocket shirts with epaulets. They share idiosyncrasies galore: they like spicy foods and sweet liqueurs, are absentminded, have a habit of falling asleep in front of the television, think it's funny to sneeze in a crowd of strangers, flush the toilet before using it, store rubber bands on their wrists, read magazines from back to front, dip buttered toast in their coffee... Bouchard professed himself struck by the similarities in their mannerisms, the questions they asked, their "temperament, tempo, the way they do things." (Holden, 1980)

Bouchard did *not* find outstanding similarities between identical twins on such standard measures as IQ tests or standardized personality tests, but he *did* find striking similarities were *mannerisms* (such as wearing rubber bands on the wrists, or reading magazines backwards), *personal choices* (such as choice of names for pets or children, or choice of clothing styles), and *expressive social behavior* (shyness or social ease, laughter, facial expressions and posture). These are exactly the sorts of things many of us refer to as *personality*, so in that sense Bouchard's findings can be interpreted as strong support for genetic influences on personality.

http://www.intropsych.com/ch11_personality/bouchards_twin_research.html

However there have been other twin studies that seem to point in an opposite direction – that of twins who are raised together in the same environment but yet develop very different and unique personalities. Even within the case of Oskar Stohr and Jack Yufe – they developed different worldviews and ideas about what is right and wrong, ethical or not ethical. Consider this excerpt:

Let's look more closely at the case of Oskar and Jack. Do they have any *differences*?

Well, for one, they had opposing political views—Jack is politically liberal, Oskar is conservative. These contrasting values certainly played a role in their relative ambivalence towards each other when they first met.

Our political views and values do appear to be strongly affected by our upbringing, as do our attitudes and religious beliefs. These are aspects of a child that a family environment can often influence. Although the effects of a shared environment are relatively small, we cannot say that genetics are the only factor contributing to our behavior.

Even so, the specific aspects of our environments that shape us are still not clear; the research is continuing to evolve. We often like to think that our experiences are what define us. We feel influenced by our surroundings, and we feel that we can influence the things around us through our personality and behavior.

Also consider this case:



“An interesting case to consider is that of Brian Dugan, a man who admitted to abducting, raping, and killing a 10-year-old girl in 1983. His defense lawyers used brain scans and the testimony of prominent neuroscientists to argue that Dugan's brain had been hardwired to commit violent acts, that it was in his nature to kill and he was unable to control his behavior.

This argument falls in sharp contrast to the idea on which laws are based, that all people have free will and can choose whether or not to follow the law. The environment may influence the ease of that choice--a starving man might have little regard for laws against theft if it means he can eat--but ultimately the individual has the ability to choose. To say that nature has the most influence in determining personality, then, provides a scapegoat for a person when he does something considered unlawful or morally unacceptable: I couldn't help it, it's in my nature. Like an animal, such an individual would exist outside the realm of right or wrong, innocent or guilty, because he would be acting purely on instinct. How could that person justifiably be punished? The converse, saying nurture has the most influence in determining personality, similarly frees a person from accountability for his actions by placing blame on those who raised him."

Developmental Issues

- ▶ Nature and Nurture: the extent to which development is influenced by biological inheritance and/or environmental experiences
 - ▶ Nature proponents argue that an evolutionary and genetic foundation produces commonalities in growth and development
 - ▶ Nurture proponents emphasize the importance of both the biological and social environment

Developmental Issues

- ▶ Stability and Change: the degree to which early traits and characteristics persist through life or change
 - ▶ Stability: traits and characteristics are seen as the result of heredity and early life experiences
 - ▶ Change: traits and characteristics can be altered by later experiences
 - ▶ Role of early and later experiences is hotly debated

Developmental Issues

- ▶ Continuity and Discontinuity: focuses on whether development is either:
 - ▶ A process of gradual, cumulative change (continuous)
 - ▶ A set of distinct stages (discontinuous)
- ▶ Evaluating Developmental Issues:
 - ▶ Most developmentalists acknowledge that development is not all-or-nothing
 - ▶ There is debate regarding how strongly each of these issues influences development

The Nature of Development

□ Conceptions of Age

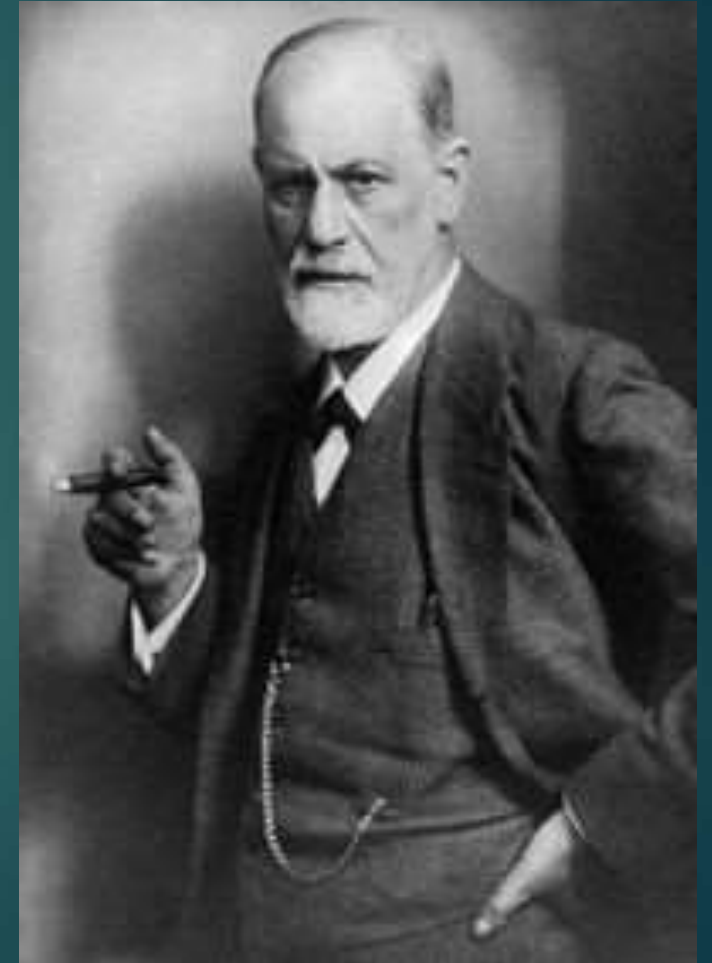
- How relevant is chronological age to understanding a person's psychological development?
- How should age be conceptualized?
 - **Chronological age:** number of years that have elapsed since birth
 - **Biological age:** a person's age in terms of biological health
 - **Psychological age:** an individual's adaptive capacities compared with those of other individuals of the same chronological age
 - **Social age:** social roles and expectations related to a person's age

DEVELOPMENTAL THEORIES



PSYCHOANALYTIC THEORIES

SIGMUND FREUD



Overview

- ▶ Focus on sex and aggression (unconscious)
- ▶ Gained popularity because of Freud's faithful followers.
- ▶ Based on clinical cases handled by Freud, personal dream analyses, and vast readings on sciences and humanities.
- ▶ Psychoanalysis was not an academic product, nor was it pure science
- ▶ Psychoanalysis is both an approach to therapy and a theory of personality



VIEW OF HUMAN NATURE



1. DETERMINISTIC

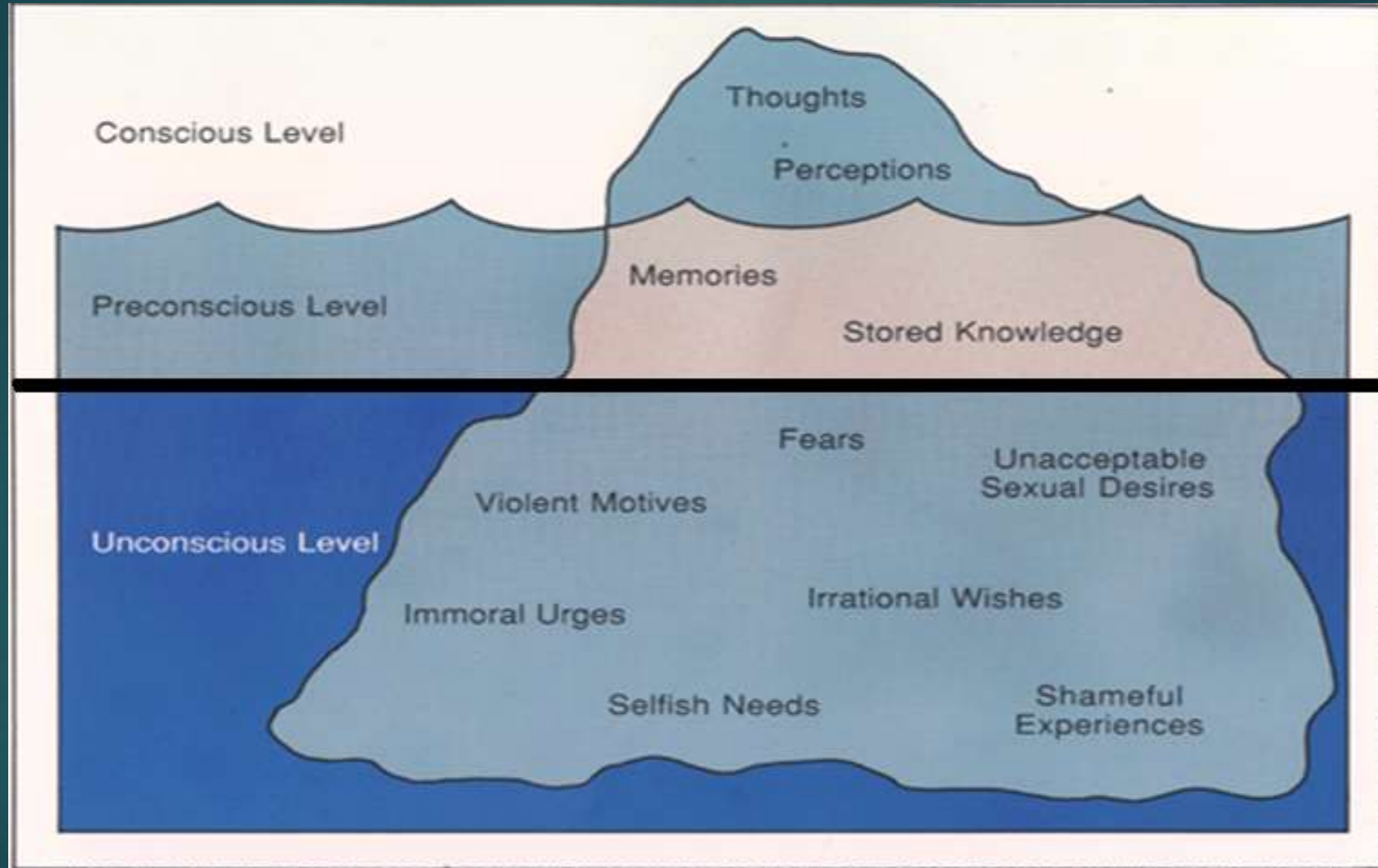
- ▶ Life is about gaining pleasure and avoiding pain

2. HUMAN AS ENERGY SYSTEM

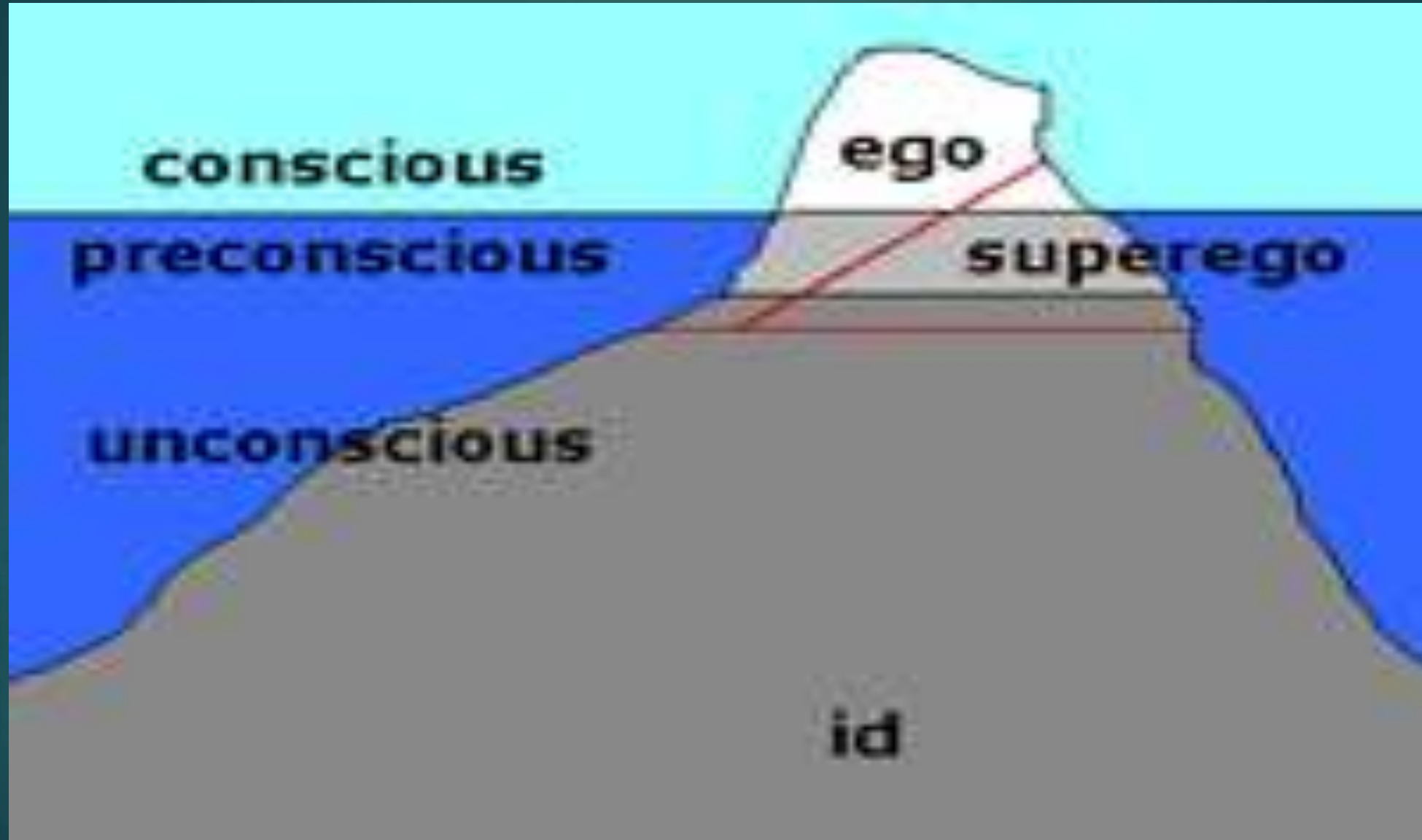
- ▶ Freud believe that human are motivated by the unconscious, where the Id is found along with the aggression and sex instincts



Levels of Mental Life



Structure of Personality

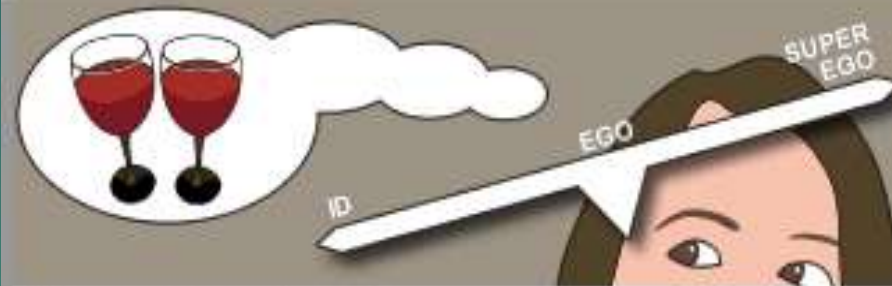


Functional or Dynamic Theory



Dynamic equilibrium

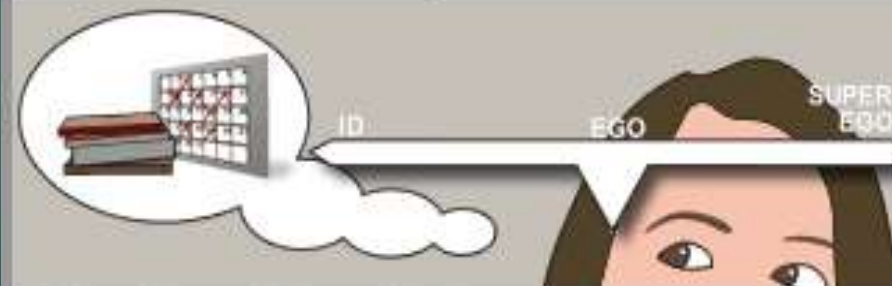
Exams are approaching. Should you



Do no revision and socialise



Do nothing but revision, having no rest or relaxation



Revise with a realistic revision timetable

► Freud's Psychosexual Stages of Development



ORAL 0-2
Infant achieves gratification through oral activities such as feeding, thumb sucking and babbling.



ANAL 2-3
The child learns to respond to some of the demands of society (such as bowel and bladder control).



PHALLIC 3-7
The child learns to realize the differences between males and females and becomes aware of sexuality.



LATENCY 7-11
The child continues his or her development but sexual urges are relatively quiet.



GENITAL 11-Adult
The growing adolescent shakes off old dependencies and learns to deal maturely with the opposite sex.



Freud's Psychosexual Stages of Development

Stage	Ages	Erogenous Zone	Key Events or Complexes	Possible problems caused by fixation
Oral	Birth to 18 mos. 	• Mouth	• Breast feeding, eating, chewing & biting 	• <u>Oral receptive</u> – smoking, drinking alcohol, overeating • <u>Oral aggressive</u> – biting nails, cursing, gossiping, overly demanding 



Freud's Psychosexual Stages of Development

Stage	Ages	Erogenous Zone	Key Events or Complexes	Possible problems caused by fixation
Anal	18 mos. to 3 yrs. 	• Anus	• Toilet training 	• <i>Anal retentive</i> – neat, stingy, rigid • <i>Anal expulsive</i> – messy & disorganized



Freud's Psychosexual Stages of Development

Stage	Ages	Erogenous Zone	Key Events or Complexes	Possible problems caused by fixation
Phallic  Anthony Rolfe Johnson as Oedipus and Yvonne Naef as Jocasta in the Vlaamse Opera production of Stravinsky's <i>Oedipus Rex</i> Photo © Johan Jacobs	3 - 6	• Genitals	• Oedipus Complex: Competition against father for mother's affection; fear of castration (castration anxiety) results in identification with father • Electra complex – female version • Penis envy – feeling of loss that generates hostility toward mother	• Going through life trying to prove manhood • Searching for mother-figure in relationships • Gender identity difficulties • Sexual difficulties • Women feeling inferior



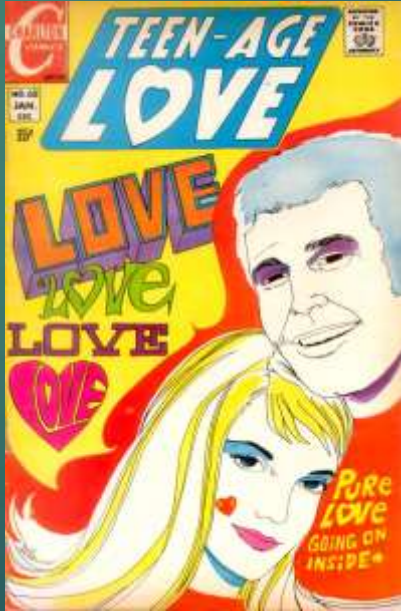
Freud's Psychosexual Stages of Development

Stage	Ages	Erogenous Zone	Key Events or Complexes	Possible problems caused by fixation
Latency	6 to 12	NA	• Repression of sexual feelings • Focus on social & intellectual skills • Same sex friendship	Continued repression of sexual urges and an increased use of defense mechanisms such as sublimation





Freud's Psychosexual Stages of Development

Stage	Ages	Erogenous Zone	Key Events or Complexes	Possible problems caused by fixation
Genital	12 & on	Genitals	• Seeking intimate relationships  A colorful comic book cover for 'TEEN-AGE LOVE'. The title is in large, stylized letters. Below the title, there's a drawing of a young man and woman. The man is in the foreground, looking towards the viewer, and the woman is behind him, looking at him. The cover has a yellow background with red and blue accents. There's a small logo in the top left corner and some text at the bottom right.	(Fixation may have occurred at earlier stages) Continued problems depending on fixation patterns

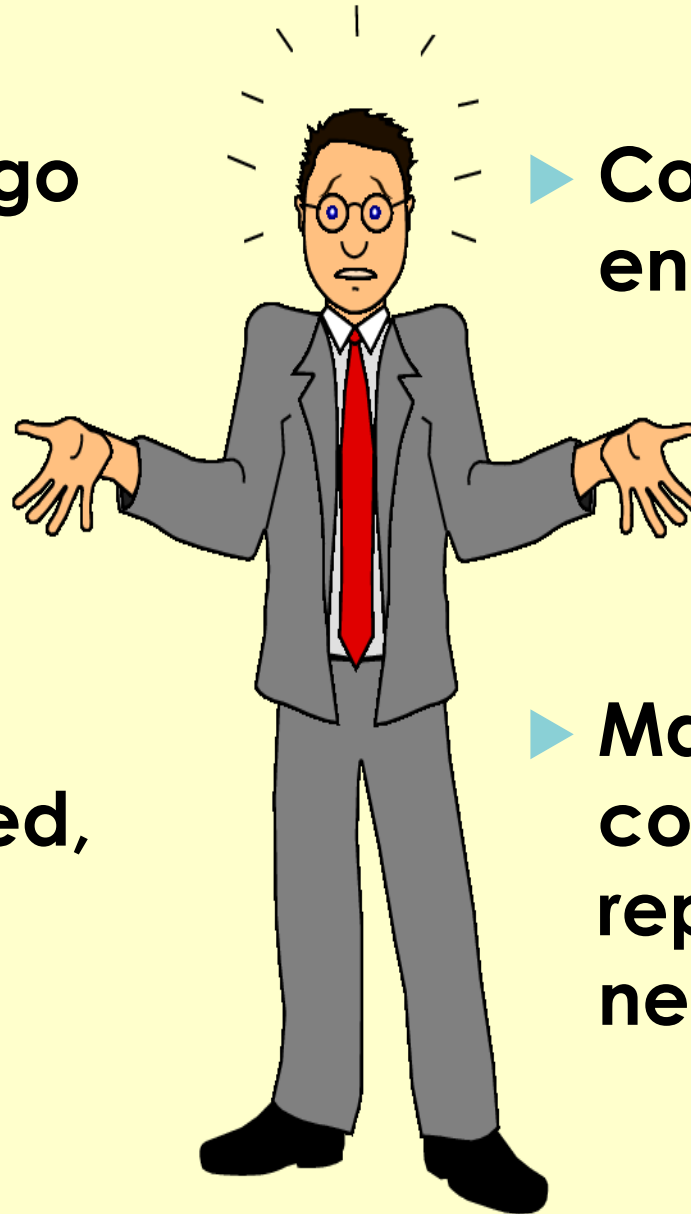
DEFENSE MECHANISMS

▶ Protects the ego from anxiety

▶ Consumes psychic energy.

▶ Normal and universally used, unless taken extremely.

▶ May lead to compulsive, repetitive, & neurotic behavior.



Defense Mechanisms

- ▶ **Projection** — In this mechanism, an individual puts the blame of his own failure upon others and some unfavourable factors of his environment. Blaming others for his mistake **e.g. a student projects that he has no romantic feelings to his classmate.**
- ▶ **Sublimation** — It is a defence mechanism in which unacceptable desire are redirected into socially accepted channels. **e.g. Anger -Kick boxing -- some people, poem writing, engage in social services etc.**
- ▶ **Repression** — Pushes threatening thoughts back into the unconscious - **Posttraumatic Stress Disorder- PTSD – Common with veterans and victims of sexual abuse**



Defense Mechanisms

- ▶ **Rationalisation** — An individual tries to justify his failure by giving some excuses e.g. A student makes use of rationalisation, when he tries to blame teachers for hard question paper.
- ▶ **Compensation** — It is an attempt to cover ones deficiency in one field by exhibiting his strength in another field e.g. If a student is not good in his studies, may show his ability in sports.





Defense Mechanisms

- **Identification** — It is a process which may operate outside and beyond conscious awareness. Hero worshipping by an individual is a sort of identification where an individual identifies himself with a **popular hero or an actor.**
- **Displacement** — An individual does something as a substitute for something else **e.g. If a wife gets angry with Husband and cannot say anything to him, she beats her child.**



Defense Mechanisms

- ▶ **Withdrawal** — Some persons withdraw themselves from the circumstances that cause tension, frustration or pain e.g. If a person is being humiliated or laughed at, he may shut himself in a room and may not need any one.
- ▶ **Day-dreaming** — It is a defence mechanism which sometimes helps in making adjustment. e.g. A young man who has been jilted in love, dreams of becoming a bride groom and feels satisfaction in the imaginary world.



Defense Mechanisms

- ▶ **Denial** – Simplest form of self defence eg. If a person is diagnosed as having cancer, they will first get shock, then start denying reality saying perhaps that the diagnosis was not proper
- ▶ **Reaction Formation** – It is the replacement in consciousness of an anxiety producing impulse or feeling by its opposite eg. A person who hates another cannot accept the painful fact of hating and so shows extraordinary love towards that person



Defense Mechanisms

- ▶ **Introjection** – taking in and accepting uncritically the values and standards of others **eg. If a child is constantly called stupid, the child thinks that it is really stupid**
- ▶ **Regression** - reverting back to an earlier stage.
- ▶ **Fixation** - remaining at the present, more comfortable psychological stage.



THEORY OF COGNITIVE DEVELOPMENT

JEAN PIAGET



Jean William Fritz Piaget



- ▶ Born: 9 August 1896
Neuchatel, Switzerland
- ▶ Died: 16 September 1980
Geneva, Switzerland (aged 84)
- ▶ University of Neuchatel;
University of Zurich
- ▶ Fields: Developmental
Psychology, Epistemology

Piaget was 10 years old when he published his first article on an albino sparrow



(Littlefield Cook & Cook, 2005/2009, p. 152)

Binet Laboratory

- ▶ Age 21: Piaget earns his PhD and heads off the work at the Binet laboratory with Theophile Simon and Alfred Binet
 - ▶ There he learned many of the key ideas in his revolutionary theory of cognitive development



BINET

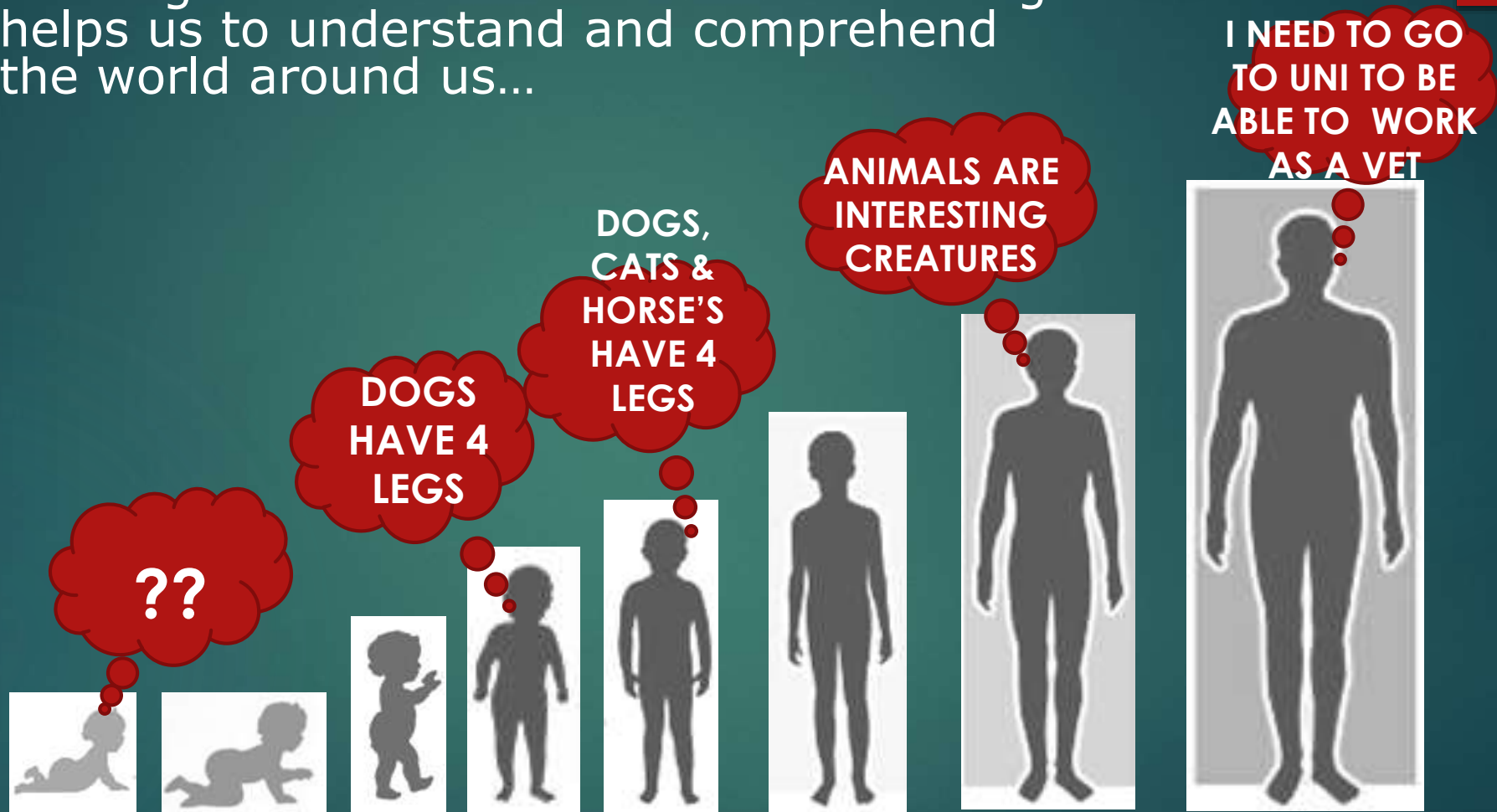


PIAGET

(Littlefield Cook & Cook, 2005/2009, p. 152)

How do we understand the world around us?

As we get older our collection of knowledge helps us to understand and comprehend the world around us...

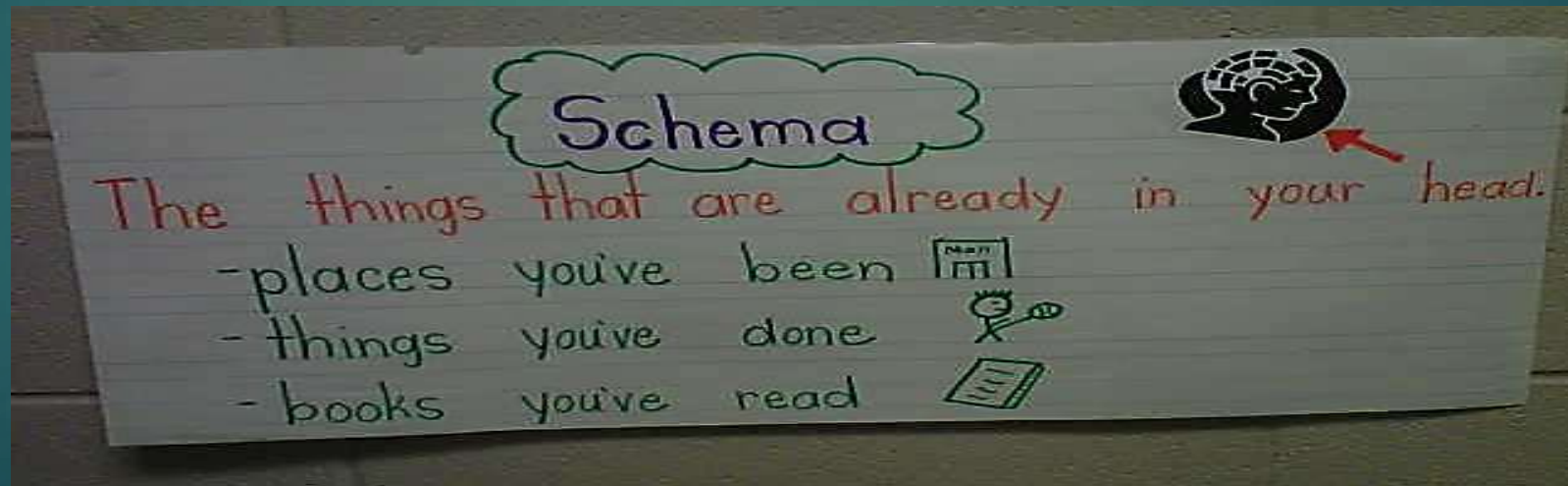


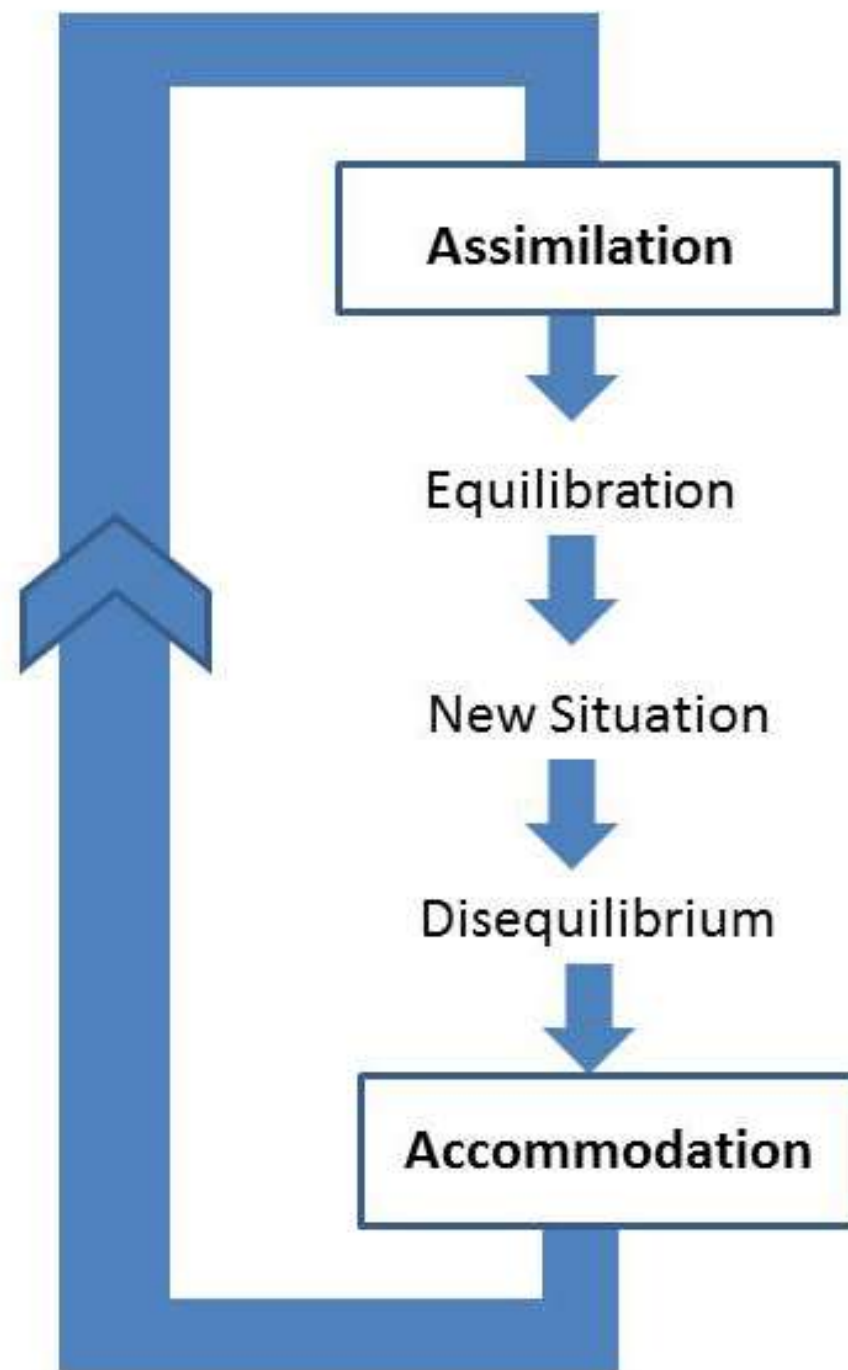


Three Basic Components to Piaget's Cognitive Theory

SCHEMA

- ▶ Basic building block of intelligent behavior.
- ▶ Units of knowledge
- ▶ Set of linked mental representations of the world, which we use both to understand and to respond to situations.





SENSORIMOTOR STAGE

The child begins to interact with the environment.



0-2

PREOPERATIONAL STAGE

The child begins to represent the world symbolically.



2-6 or 7

CONCRETE OPERATIONAL STAGE

The child learns rules such as conservation.



7-11 or 12

FORMAL OPERATIONAL STAGE

The adolescent can transcend the concrete situation and think about the future.



12-Adulthood



4 stages of cognitive development

Stage 1: Sensorimotor Thought (birth-2 years)

Babies are stuck in the HERE AND NOW world

they “know the world only in terms of their own sensory input (what they see, smell, taste, touch, and hear) and their physical or motor actions on it (e.g. sucking, reaching, grasping)



(Littlefield Cook & Cook, 2005/2009, p. 157)

Babies lack REPRESENTATIONAL THOUGHT or ability to think through the use of symbols

CAN YOU THINK WITHOUT WORDS??

(Littlefield Cook & Cook, 2005/2009, p. 158)

NO! of course not!

That's why Piaget says babies cannot think!

- ▶ Evidence of representational thought emerges from the use of language and

ObJeCt PeRmAnEnCe

- ▶ “the fact that objects, events, or even people continue to exist when they are not in the infants direct line of sensory or motor action” (Littlefield Cook & Cook, 2005/2009, p. 159)

The understanding of object permanence marks the change into...





PREOPERATIONAL THOUGHT

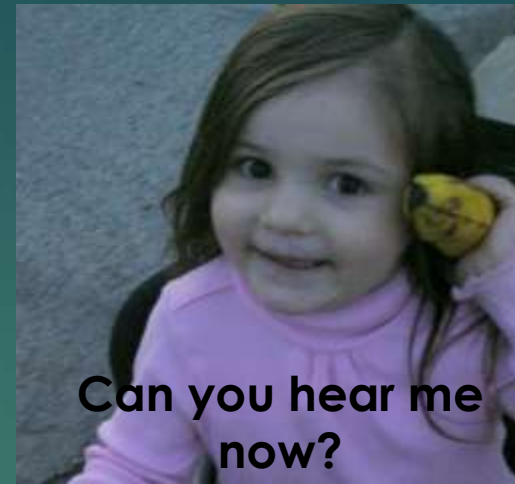
STAGE 2 (2-7 YEARS)

Preoperational Thought is characterized by:

- ▶ Intuitive Thought – logic bases only on experiences
- ▶ Symbols in *play*
- ▶ Egocentrism
- ▶ Lack of conservation
- ▶ Animism

Symbols in Play

- ▶ Symbolic play: use one object to stand for another
- ▶ Fantasy play: pretend to be something, or pretend activities that are impossible
- ▶ Make-believe play: use toys as props (also animism)



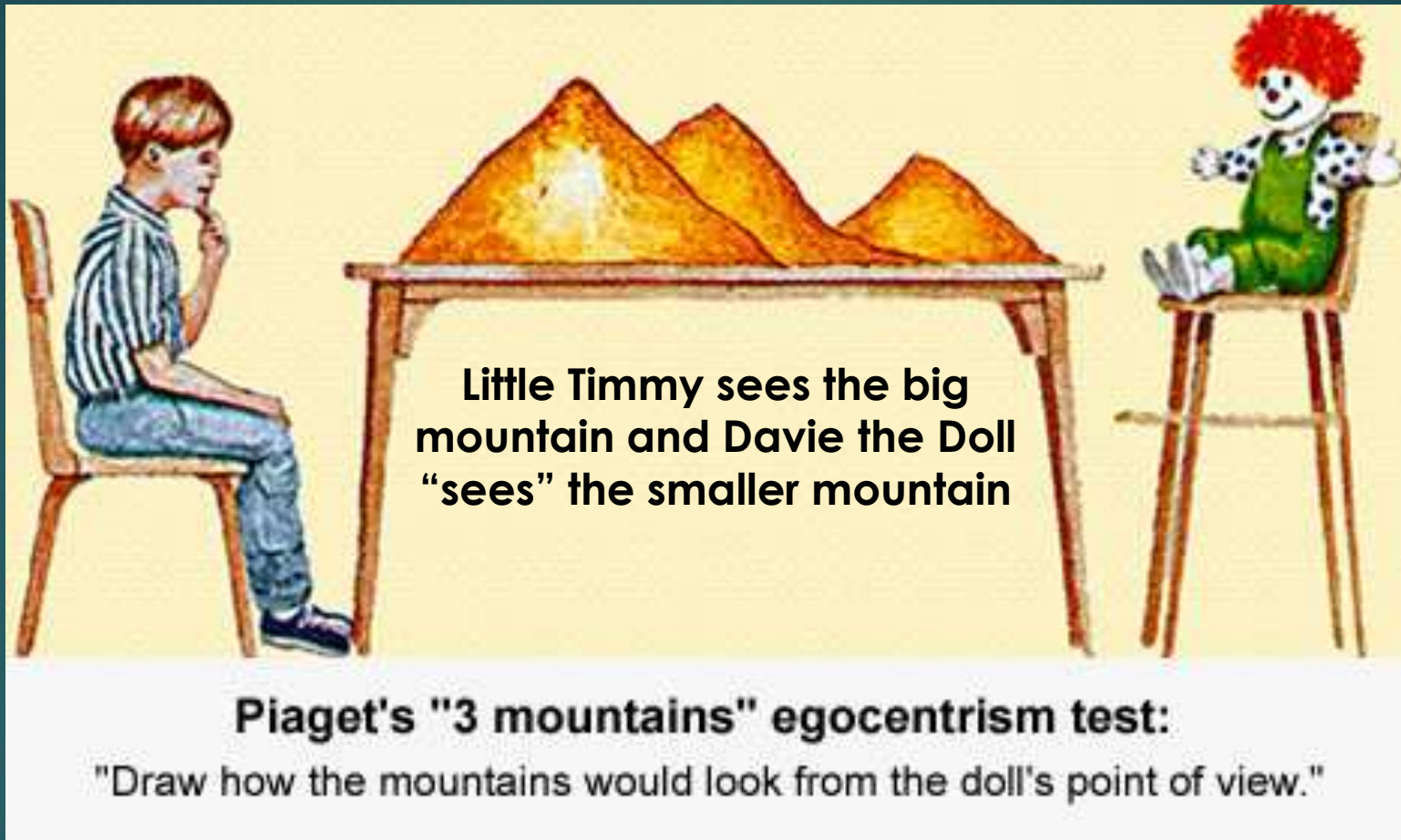


EGOCENTRICISM:

“CHILD’S INABILITY TO TAKE IN OTHERS
PERSPECTIVE”

(LITTLEFIELD COOK & COOK, 2005/2009, P. 163)

Three Mountain Test



Timmy's egocentrism prevents him from seeing Davie's perspective... Timmy would draw the big mountain



CONSERVATION

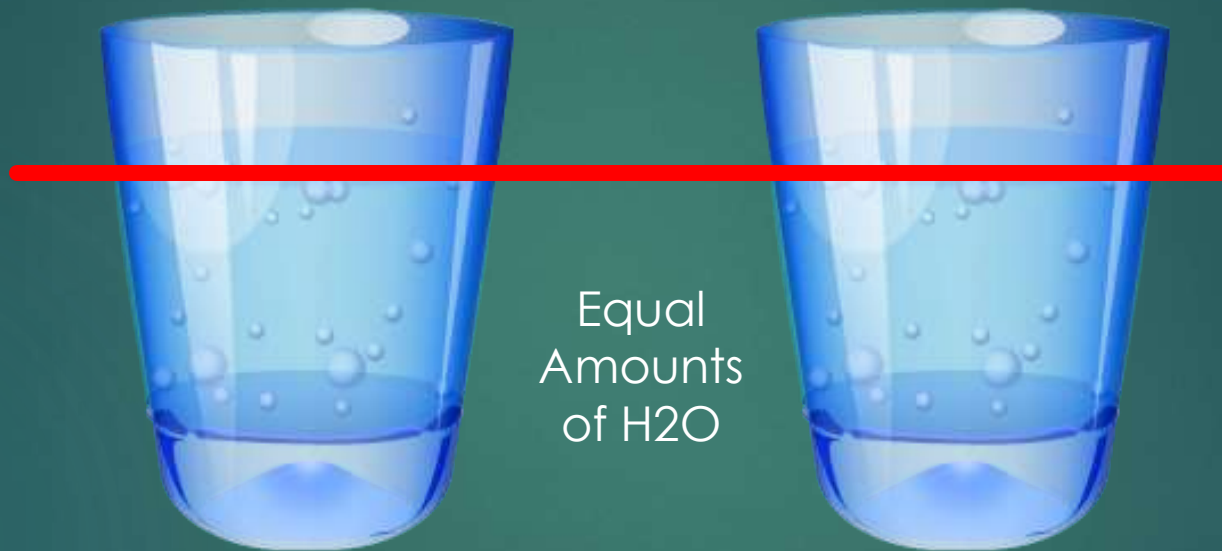
ACCORDING TO THIS GUY:

Operations = reversible mental
actions...

► Thus, the preoperational
Stage is marked by children's
lack of conservation - “**concept that certain basic
properties of an object (e.g. volume, mass, and
weight) remain the same even if its physical
appearance changes**”

(Littlefield Cook & Cook, 2005/2009, p. 164)

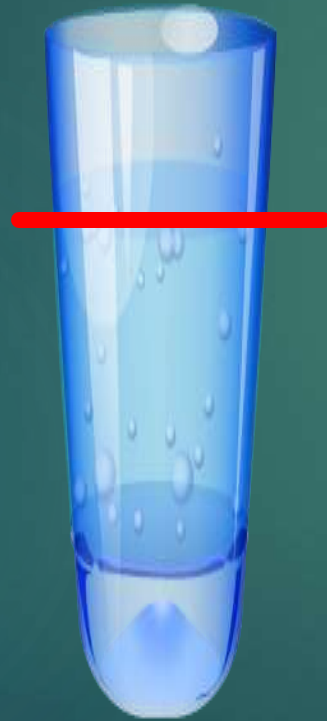
FAMOUS CONSERVATION TEST



The FIRST step in the experiment is to show the child 2 cups with equal amount of water

Step 2

Pour one cup into a tall, skinny cup and the other into a short, fat cup



Which
has
MORE??



Step 3

a **preoperational** child would conclude that the tall skinny class had more water because the level of water was higher.

THE UNDERSTANDING OF CONSERVATION
PRINCIPLES SENDS THAT PRECIOUS LITTLE
CHILD RIGHT INTO THE WORLD OF...



CONCRETE OPERATIONAL THOUGHT

STAGE 3
(7-11 YEARS)

Logic is “still tied closely to concrete materials, contexts, and situations”

(Littlefield Cook & Cook, 2005/2009, p. 166)

- ▶ Characterized by:
 - ▶ ytilibisreveR
 - ▶ Logical abilities: class inclusion

REVERSIBILITY

Relates to the CONSERVATION EXPERIMENT

children in the concrete operational stage understand that if you reverse the action (pour the water back into the same size cups), then the water amount REMAINS THE SAME

ITS LIKE MAGIC... but not really.

(Littlefield Cook & Cook, 2005/2009, p. 165)



LOGICAL ABILITIES: class inclusion



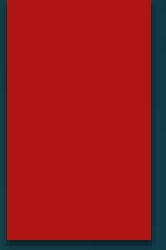
ARE THERE MORE DOGS
OR ANIMALS?



Through understanding class
inclusion, children in the concrete
operational stage know that dogs
belong to the larger CATEGORY of
animals

So they would answer:

ANIMALS



FORMAL OPERATIONAL THOUGHT

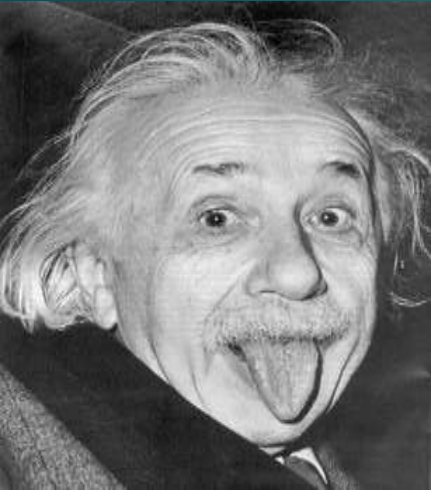
STAGE 4 (AGE 12 AND UP)

5 IMPORTANT HIGHER-LEVEL
COGNITIVE ABILITIES

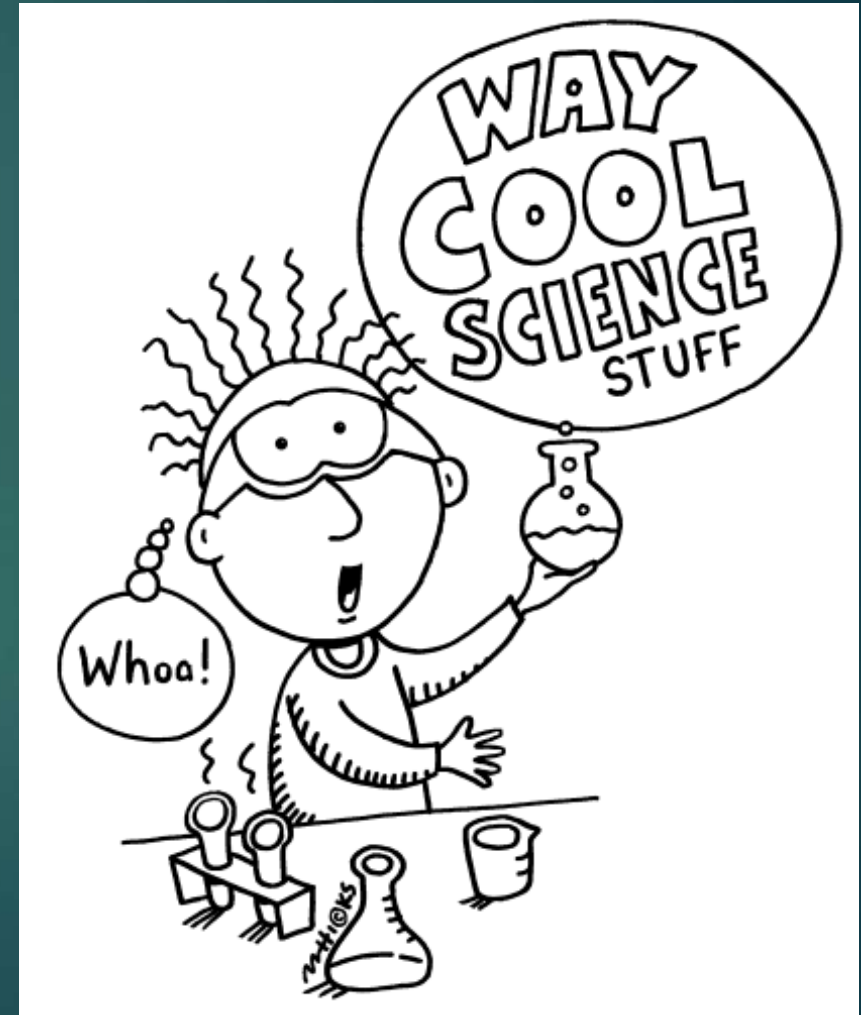
1. Hypothetico-deductive reasoning

- ▶ “ability to plan systematic tests to explore multiple variables”... HUH?

IT MEANS SCIENTIFIC
REASONING!!!



(Littlefield Cook & Cook, 2005/2009, p. 167)



2. Abstract thought

“Thought about things that are not real or tangible” (Littlefield Cook & Cook, 2005/2009, p. 167)



**What do you
think this
Painting is saying?**

3. Separating Reality from Possibility

“direction of thinking about reality and possibility reverses: ... reality is thought of as only one of many possible outcomes”

How things *could* be



(Littlefield Cook & Cook, 2005/2009, p. 167)

4. Combinational logic

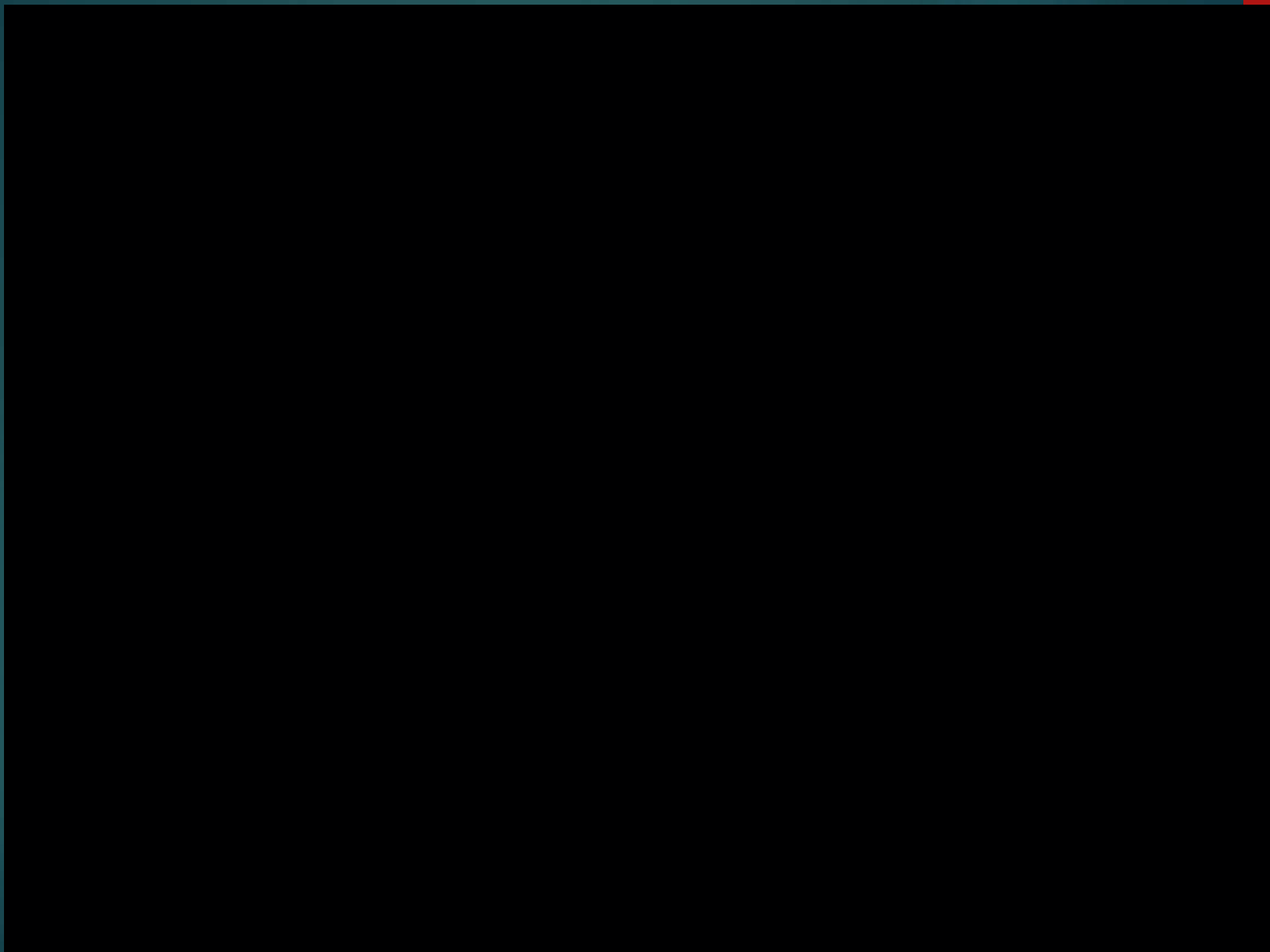
Thinking about multiple aspects and combining them logically to solve problems



5. Reflective Thinking

Thinking about your own thinking





Piaget's 4 Stages of Cognition (3)

by clarmandpiaget



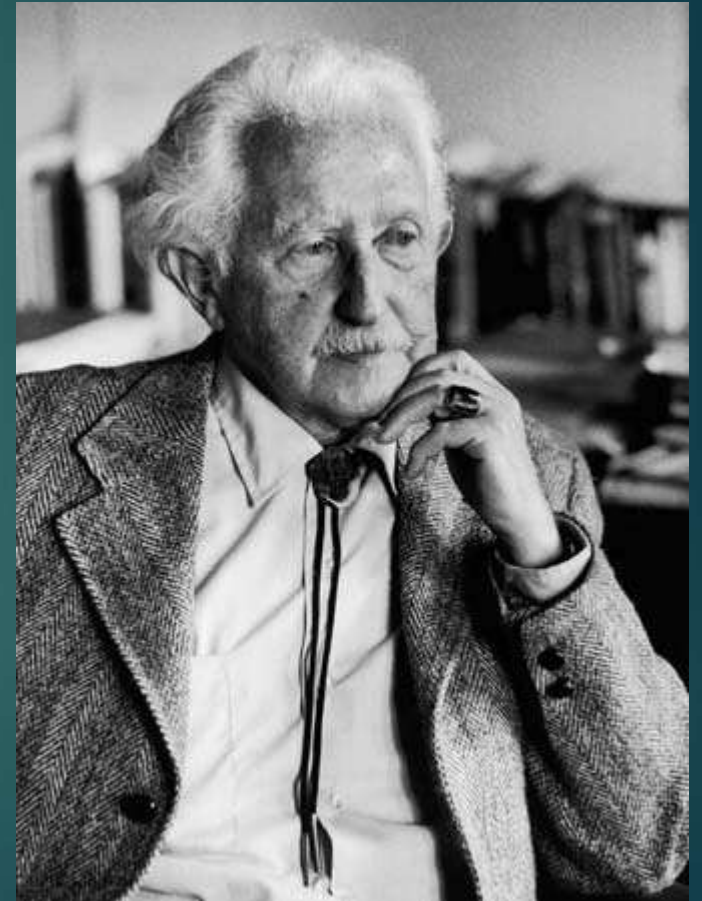


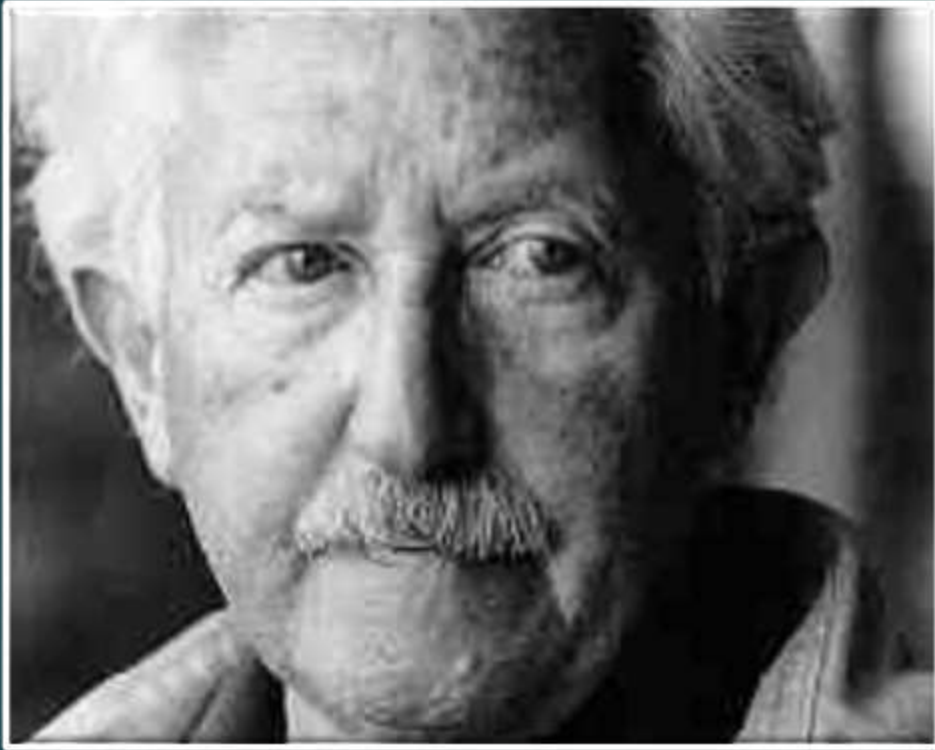
“The principle goal of education in the schools should be creating men and women who are capable of doing new things, not simply repeating what other generations have done.”

– JEAN PIAGET

PSYCHOSOCIAL THEORY OF DEVELOPMENT

ERIK ERIKSON





Erik Erickson

The first published his eight theory of human development in his 1950 book, *Childhood and Society*.

The stages were included in the chapter entitled *The Eight Ages of Man*. He expanded and refined his theory in later books and revisions, mainly: *Identity the Life Cycle*(1951); *Insight and Responsibility*(1964); *The Life Cycle Completed; A Review*(1982, revised 1996 by Joan Erickson); and *Vital Involvement in Old Age*(1989).

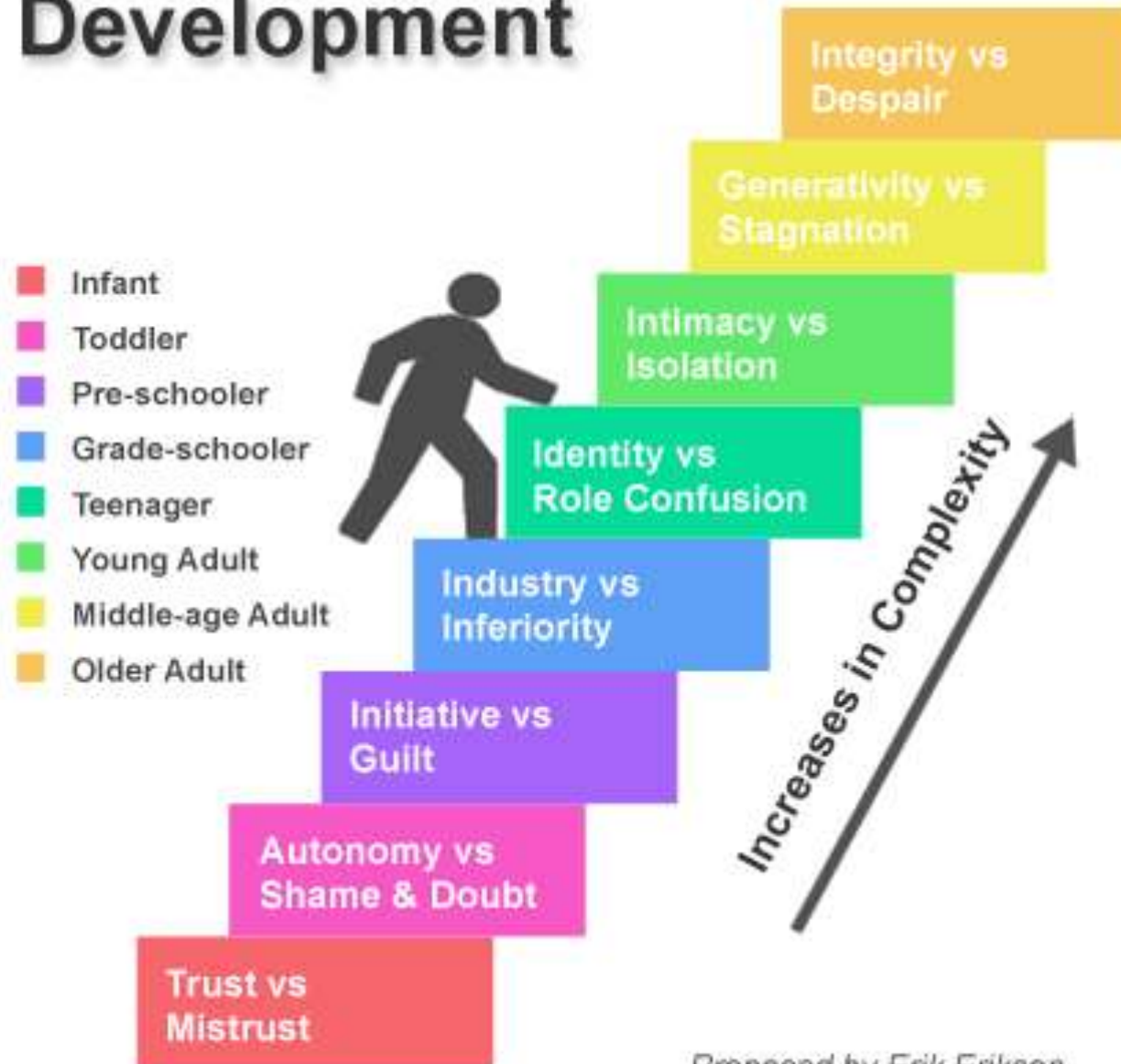
overview of the 8 stages:

- ▶ Psychosocial ('psycho' relating to the mind, brain, personality, etc. and 'social' which means the external relationships and environment). Biopsychosocial, in which "bio" refers to life as in biological.

- 
- ▶ If a stage is managed well, we carry away a certain virtue or psychosocial strength which will help us through the rest of the stages of our lives.

- 
- ▶ **Syntonic** – for the first listed “positive” disposition in each crisis.
 - ▶ **Dystonic** – for the second listed “negative” disposition.

Stages of Psychosocial Development



Proposed by Erik Erikson

ERIKSON'S PSYCHOSOCIAL THEORY & STAGES

**BY: NAVON HUNT
DR. SAENZ**

ERICKSON'S PSYCHOSOCIAL STAGES

<i>Stages</i>	<i>Crisis</i>	<i>Favorable Outcome</i>	<i>Unfavorable Outcome</i>
Childhood			
1st year of life	<i>Trust vs. Mistrust</i>	Faith in the environment and future events	Suspicion, fear of future events
2nd year	<i>Autonomy vs. Doubt</i>	A sense of self-control and adequacy	Feelings of shame and self-doubt
3rd through 5th years	<i>Initiative vs. Guilt</i>	Ability to be a "self-starter," to initiate one's own activities.	A sense of guilt and inadequacy to be on one's own
6th year to puberty	<i>Industry vs. Inferiority</i>	Ability to learn how things work, to understand and organize.	A sense of inferiority at understanding and organizing.

Transition years

Adolescence	<i>Identity vs. confusion</i>	Seeing oneself as a unique and integrated person.	Confusion over who and what one really is.
Adulthood			
Early adulthood	<i>Intimacy vs. isolation</i>	Ability to make commitments to others, to love.	Inability to form affectionate relationship.
Middle age	<i>Generativity vs. self-absorption</i>	Concern for family and society in general.	Concern only for self— one's own well-being and prosperity.
Aging years	<i>Integrity vs. despair</i>	A sense of integrity and fulfillment; willingness to face death.	Dissatisfaction with life; despair over prospect of death.

- 
- ▶ **Malignancy** – it involves too little of the positive and too much of the negative aspects of the tasks, such as a person who can't trust others.
 - ▶ **Maladaptation** – is not quite as bad and involves too much of the positive and too little of the negative such as a person who trusts too much.

Stage	Crisis	Maladaptation	Malignancy	Virtue
Infancy	Trust vs. Mistrust	Sensory Distortion	Withdrawal	Hope
Early Adulthood	Autonomy vs. Shame & Doubt	Impulsivity	Compulsion	Will Power
Pre-school	Initiative vs. Guilt	ruthlessness	Inhibition	Purpose
School Age	Industry vs. Inferiority	Narrow Virtuosity	Inertia	Competence



Adolescence	Identity vs. Role Confusion	Fanaticism	Repudiation	Fidelity
Young Adulthood	Intimacy vs. Isolation	Promiscuity	Exclusivity	Love
Middle Adulthood	Generativity vs. Stagnation	Over extension	Rejectivity	Care
Maturity	Ego Integrity vs. Despair	Presumption	Disdain	Wisdom

THEORY OF MORAL DEVELOPMENT

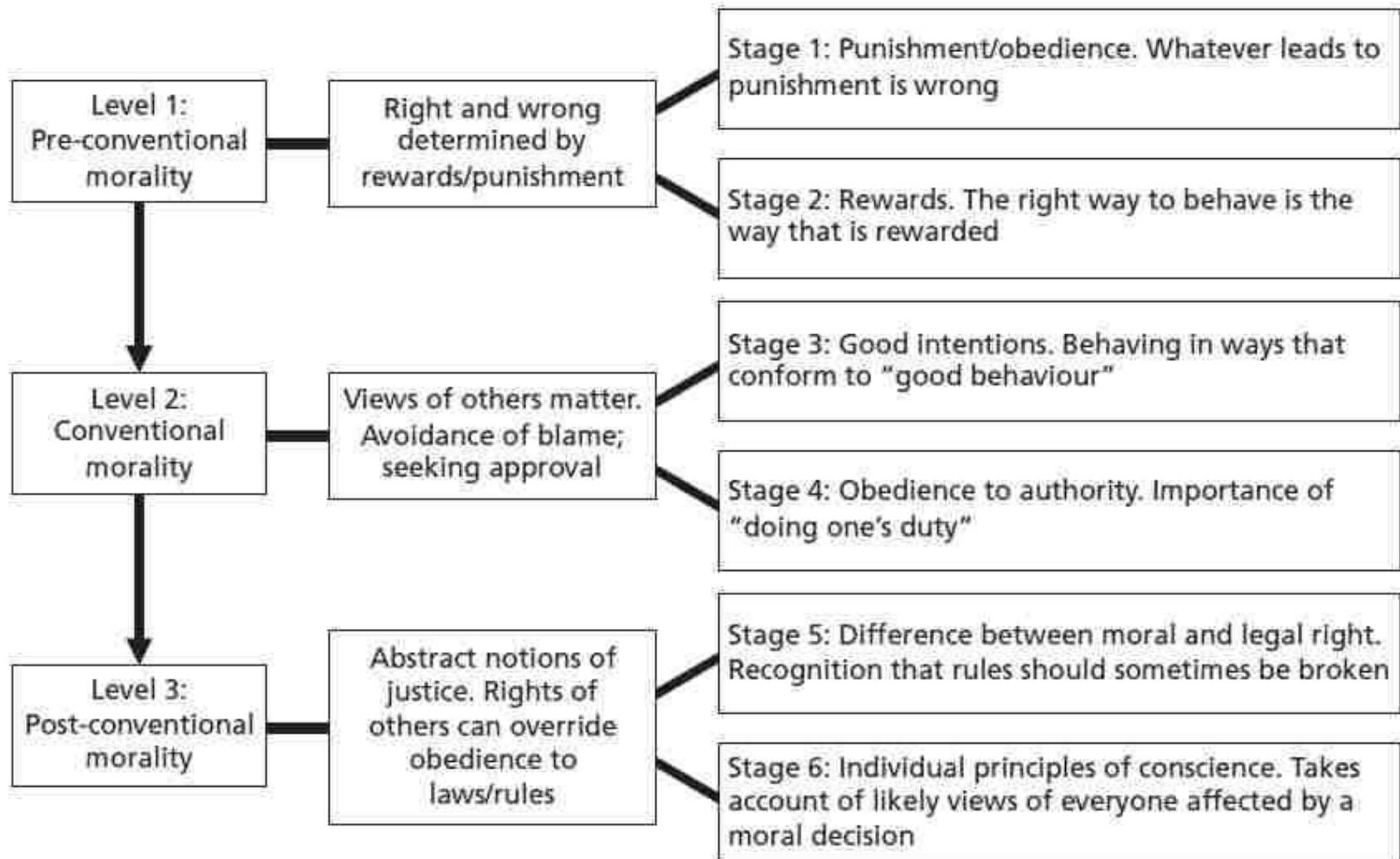
LAWRENCE KOHLBERG



AGREE OR DISAGREE?

Cheating is a persistent problem in classrooms. How students think about this problem and how teachers should respond to it depend on students' levels of moral development.





SOCIO-CULTURAL THEORY

LEV VYGOTSKY



About the Theory

- ▶ Vygotsky suggested that children's cognitive development is fostered by interactions with **people who are more capable or advanced in their thinking – teachers and parents.**
- ▶ He focused on the connections between people and the **sociocultural context** in which they act and interact in shared experiences.
- ▶ According to Vygotsky, humans use tools that develop from a culture, such as speech and writing, to mediate their social environments.

Social Interaction

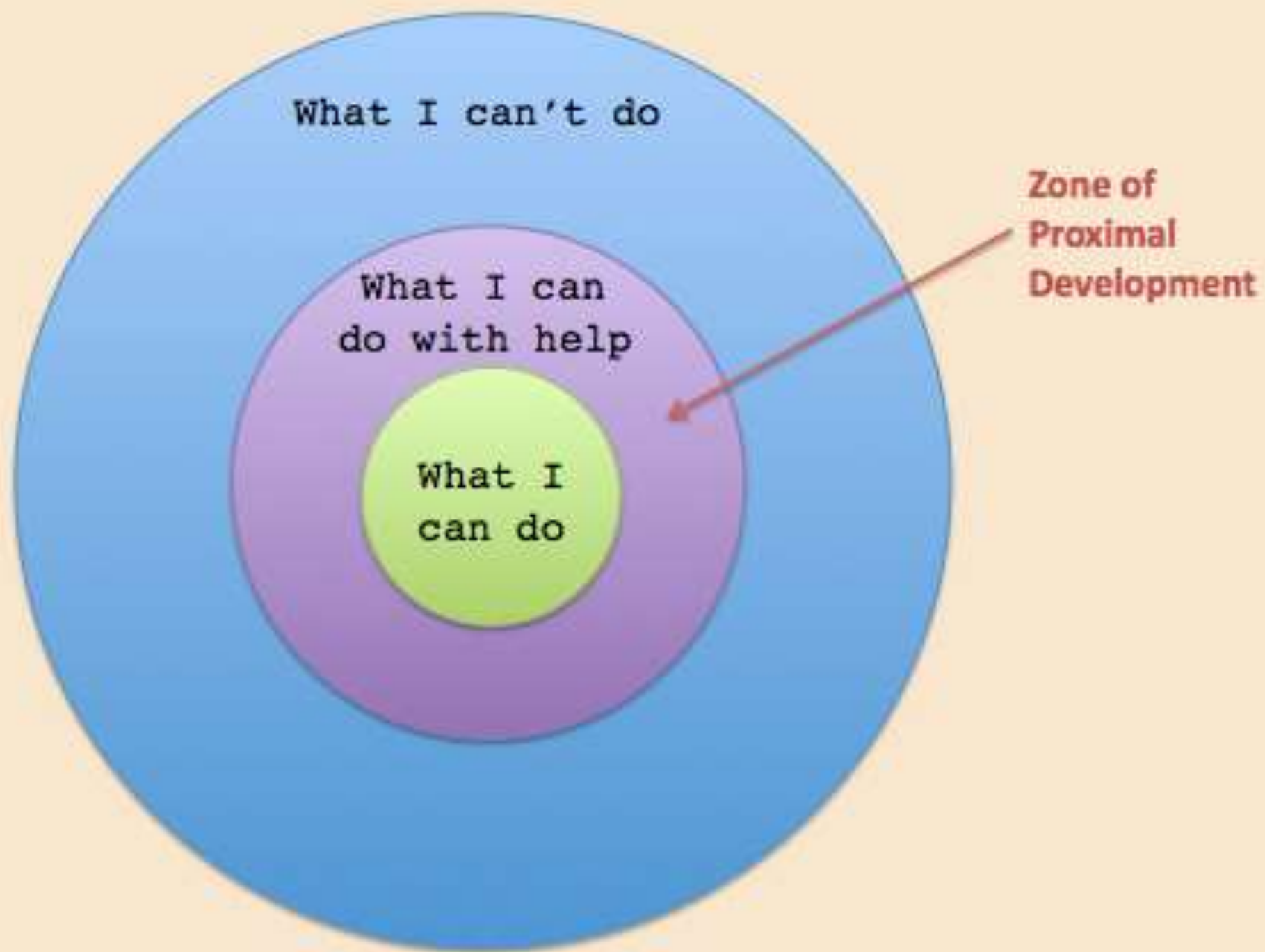
- ▶ Vygotsky felt social learning anticipates development. He states:

“Every function in the child’s cultural development appears twice: first, on the social level, and later, on the individual level; first, between people (**interpsychological**) and then inside the child (**intrapsychological**).”

Language

- Thought and language are interdependent
 - ***self-talk***
 - ▶ children talk to themselves out loud
 - ***inner speech private speech***
 - ▶ children talk to themselves mentally
 - ▶ language transformed into inner verbal thought





Zone of Proximal Development (ZPD)

- ▶ What the learner **already knows** (do not teach; **too boring**)
- ▶ What the learner is **not yet ready** or able to learn (**do not teach; too difficult**)
- ▶ **ZONE OF PROXIMAL DEVELOPMENT** – what the learner could understand with guidance (do teach; exciting, challenging)
- ▶ **Teaching in the Magic Middle** –ZPD is the teaching space between the boring and the impossible. It is here that **scaffolding** from the teacher or a peer can support learning.

Scaffolding

- ▶ Assistance provided by **more competent** peers or adults to enable the task to be done successfully
- ▶ Scaffolded instruction allows the learner to **move through the ZPD**
- ▶ Modelling; feedback; instruction; questioning; encouragement; task structuring; chunking; breaking the problem down
- ▶ Scaffolding is **gradually withdrawn**



The More Knowledgeable Other (MKO)

- ▶ MKO refers to **someone who has a better understanding or a higher ability level than the learner**, with respect to a particular task, process, or concept.
- ▶ For example: **Teachers, Other adults, Advanced students, sometimes even computers.**
- ▶ Many times, a child's peers or an adult's children may be the individuals with more knowledge or experience.

This is an example of how ZPD can work in the life of a child



- ▶ Like all children, Ray is constantly learning and exploring the world around him.

This is an example of how ZPD can work in the life of a child



For our example, we will look at Ray's love of games.

Over the years, Ray has developed skills and knowledge that enable him to play a variety of games.

For each game, he is able to successfully strategize and solve problems independently.

This is an example of how ZPD can work in the life of a child



There is one game, however, that Ray has never learned. It's the card game Yu-Gi-Oh. Ray knows his brother plays it very well.

Ray would like to learn, but is unsure where to start.

This is an example of how ZPD can work in the life of a child



Ray finally asks his brother Ali for help. Ali agrees, and begins working with Ray in learning the game of Yu-Gi-Oh.

Ray is learning in the region Vygotsky would call ZPD.

This is an example of how ZPD can work in the life of a child



In ZPD, Ray is doing something requiring the help of someone more capable. Without Ali's help, Ray would be unable to play the game.

Eventually, Ray will learn the game well enough to play the game by himself.

This is an example of how ZPD can work in the life of a child

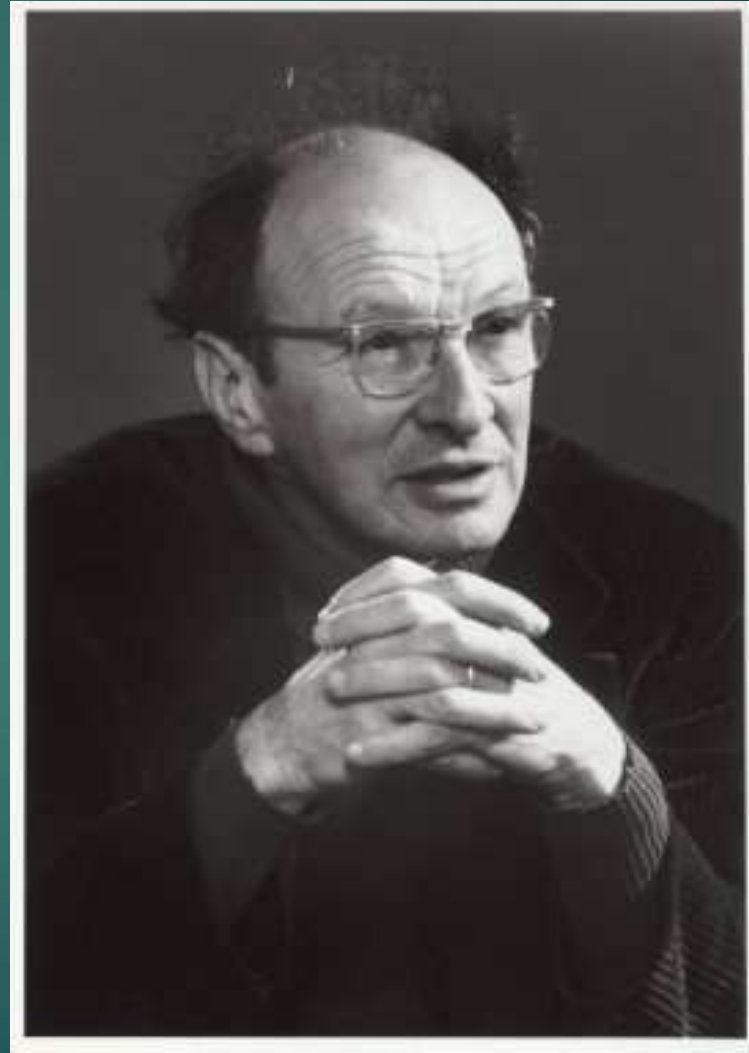


Once Ray learns Yu-Gi-Oh, the skill moves out of the ZPD region and is added to all the other games Ray plays independently.

In time, Ray becomes the more capable player, and begins to teach his sister .

ECOLOGICAL SYSTEMS THEORY

URIE BRONFENBRENNER



Conceptual Model

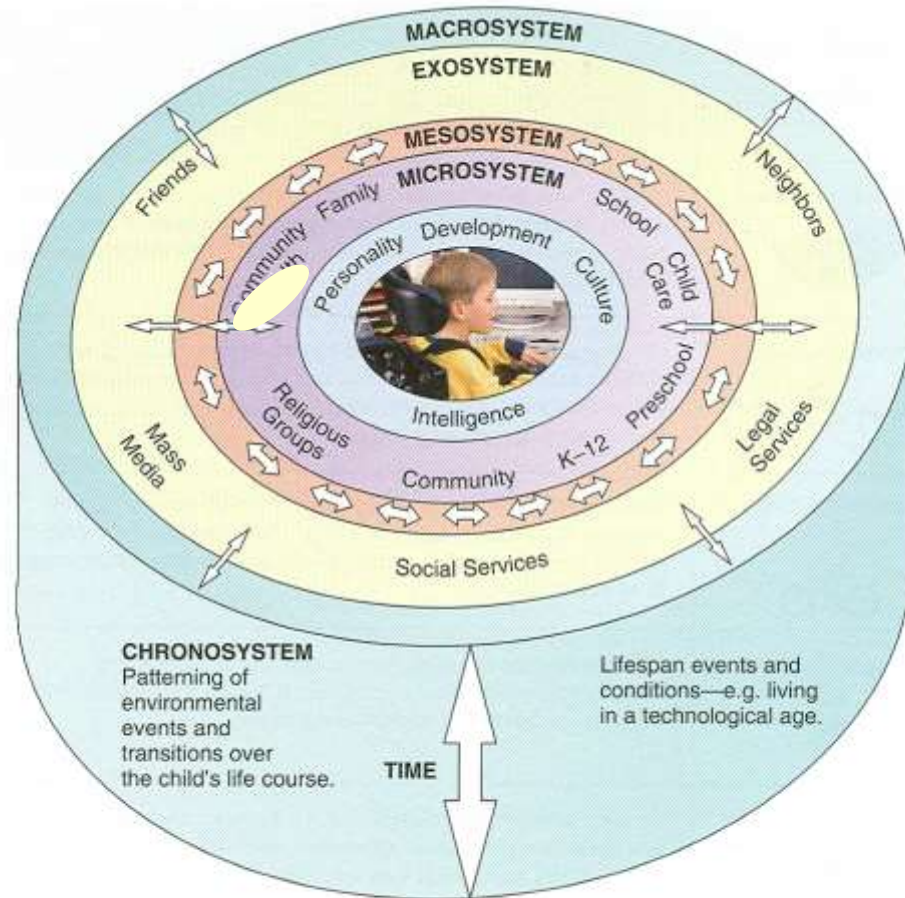
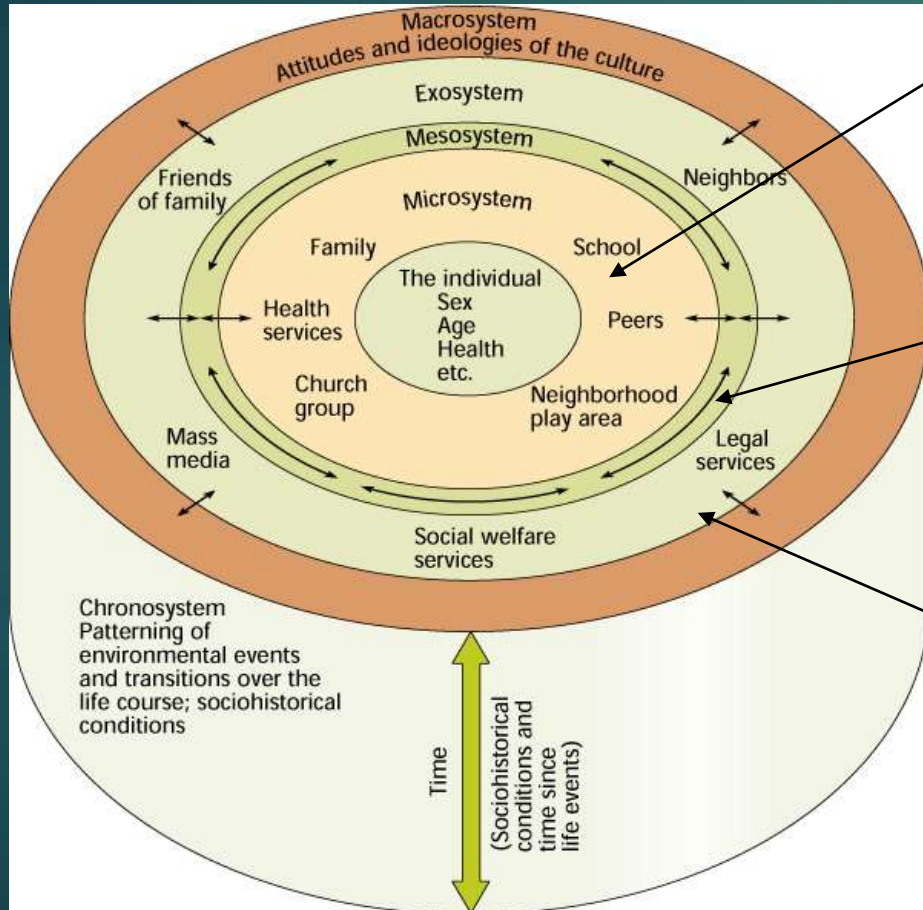


FIGURE 5.10 Ecological Influences on Development

As you can see, there are many influences on children's development. Currently there is a lot of interest in how these influences shape children's lives and what parents and educators can do to enhance positive influences and minimize or eliminate negative environmental influences and negative social interactions.

- Series of systems – each is contained within the other so that literally children are affected by all systems but some are more immediate to the children and are able to be influenced by them

Model

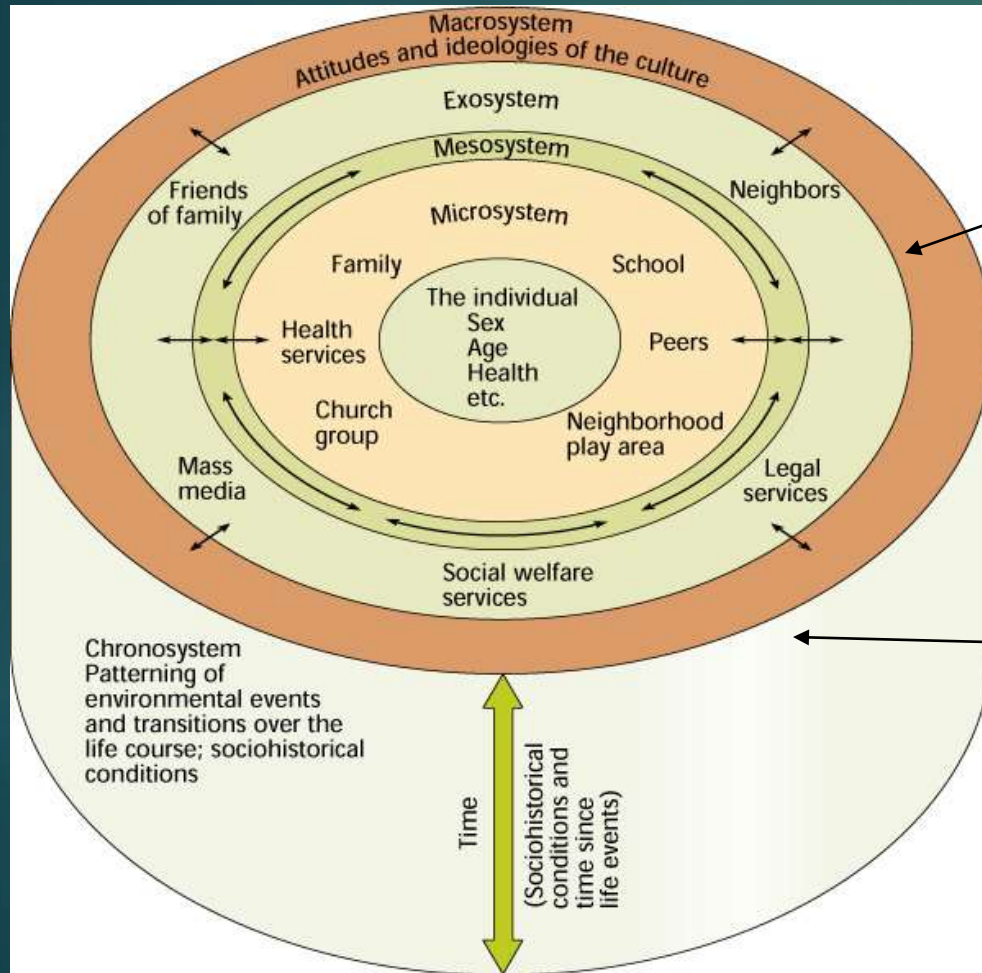


Microsystem: Within this system the student has *direct interactions* with parents, teachers, peers, and others.

Mesosystem: This system involves the *linkages* between microsystems such as family and school, and relationships between students and peers.

Exosystem: This system works when settings in which a child *does not* have an active role influence the student's experiences.

Model (cont)



Macrosystem: This system involves the broader culture in which students and teachers live.

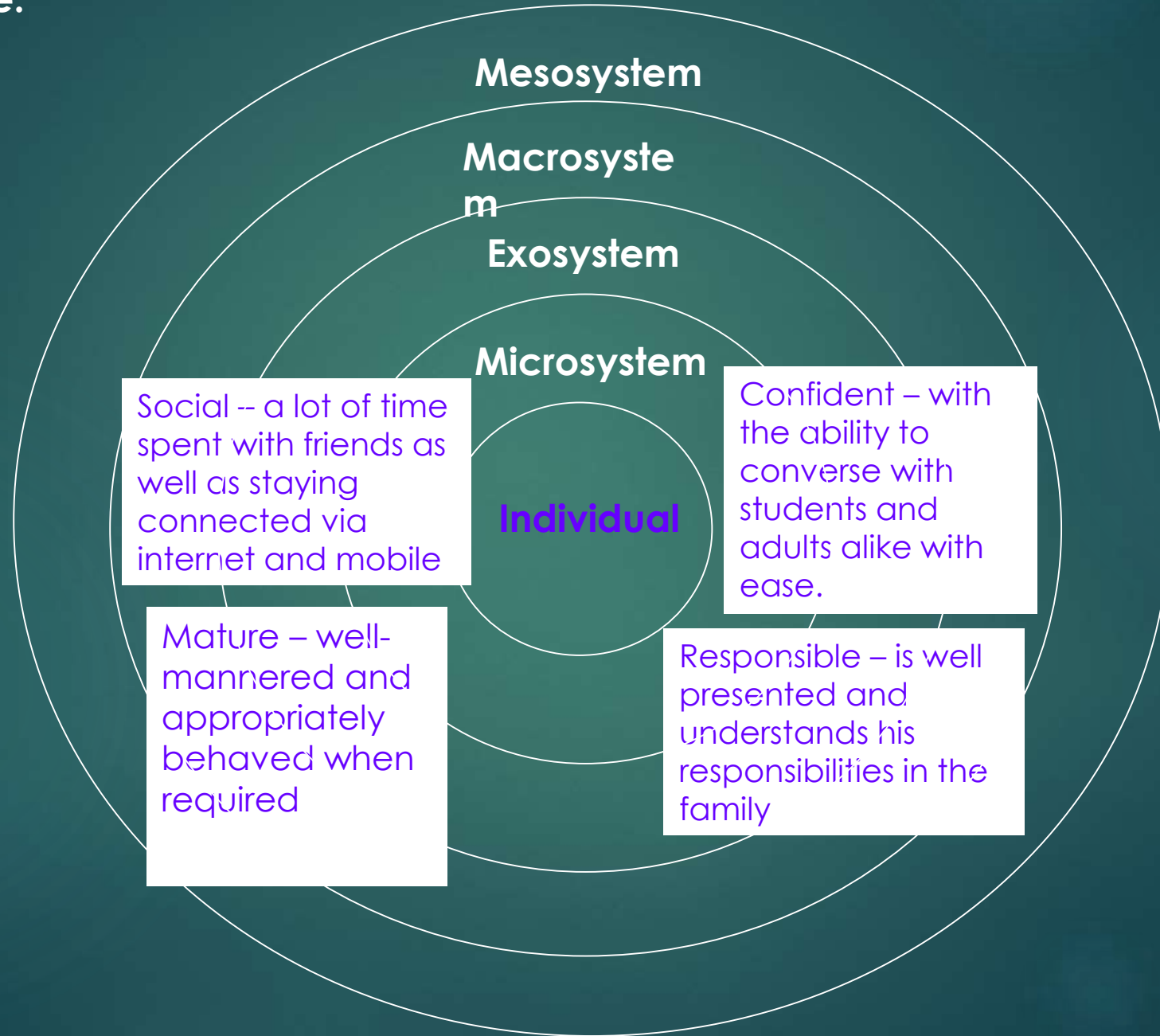
Chronosystem: The sociohistorical conditions of a student's development.

Sample Individual

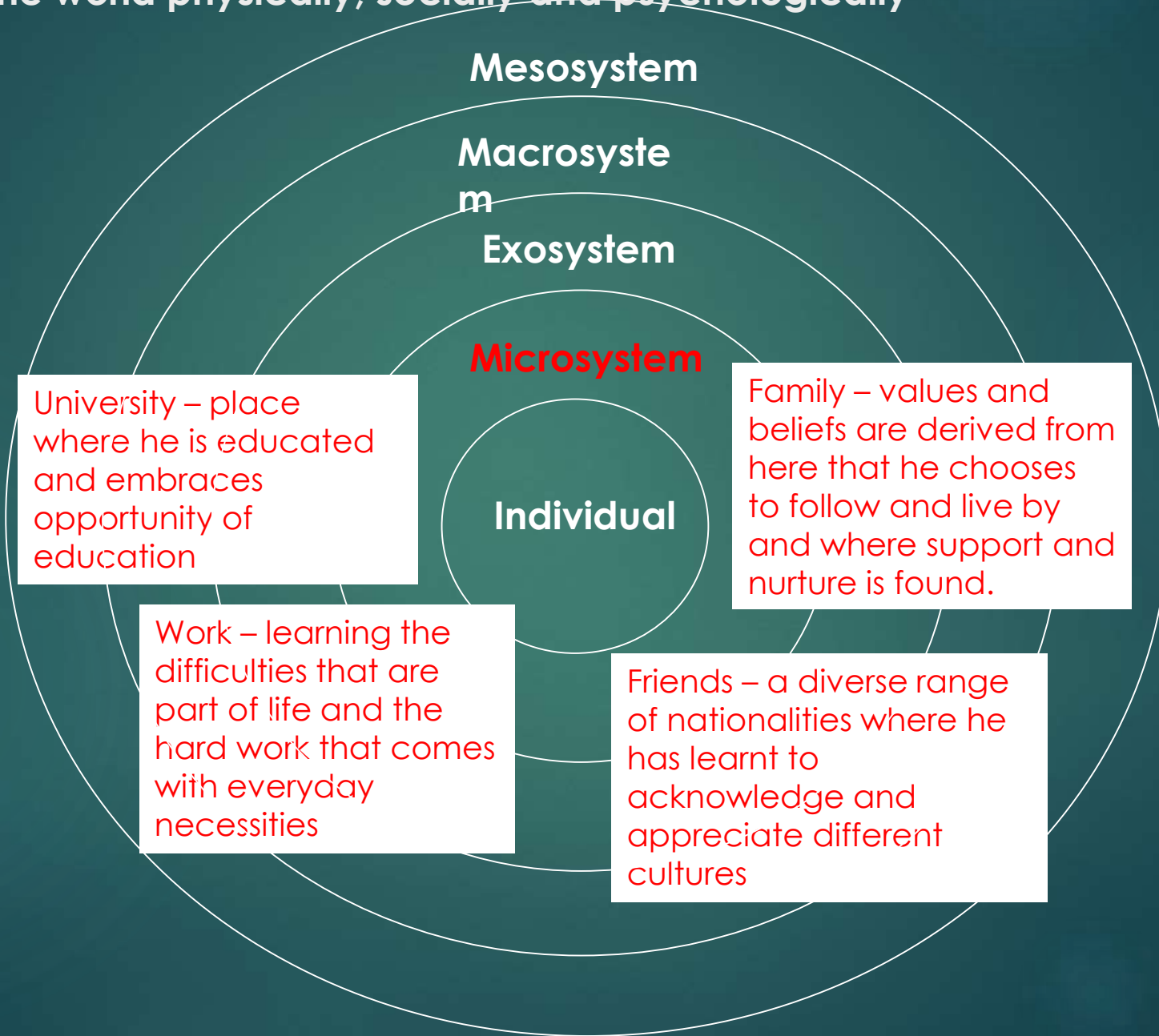


- Age: 18
- Gender: **Male**
- SES: Graduate parents, a pharmacist and banker
- Family: only child
- Nationality: **Filipino - Italian**
- Religion: Roman-Catholic
- Interests: Sports oriented
- Region: South-west Sydney

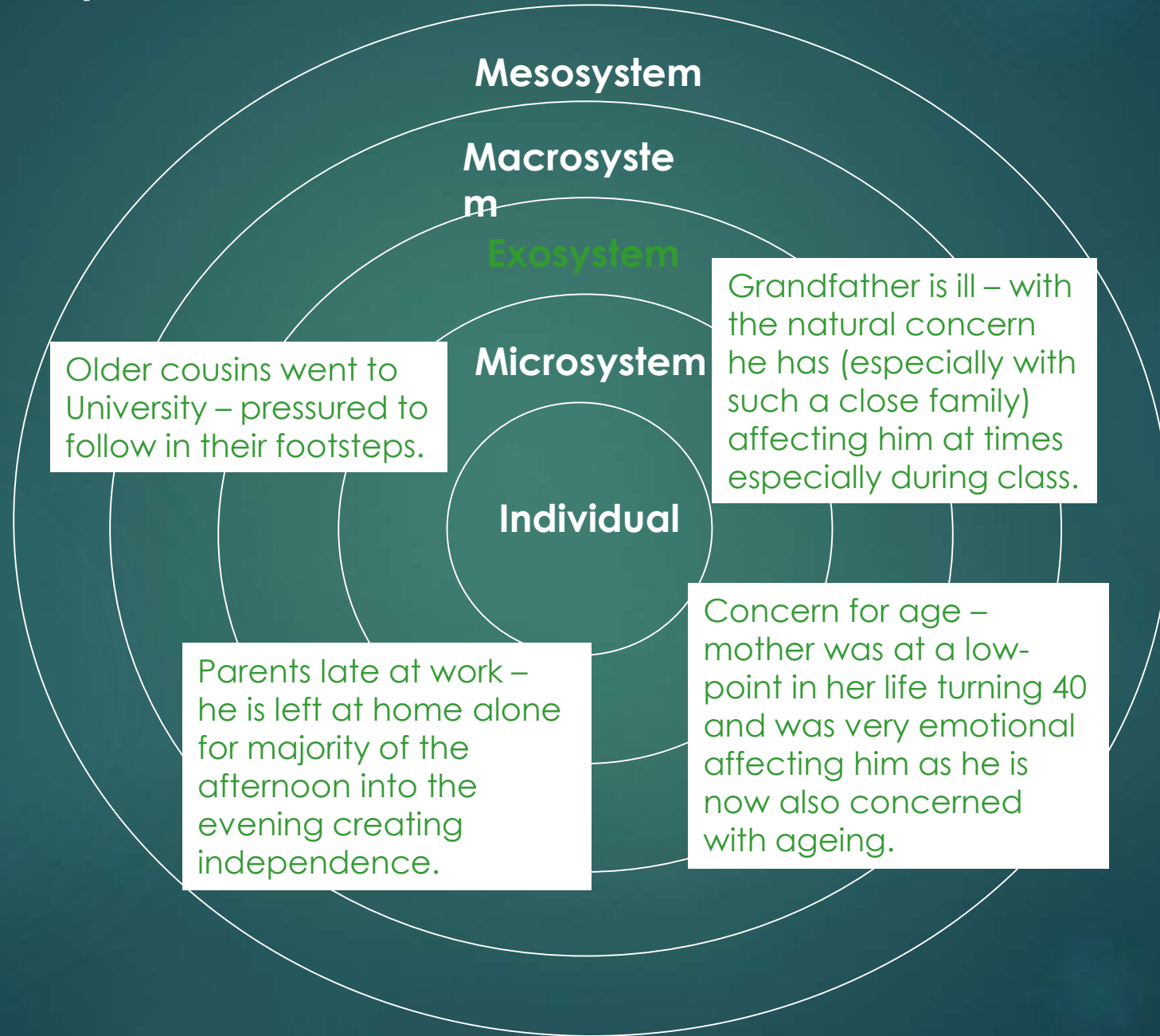
The Individual – the elements of temperament, the child's personality and attitude.



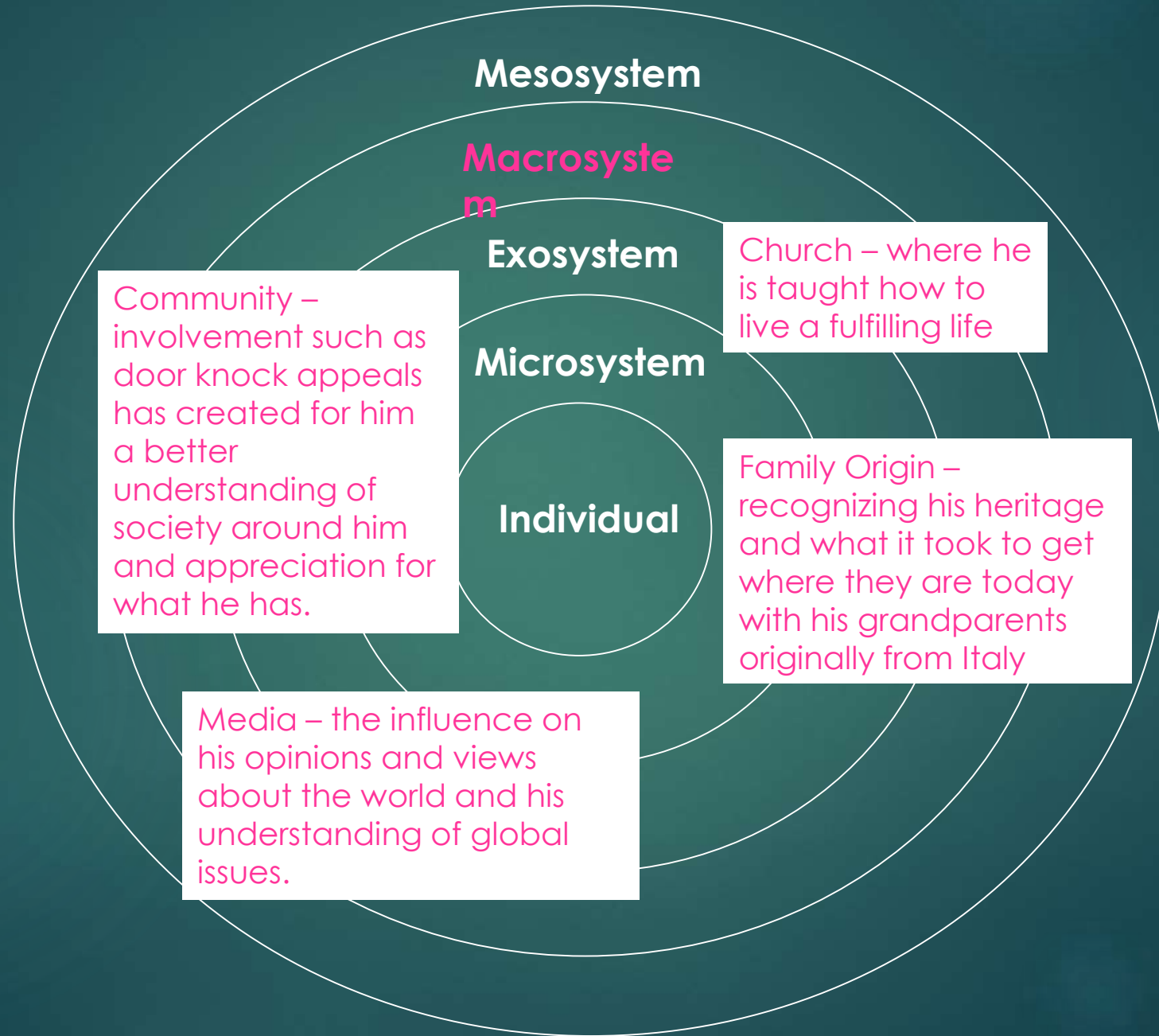
Microsystem – the environment closest to the child where he is educated about the world physically, socially and psychologically



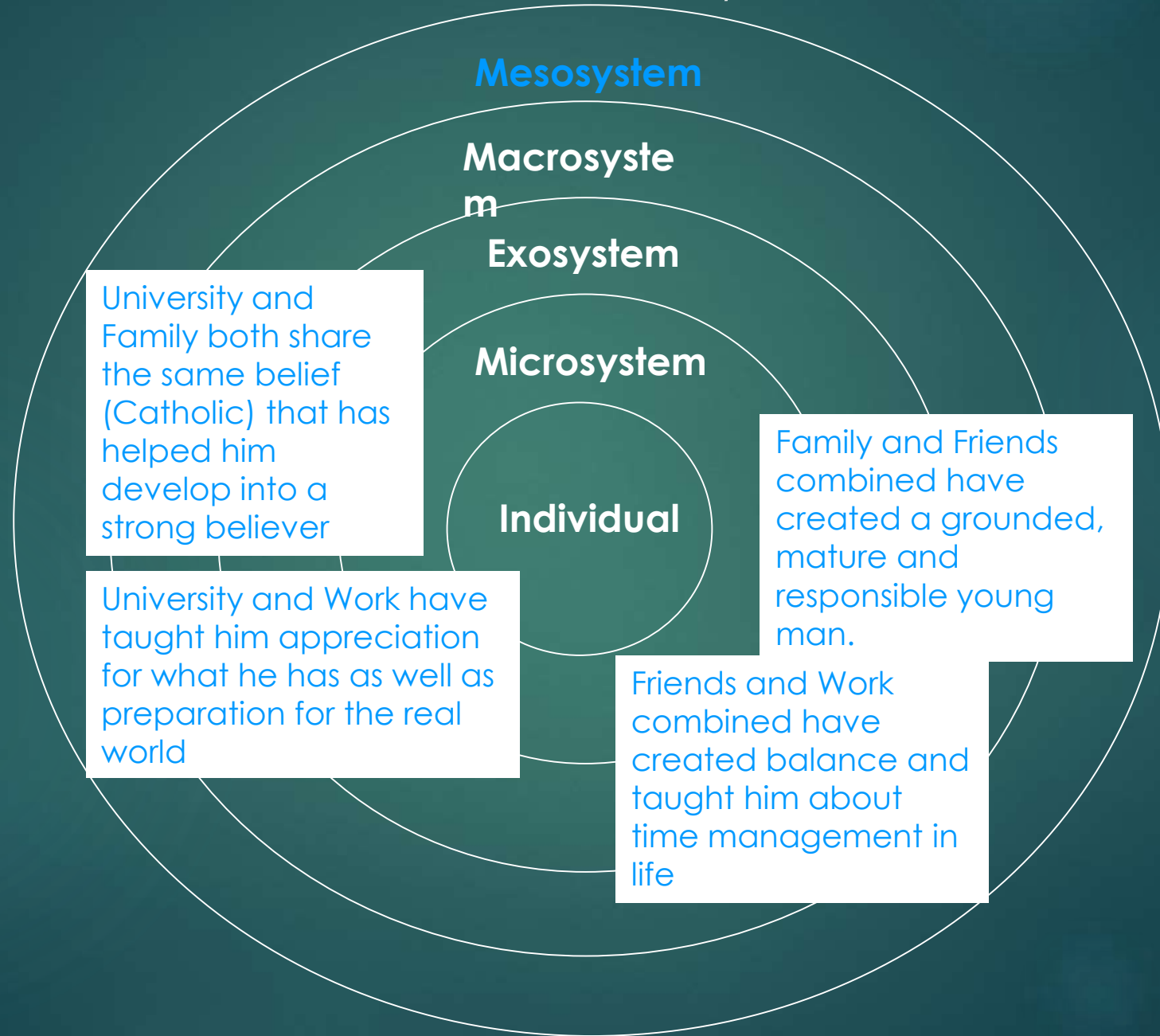
Exosystem – where the child is affected directly but is experienced vicariously



Macrosystem – the most distant layer in the child's environment

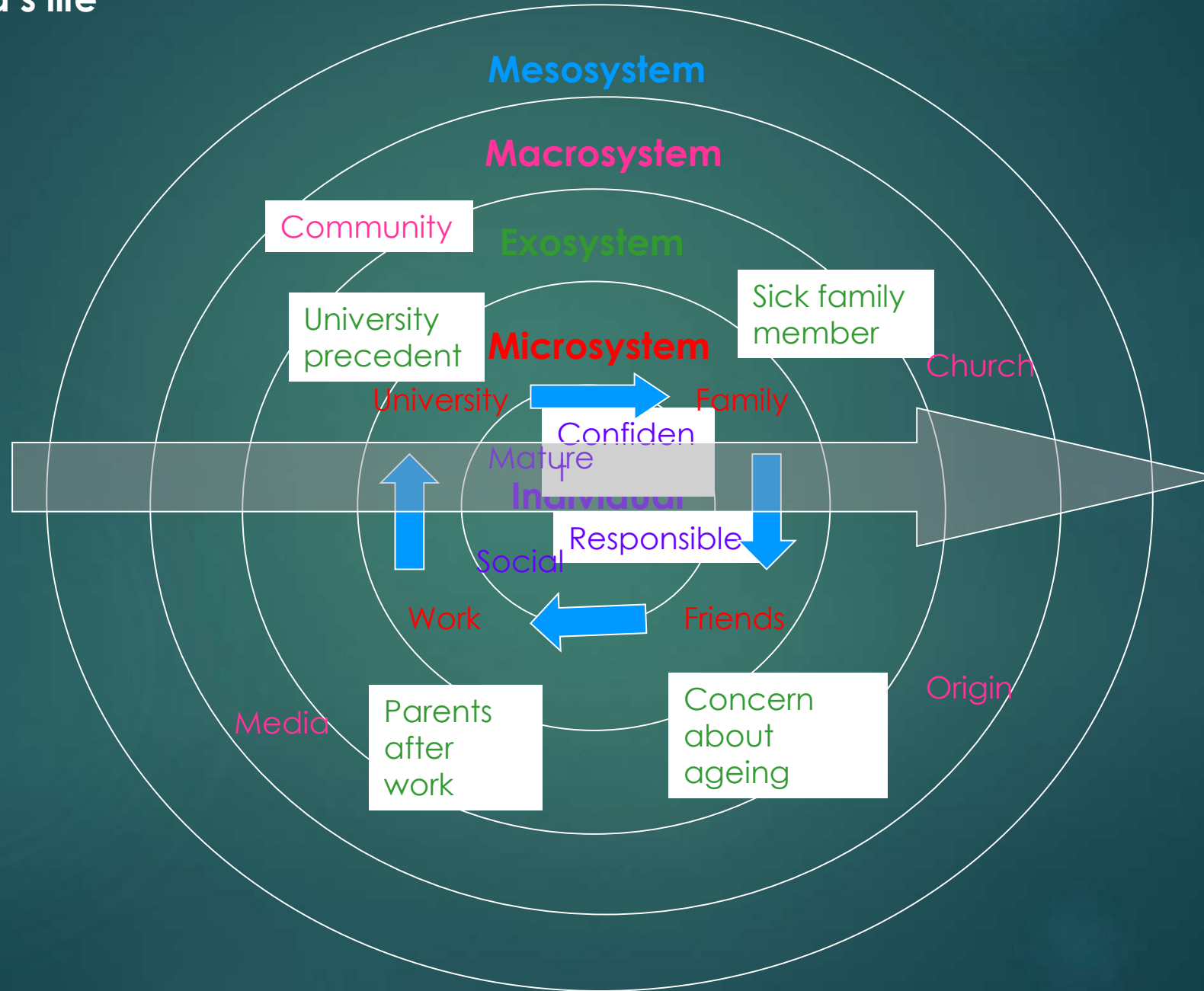


Mesosystem – the connection of the systems, mainly the connection of elements within the microsystem



Chronosystem
child's life

– the timing of different elements within the systems of the



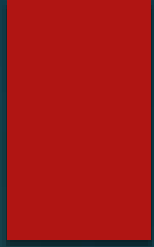


FACILITATING LEARNING

A META-COGNITIVE PROCESS

TOPIC OUTLINE

- ▶ Meaning of Learning and Learner
- ▶ Student Diversity (Individual Differences, Learning Styles, Multiple Intelligences, & Learners with Exceptionalities)
- ▶ Behaviorism Theories (Classical Conditioning, Operant Conditioning, and Social Learning)
- ▶ Cognitivism / Constructivism Theories (Meaningful Reception Learning, Insight Learning, Field Theory, Gestalt Psychology, Information Processing, and Network Model)
- ▶ Metacognition
- ▶ Motivation Theories (Hierarchy of Needs, ERG Theory, Theory X and Y, McClelland's Motivational Needs)



“ If you teach a person what to learn, you are preparing that person for the past. If you teach a person how to learn, you are preparing that person for its future. ”

- CYRIL HOULE

- 
- ▶ Teaching – giving / transferring
 - ▶ Learning – acquiring / accepting

EDUCATIVE PROCESS

- ▶ Learner – given focus
- ▶ Teacher – prime mover
- ▶ Learning Environment - headway

Learner – embodied spirit

Sentient body

Experiencing sensation

-rational soul

-self-reflection

-free will

-intellectual abstractions

cognitive & affective faculties

Instinct

Imagining

Intellect

memory

Feelings

Emotions

Free rational

Volition

Influences / different factors to become different to each other

- ▶ Environment / home
- ▶ Biological aspect
- ▶ Teacher
- ▶ Personal attributes
- ▶ Professional attributes

Badge of Profession – sense of service (teacher)

Effective – doing the right thing

Efficiency – time, effort, motivation, method

STUDENT DIVERSITY



INDIVIDUAL DIFFERENCES

FACTORS that bring about Student Diversity



Socioeconomic Status

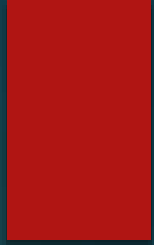
Thinking / Learning Style

Exceptionalities

Learning / Thinking Styles

- ▶ refer to the preferred way an individual processes information.





Learning Style: Dominant Visual

Visual Learners learn through seeing.

They like written instructions and prefer to take detailed notes as they listen to lectures.



Visual/Seeing

They often like to have a quiet place to study.

They like to use colorful highlighters to mark notes and texts.

Visual learners remember visual details with ease and prefer to see what they are learning.



They learn best with visual materials such as pictures, charts, videos, illustrated textbooks, and handouts.

```
graph TD; A[visual] --> B[iconic]; A --> C[symbolic];
```

visual

iconic

symbolic

Learning Style: Dominant Auditory

Auditory Learners
learn through
listening.



Auditory/Hearing

They like to have
music on while
they study.

They like to read aloud and often like to talk
to themselves or create musical jingles to
help them learn new material.

Talking
through a
problem

They remember by talking out loud and they like
to have things explained orally rather than
through written instructions.




```
graph TD; auditory[auditory] --- line1[ ]; line1 --- listener[listener]; line1 --- talker[talker];
```

auditory

listener

talker

Learning Style: Dominant Kinesthetic

Kinesthetic learners learn through moving, doing, and touching.



Kinesthetic/Hands-On

Kinesthetic learners like to move around when talking or listening.

They need to take frequent study breaks.

They are explorers at heart and like to learn through active participation in what they are learning.

They like to chew gum or to snack while they study.

They like to stand, rather than sit, when learning something new.



See it



Visual

Say it



Auditory

Do it

Kinesthetic



Global

- Linked to right-hemisphere dominance in the brain
- Takes in information holistically
- Begins with understanding concepts first, with mastery of details to follow
- Prefer music or other background noise.

Analytical

- Grounded in left-hemisphere dominance in the brain
- Take in information sequentially, step-by-step
- Preferring to learn a series of facts that lead toward an understanding of a larger concept
- Requires orderly, quiet surroundings

I AM THE
LEFT BRAIN

Decisive!
011001011 LOGIC

Accurate
ANALYTIC

REASON

1234567
245879

PRACTICAL
Strategic

CONTROL

SCIENCE

Realistic
Bliss

WWW.CARTOONADAY.COM

I AM the
Right Brain!

Intuition
Love LOVE Love
Love thou art
Poetry

FREE DOM

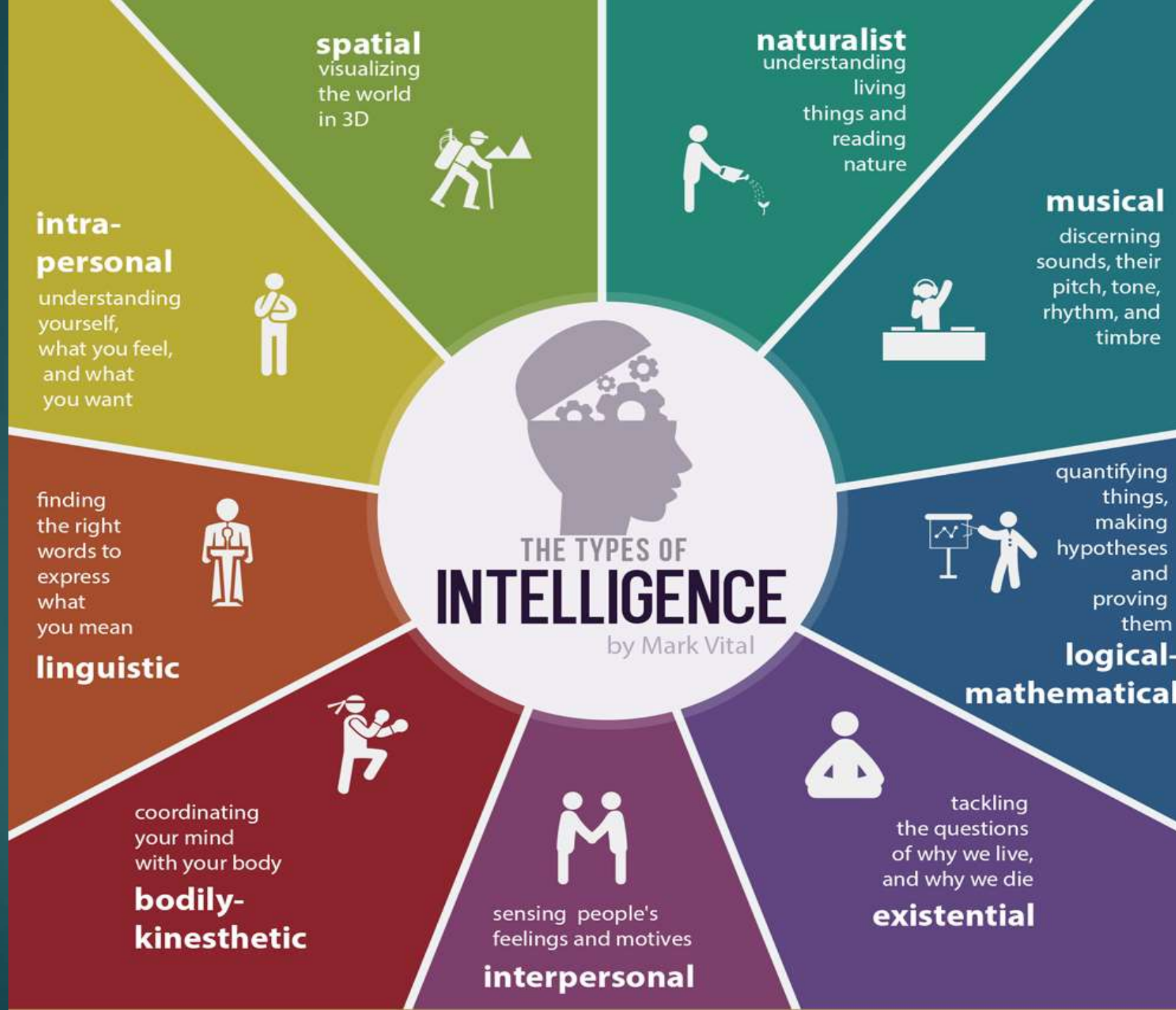
Passion
vivid

creative

YEARNING

PEACE





LEARNERS WITH EXCEPTIONALITIES

SPECIAL EDUCATION means...



Individually planned

Systematically
implemented

Carefully evaluated
instruction

help exceptional children achieve
the greatest possible **personal self-
sufficiency** and in present and
future environments (Heward,
2003)

Impairment or Disability- refers to reduce function or loss of a specific function or body organ. Examples would be: MR, LD, deafness, hearing impairment etc.





Handicap- refers to a problem with a disability or impairment encounters when interacting with people, events and physical aspects of the environment



Developmental Disability- refers to a severe chronic disability of a child five years of age or older



At risk- refers to children who have greater chances than other children to develop a disability.

Established risk- conditions that started during the pregnancy

Biological Risk- those who are born prematurely, underweight at birth, whose mother contracted diabetes or rubella during the first trimester of pregnancy.

Environmental Risk- results from extreme poverty, child abuse, limited social nurturance etc.

Giftedness – unusually high ability or aptitude in one or more of the aspects: intellectual ability, aptitude in academic subjects, creativity, visual or performing arts, or leadership.



SIX PRINCIPLES OF SPECIAL EDUCATION

217

1. Zero reject
2. Appropriate education
3. Nondiscriminatory evaluation
4. Least restrictive environment
5. IEP
6. Parent and student participation
7. Procedural due process

PEOPLE-FIRST LANGUAGE



- ▶ Avoiding generic labels
- ▶ Emphasizing abilities, not limitations
- ▶ Avoiding euphemisms
- ▶ Avoiding implying illness or suffering

CATEGORIES OF DISABILITIES

Specific Learning Disability

- ▶ a disorder in one or more of the basic psychological processes involved in understanding or in using language, spoken or written, that may manifest itself in the imperfect ability to listen, think, speak, read, write, spell, or to do mathematical calculations.

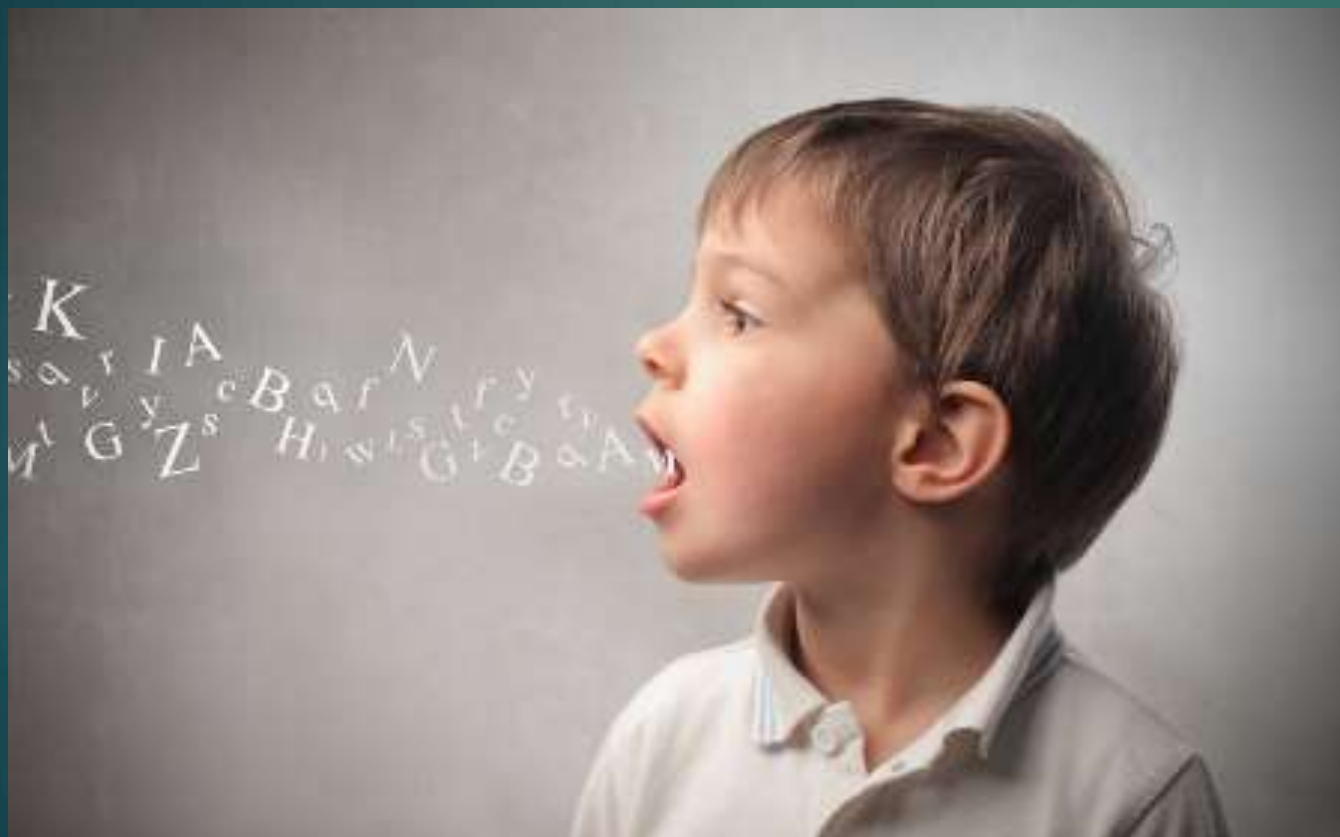


Onegay, Jo^hn anp Bop^w_et froa^w_a k. "W^hatwo^l_ulpy ouⁱ_ke
 t^o_op^a_t op^y? , Boq^a_ske John. "I do^{n't}_kow, J^r_onh^e_dli ed,
 hatwo^l_ulpyo^k_u li e ot go?" It^h_in^k_h I mi^g_ht en^j_yw at^c_hin^g_a
 oⁱ_nTV, e^s_d c^l_y fiw^e_canh^o_dc av es me do ron. "Wow,"
 saip Jonh, "Po^q_corⁿ! hat^g_eati^e_k Let's^c_th^e
 uq^r_t e^e_fim y m the^g_hs^o_{met} he stalt im^e_wetn^{esh}
 s^h_odbⁱ_g." "Look," hey e^l_lep, "af^u_lpoxaⁿ_{pit's} rⁱ_le^o_vl
 R^e_pqenqocker! M^y_fa^r_ote!" "eat!" Bopsho^u_teq, "Le^{t's}_co^k_ot.
 udiⁱ_rwa^c_o nqs^e_woh ti truns^{tou}."

Trying to read this passage, you will experience the kind of difficulty a dyslexic reader faces when deciphering normal typeface (Almeida).

Speech or Language Impairment

- ▶ a communication disorder such as stuttering, impaired articulation, a language impairment, or a voice impairment that adversely affects a child's educational performance.



Autism

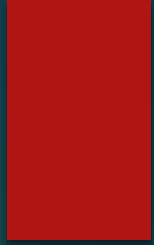
- ▶ a developmental disability significantly affecting verbal and nonverbal communication and social interaction, generally evident before age three, that adversely affects a child's educational performance.



Emotional Disturbance

- ▶ a condition exhibiting one or more of the following characteristics over a long period of time and to a marked degree that adversely affects a child's educational performance:*(a) An inability to learn that cannot be explained by intellectual, sensory, or health factors.(b) An inability to build or maintain satisfactory interpersonal relationships with peers and teachers.(c) Inappropriate types of behavior or feelings under normal circumstances.(d) A general pervasive mood of unhappiness or depression.(e) A tendency to develop physical symptoms or fears associated with personal or school problems.*





Deaf-Blindness

- ▶ concomitant [simultaneous] **hearing and visual impairments**, the combination of which causes such severe communication and other developmental and educational needs that they **cannot be accommodated in special education programs solely for children with deafness or children with blindness.**



Deafness

- ▶ a hearing impairment so **severe** that a child is **impaired** in **processing** linguistic information through hearing, with or without amplification, that adversely affects a child's educational performance.



Hearing Impairment

- ▶ an impairment in hearing, whether permanent or fluctuating, that adversely affects a child's educational performance but is not included under the definition of "deafness."



Intellectual Disability

- ▶ significantly **subaverage general intellectual functioning**, existing concurrently [at the same time] with deficits in **adaptive behavior** and manifested during the developmental period, that adversely affects a child's educational performance.



Multiple Disabilities

- ▶ concomitant [simultaneous] impairments (such as intellectual disability-blindness, intellectual disability-orthopedic impairment), the combination of which causes such severe educational needs that they cannot be accommodated in special education programs solely for one of the impairments.



Orthopedic Impairment

- ▶ a severe orthopedic (spines and joints) impairment that adversely affects a child's educational performance.



Other Health Impairment

- ▶ having limited strength, vitality, or alertness, including a heightened alertness to environmental stimuli, that results in limited alertness with respect to the educational environment, that—(a) is due to chronic or acute health problems such as asthma, attention deficit disorder or attention deficit hyperactivity disorder, diabetes, epilepsy, a heart condition, hemophilia, lead poisoning, leukemia, nephritis, rheumatic fever, sickle cell anemia, and Tourette syndrome; and (b) adversely affects a child's educational performance.



Traumatic Brain Injury

- ▶ an acquired injury to the brain caused by an external physical force, resulting in total or partial functional disability or psychosocial impairment, or both, that adversely affects a child's educational performance.



Visual Impairment Including Blindness

- ▶ an impairment in vision that, even with correction, adversely affects a child's educational performance. The term includes both partial sight and blindness.



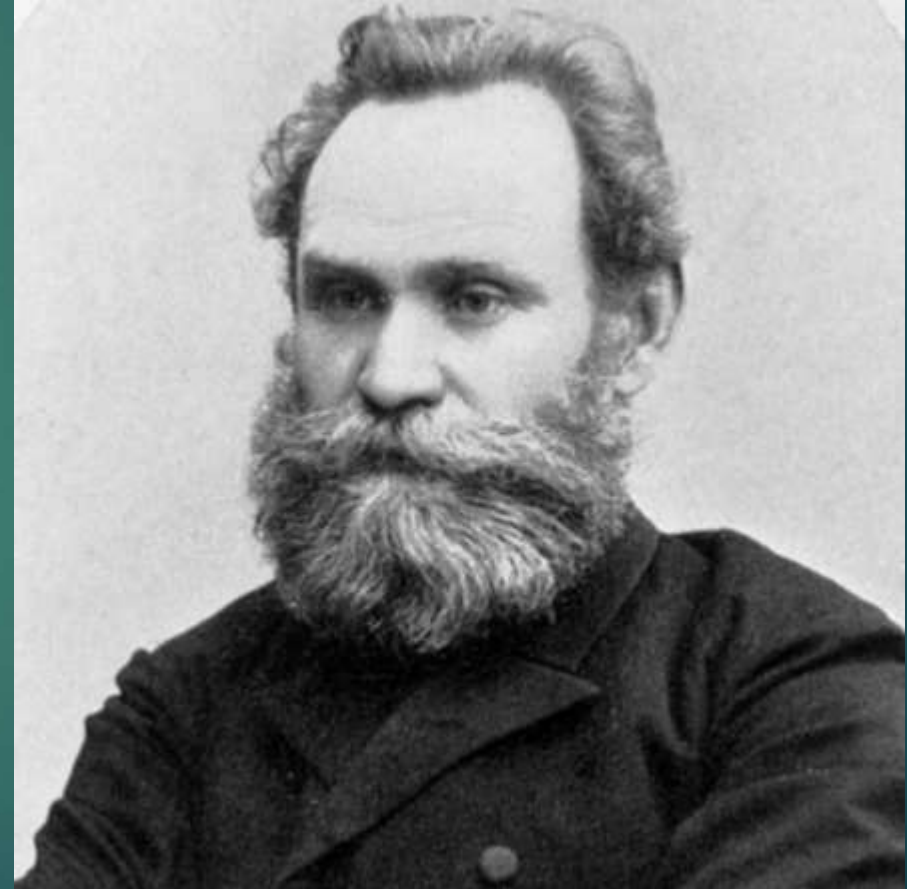


Behaviorism

- is a theory of learning based upon the idea that all behaviors are acquired through conditioning. Conditioning occurs through interaction with the environment.
- also known as *Behavioral Psychology*.

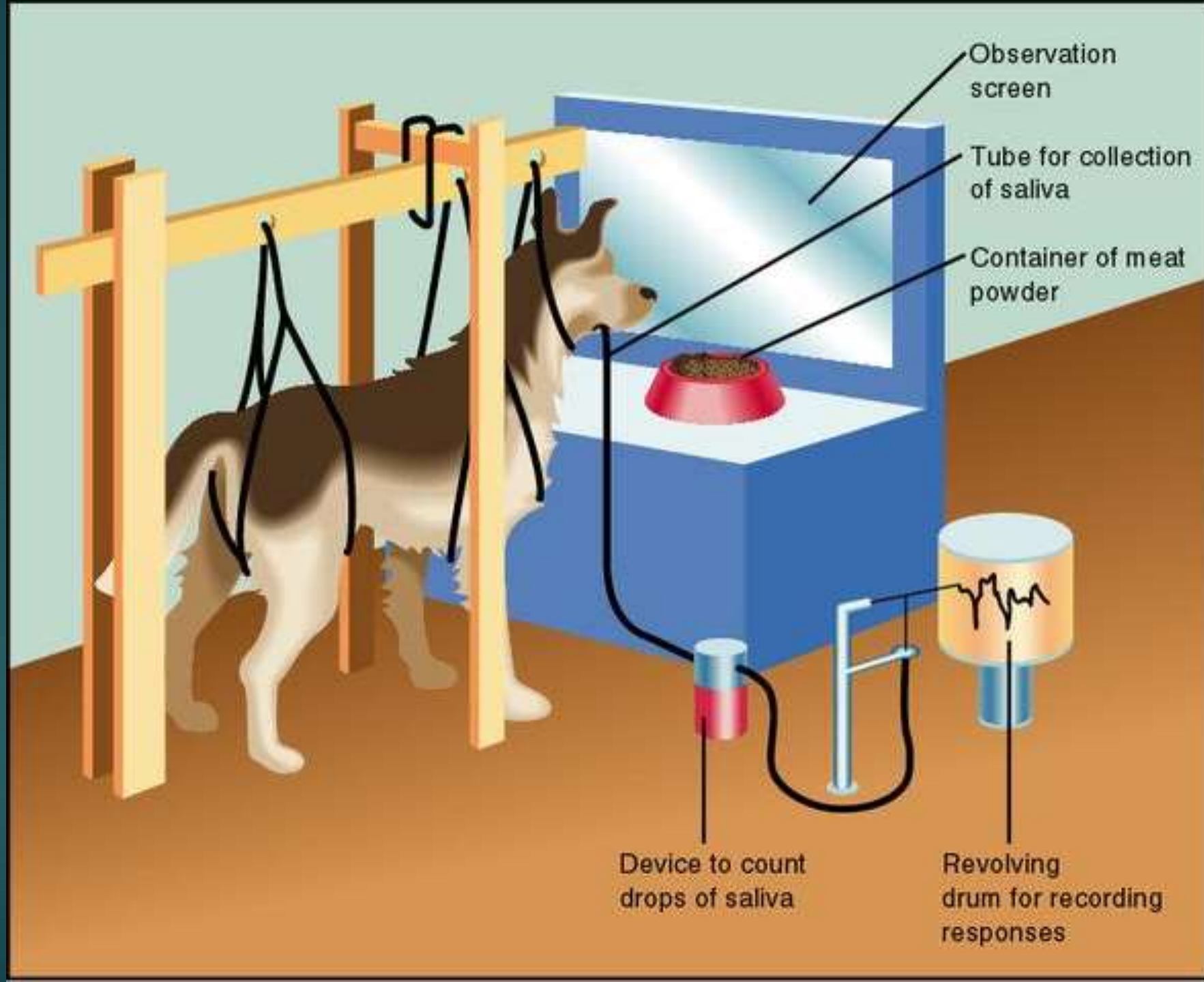
CLASSICAL CONDITIONING

IVAN PAVLOV



OVERVIEW

- ▶ **Classical** = “in the established manner”
- ▶ Believes that individual learns when a previously **neutral stimulus is paired with an unconditioned stimulus** until the neutral stimulus evokes a conditioned response
- ▶ Dog and bell experiment
- ▶ The “Adhesive” principle – a response is attached to stimulus



CLASSICAL (PAVLOVIAN) CONDITIONING

BEFORE

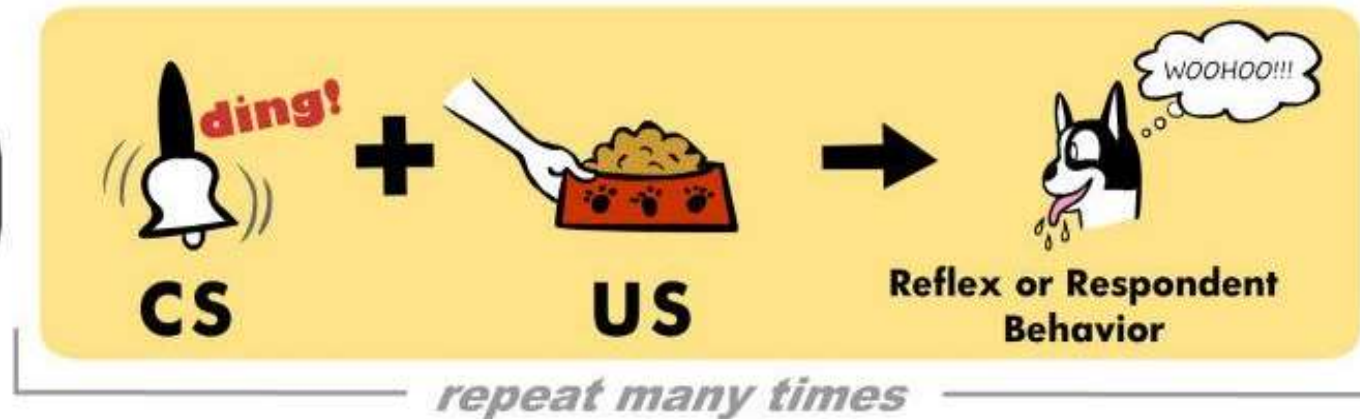


CS = Conditional Stimulus
Neutral signal
No emotional meaning
no response

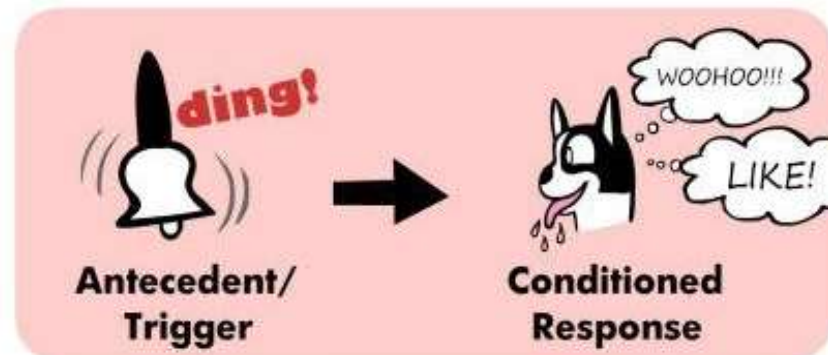


US = Unconditioned Stimulus
Trigger for hardwired emotions
related to survival
reflex =
salivation

DURING



AFTER



FINDINGS:

- ▶ **Stimulus Generalization** (salivate with other similar sounds)
- ▶ **Extinction** (stop pairing the bell, salivation stops)
- ▶ **Spontaneous Recovery** (easy to recover, easy to extinguish)
- ▶ **Discrimination** (can identify which bell would bring food)
- ▶ **Higher Order Conditioning** (use of flash light)

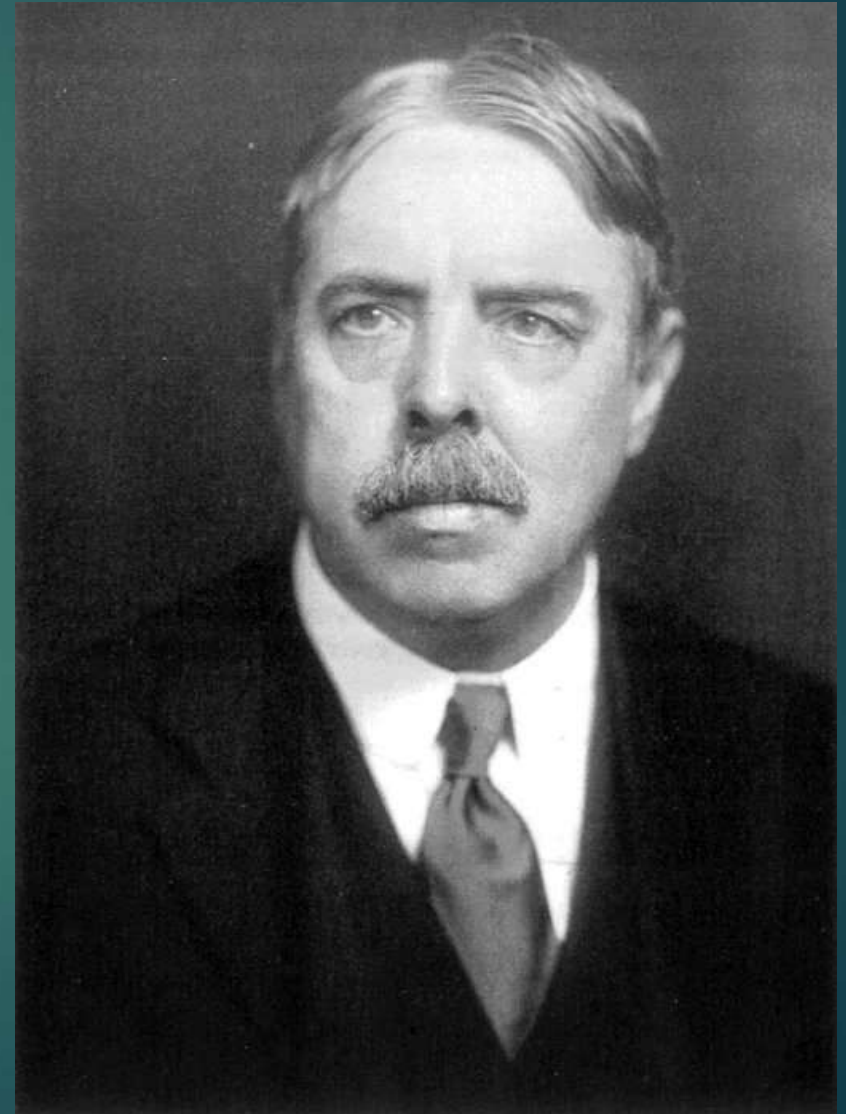


Don't become a mere recorder
of facts, but try to penetrate the
mystery of their origin.

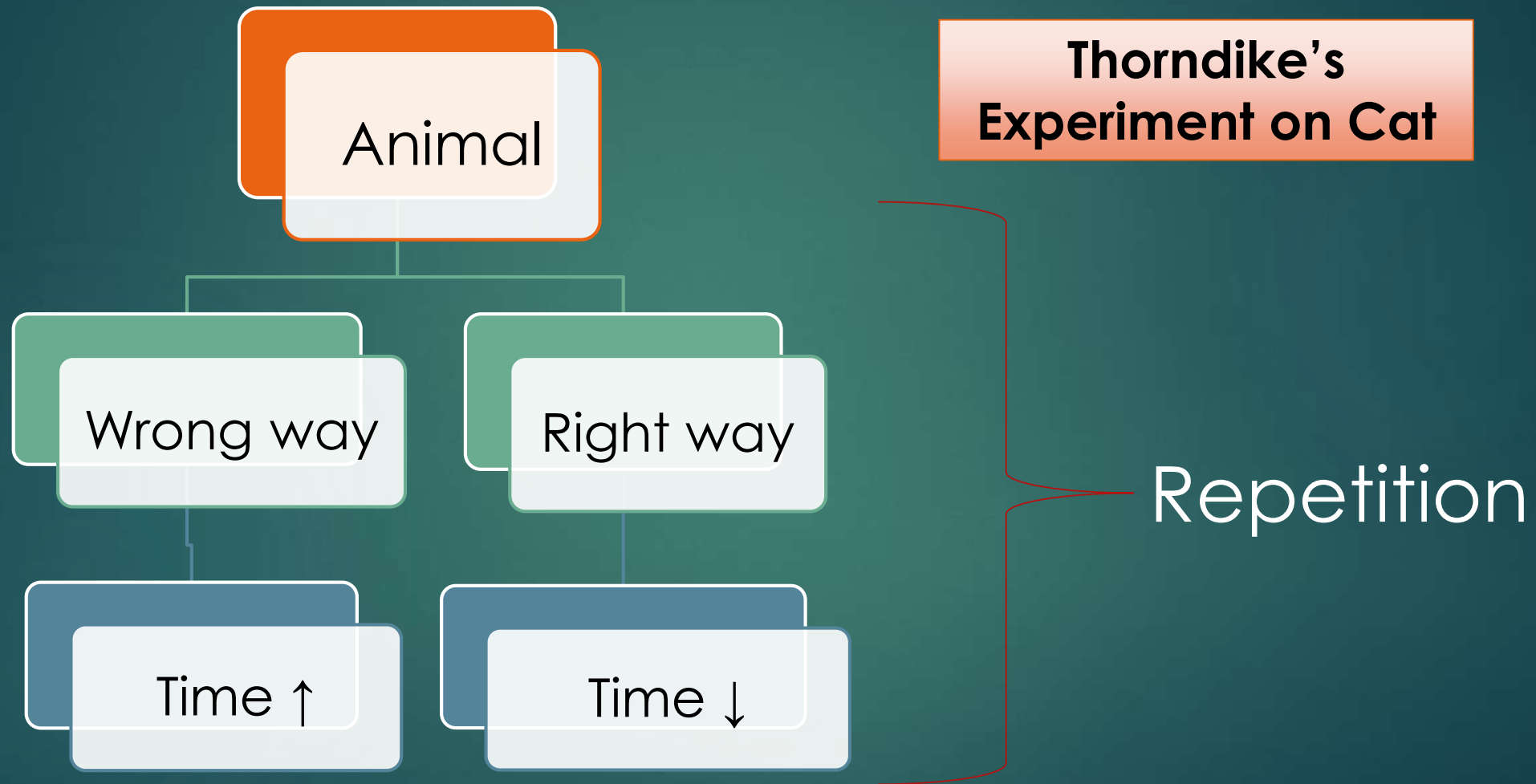
~ Ivan Pavlov

CONNECTIONISM THEORY

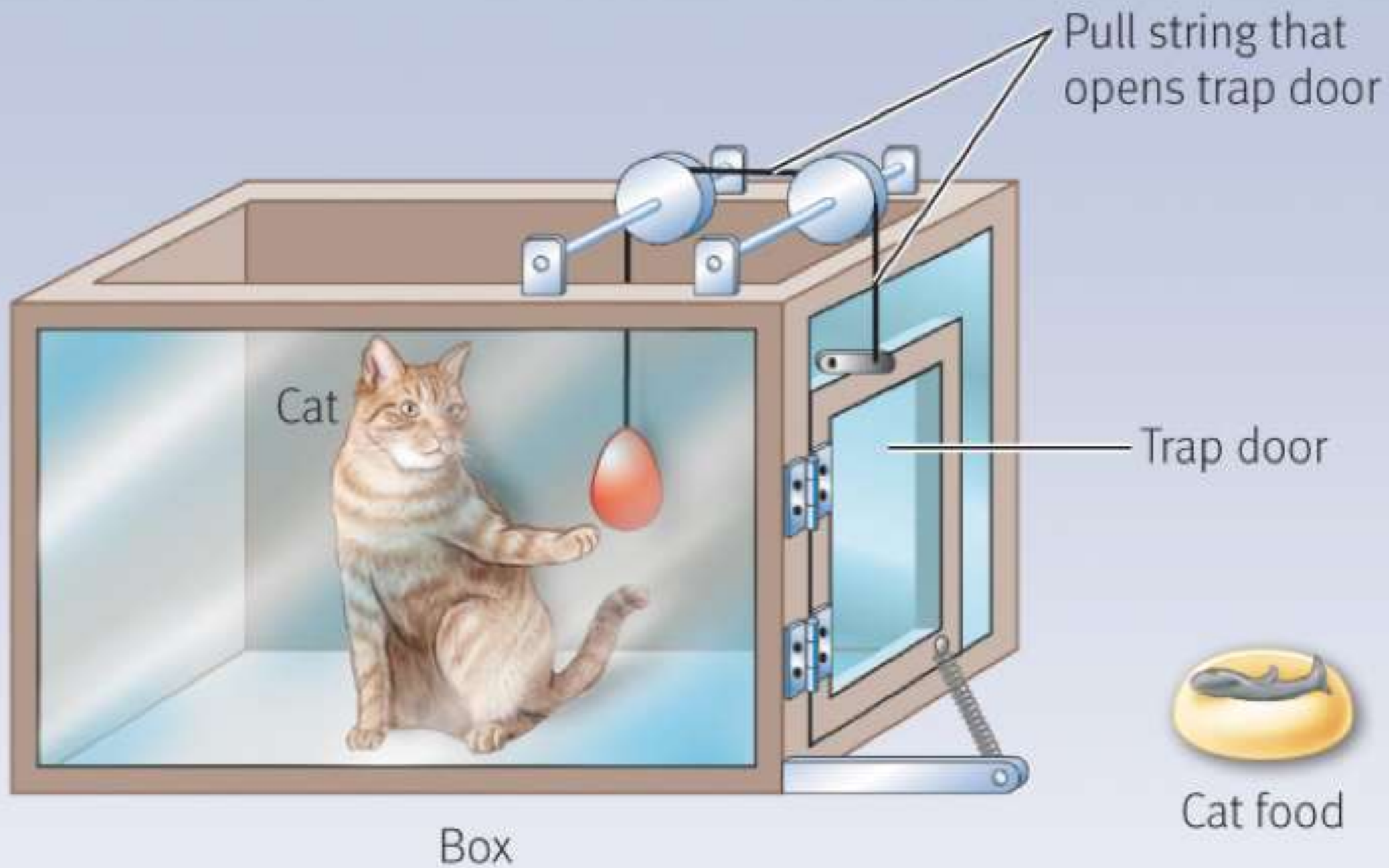
EDWARD LEE THORNDIKE



Connectionism = Stimulus + Response

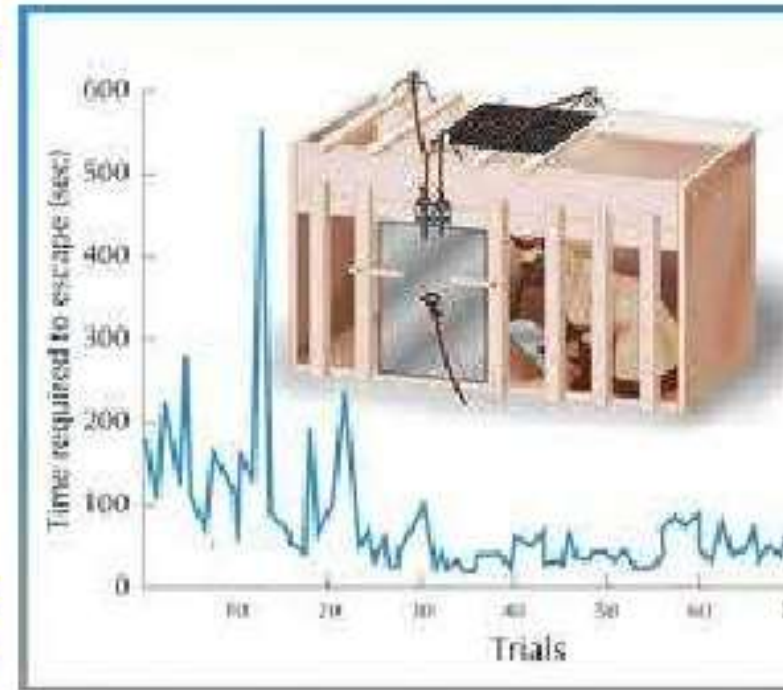


Eg: Hungry Cat



Thorndike's Puzzle-box experiment

- Thorndike put a hungry cat in a **'puzzle box'** & placed fish, just out of reach
- 1. At first the cat to escape from the box through **trial & error** (random voluntary movements)
- 2. Eventually the cat accidentally pulled the string, escaped from the box so that it could reach its **reinforcement** (the fish)
- When the cat was put back in the box, once again it went through a series of incorrect responses before pushing the lever
- The cat became **progressively** quicker at escaping (and had fewer incorrect behaviours)



1. Law of Use
THE STRENGTH COACH

2. Law of Disuse
weakness

*learn by
doing

Law of
Exercise

Trial & Error
Learning

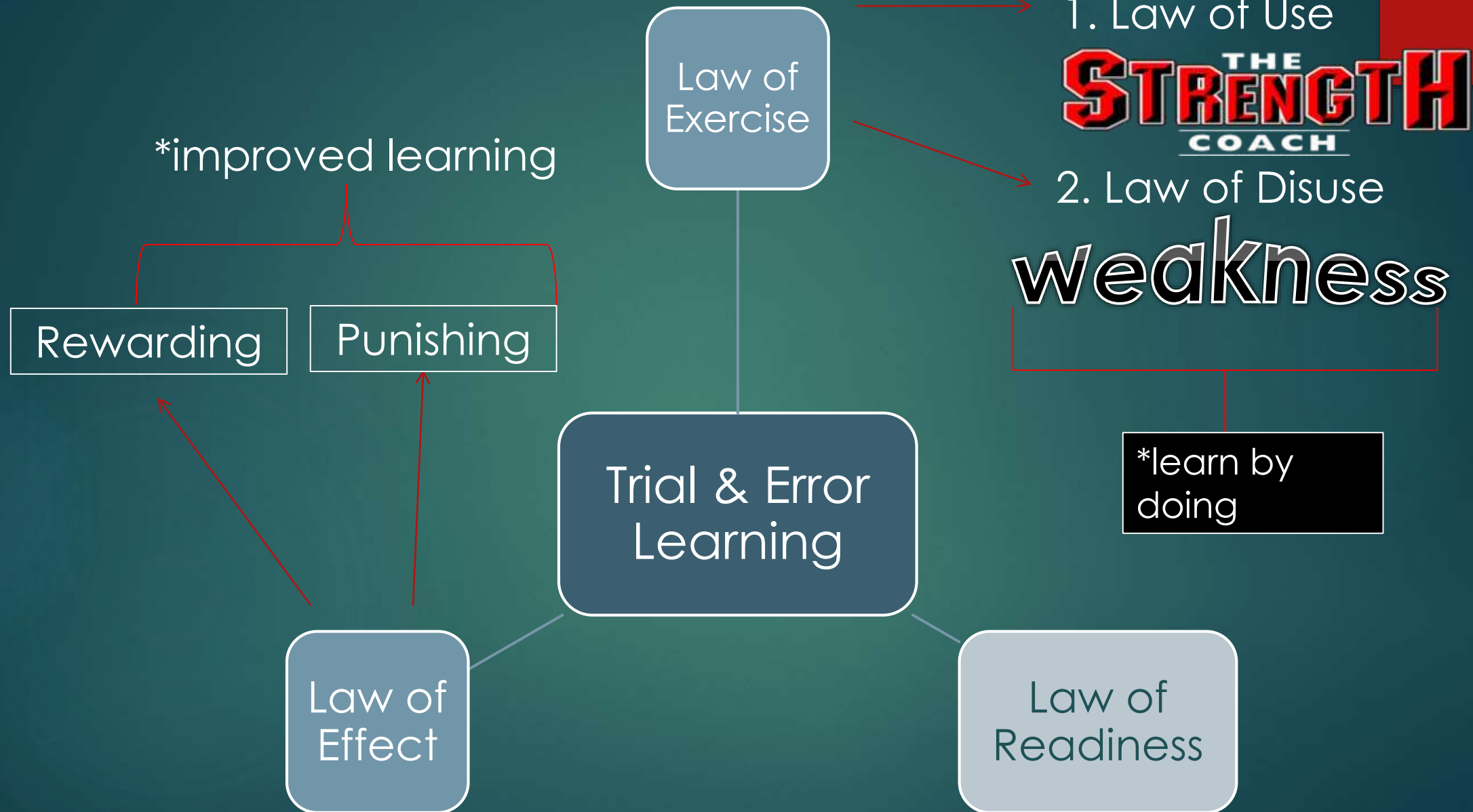
Law of
Readiness

Law of
Effect

Rewarding

Punishing

*improved learning



Three major laws of learning

- ▶ **Law of Readiness** (learner may be satisfied or frustrated depending on his/her stage of readiness)
- ▶ **Law of Exercise** (connection is strengthened in proportion to the number of times it occurs)
- ▶ **Law of Effect** (when learner's response is accompanied or followed by a satisfactory state, the strength of connection is increased)

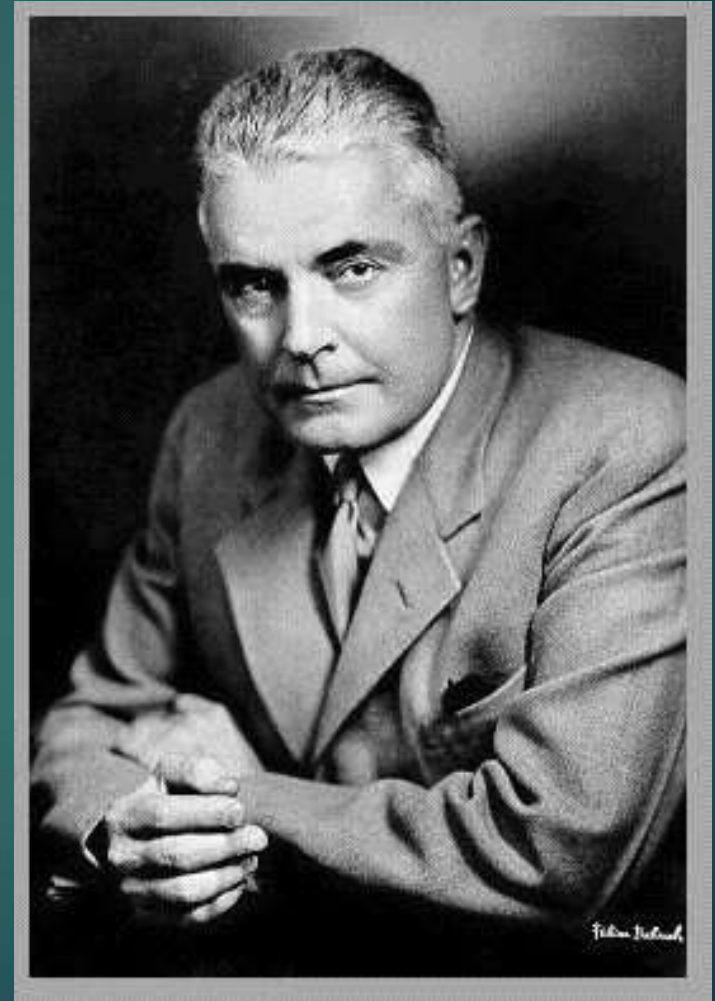


Identify what law of learning (readiness, effect, practice) is characterized in these quotes:

1. Constant correct practice makes perfect
2. Successes in learning will be accompanied by further learning
3. The learner should be biologically prepared to do related stuff.

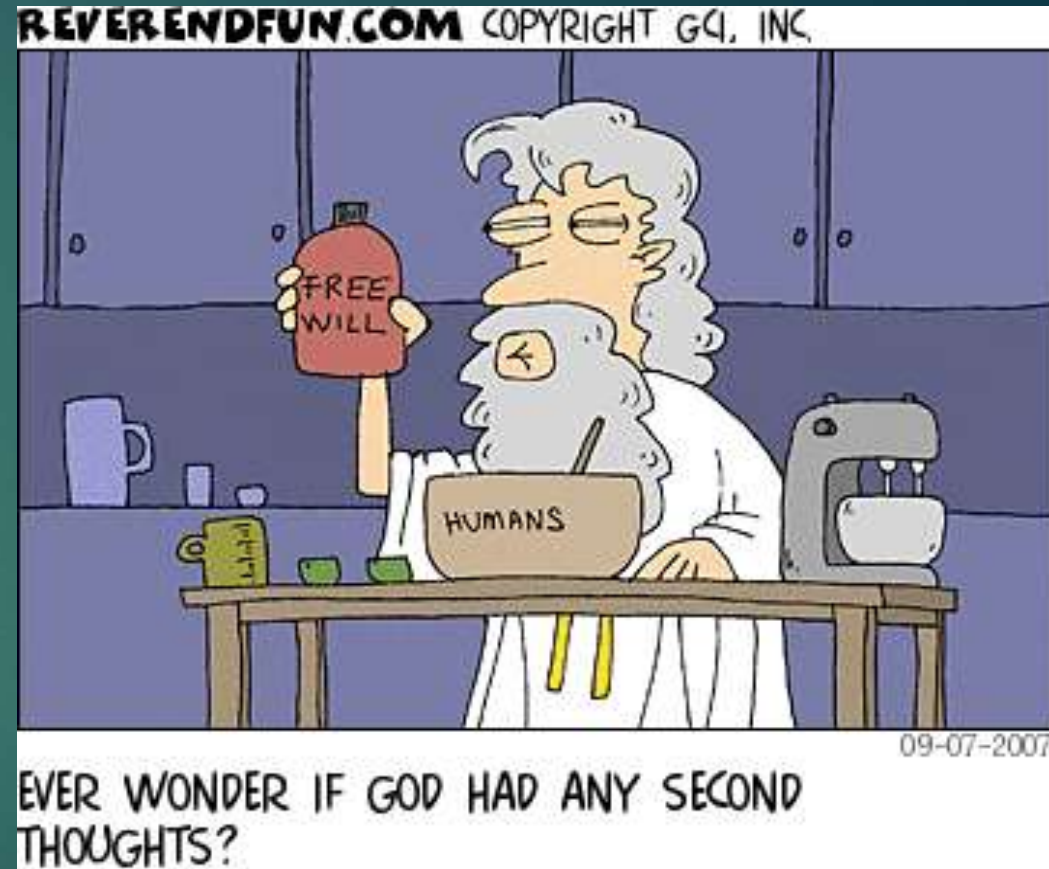
Behaviorism (Reflex Conditioning)

JOHN BROADUS WATSON



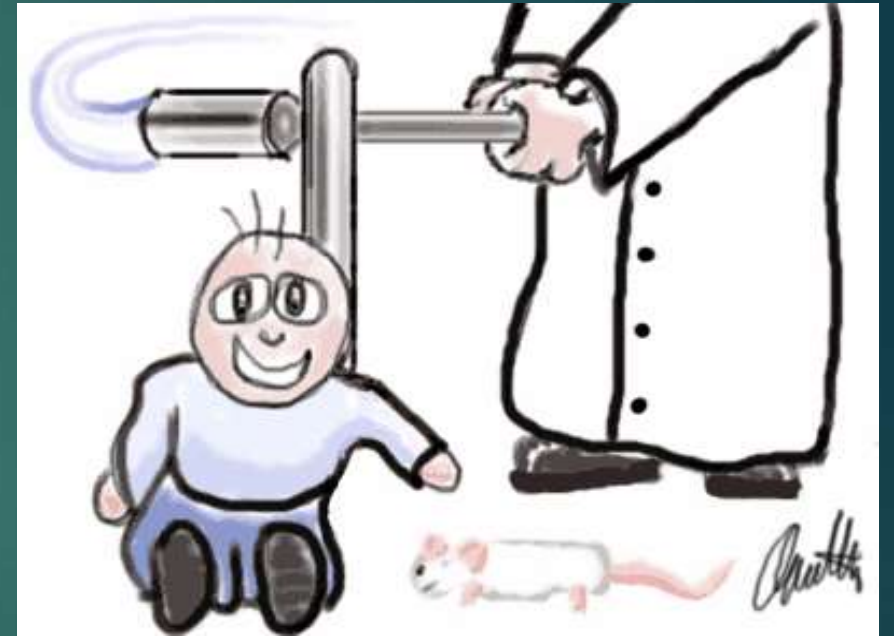
Watson rejected free will

- ▶ Humans learn by a process of conditioning.
- ▶ He believed you could take any child and turn them into whoever you wanted: “I’ll guarantee to take anyone at random and turn them into the specialist which I select”.



J.B. Watson and little Albert

- ▶ Watson attempted to show the same conditioning in humans.
- ▶ He took an 8 month old baby, little Albert, who previously had no fear of rats.
- ▶ His neutral stimulus was a loud sound just behind Albert's head, which upset him.
- ▶ Every time the rat appeared, Watson made the loud sound.







Watson's conclusion

- ▶ Watson presented Albert with the rat on his own, with no noise of a metal bar, and Albert started crying.
- ▶ Albert had learnt to associate the rat with the upsetting noise, producing a *conditioned reflex* (crying).
- ▶ He also found that Albert *generalised* his fear to all white furry objects.

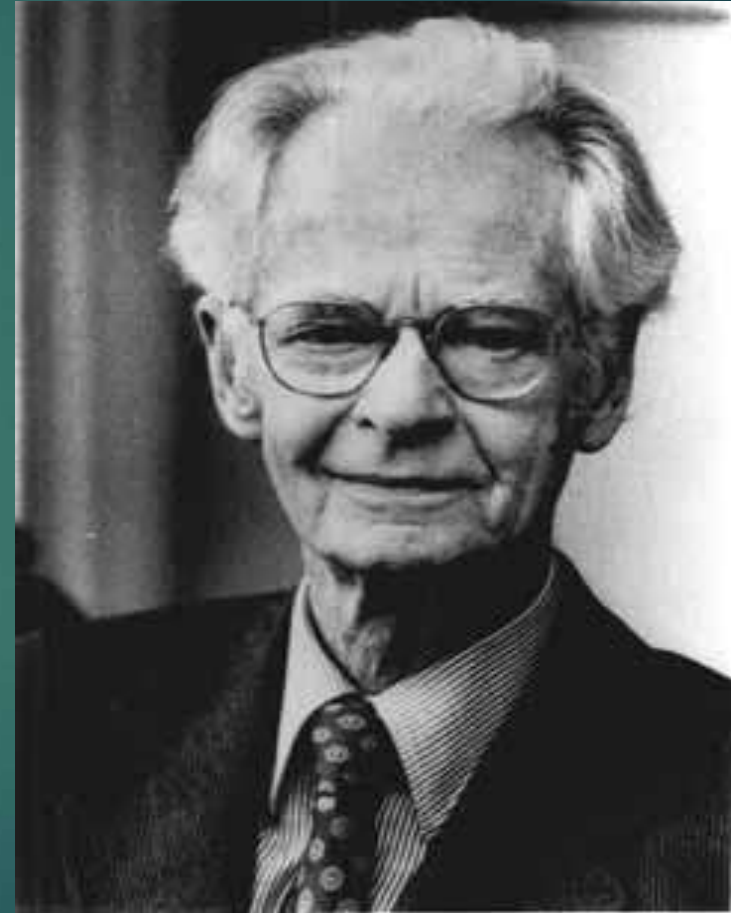
Conditioning and phobias

- ▶ If a child is bitten by a dog, the normally neutral stimulus (dogs) has produced an association of pain and fear.
- ▶ This can be generalised to every dog, so the sight of any dog produces anxiety and fear.
- ▶ The neutral stimulus has become a conditioned stimulus to a conditioned reflex, fear.



Operant Conditioning Theory

BURRHUS FREDERIC SKINNER



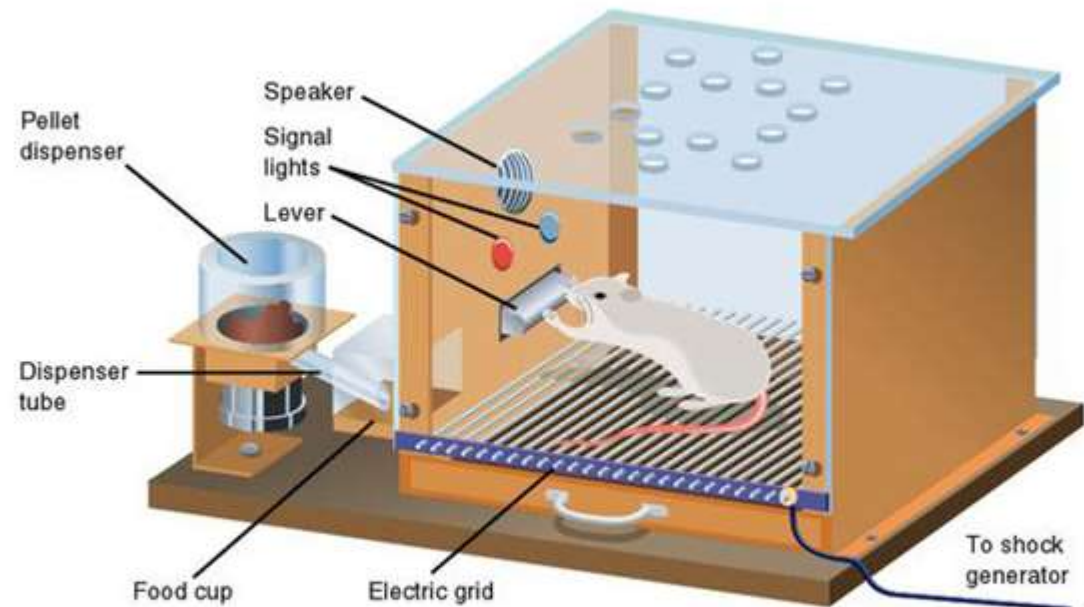
OVERVIEW

- ▶ Skinner believed that we both predict & control behaviour.
- ▶ Thus by manipulating the environment, you can manipulate behaviour
- ▶ A learning process in which the consequences which follow a response determine whether the behaviour will be repeated
- ▶ Behaviour will likely be repeated which has reinforced
- ▶ And tend not to be repeated with punishment

Skinner Box >> Rats

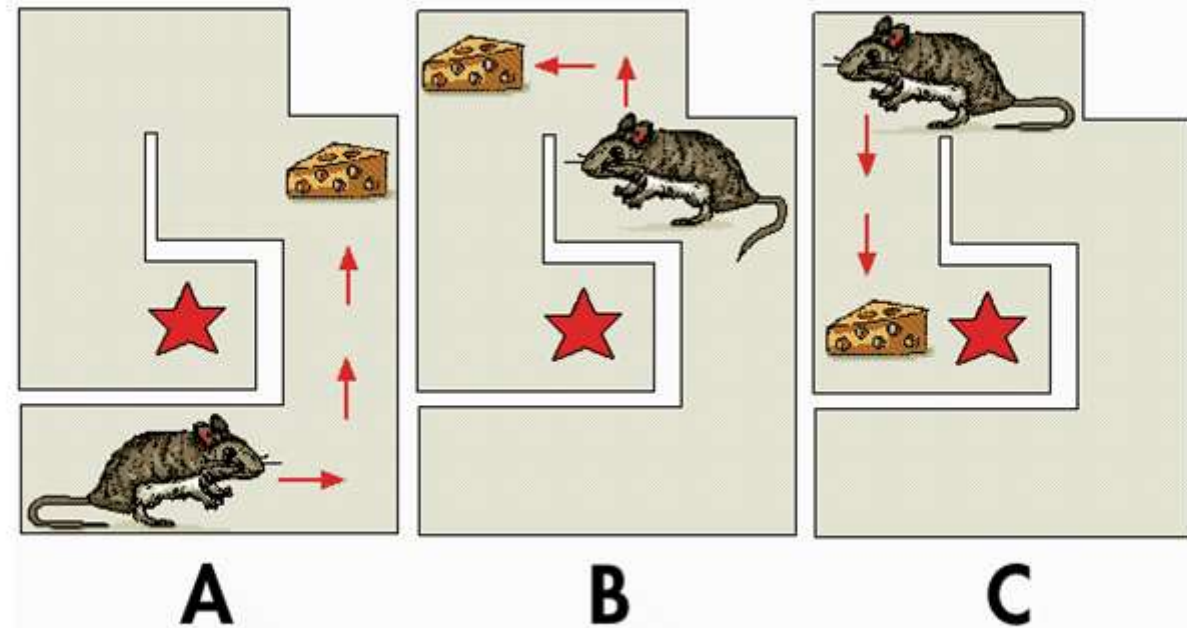
Rats were placed in metal cages with a number of levers. At first the rats would nose around the cage and accidentally press the levers, an action that would cause food or water to drop into a dish. After repeating the action, the rats saw that they could receive food and water by pressing the lever. (Learned this behavior)

So, when the rats were rewarded they were conditioned to repeat this positive action to continue being rewarded



Rat Maze

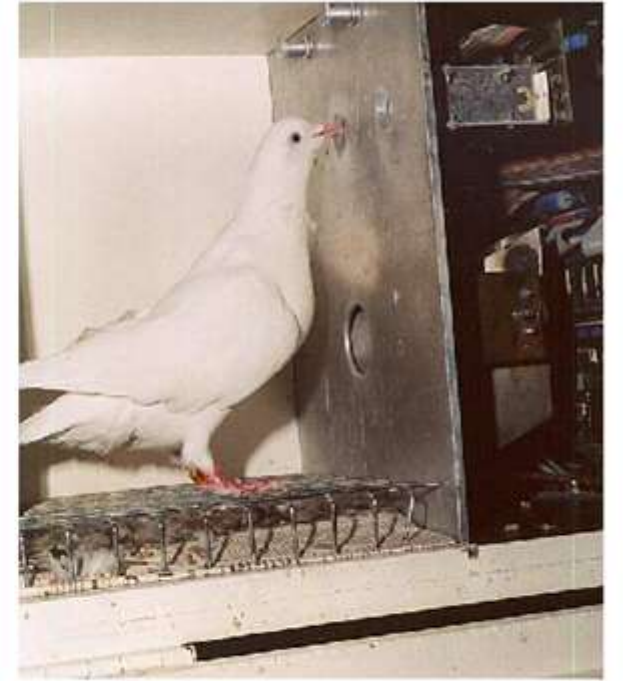
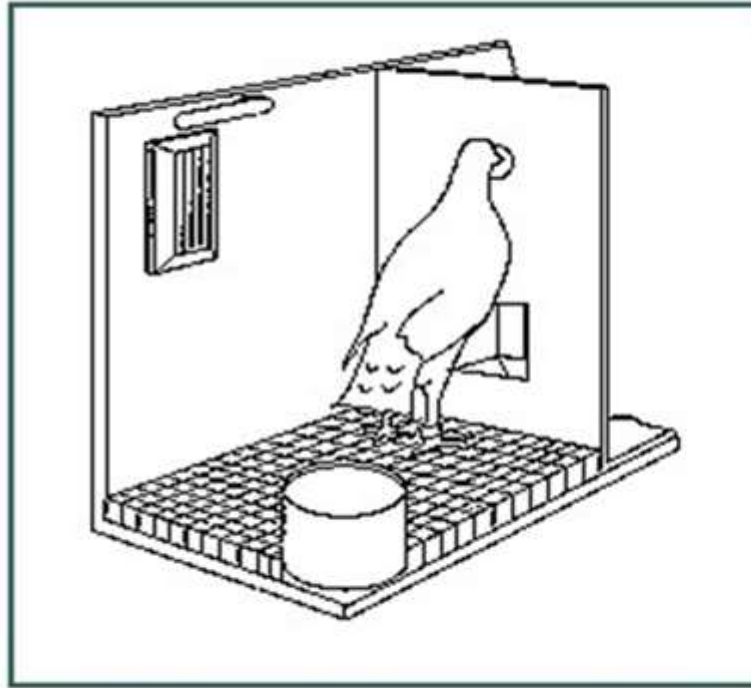
The illustration shows how a mouse can learn to manoeuvre through a maze. the mouse is rewarded with food when it reaches the first turn in the maze (A). Once the first behavior becomes ingrained, the mouse is not rewarded until it makes the second turn (B). After many times, the mouse must reach the end of the maze to receive its reward.



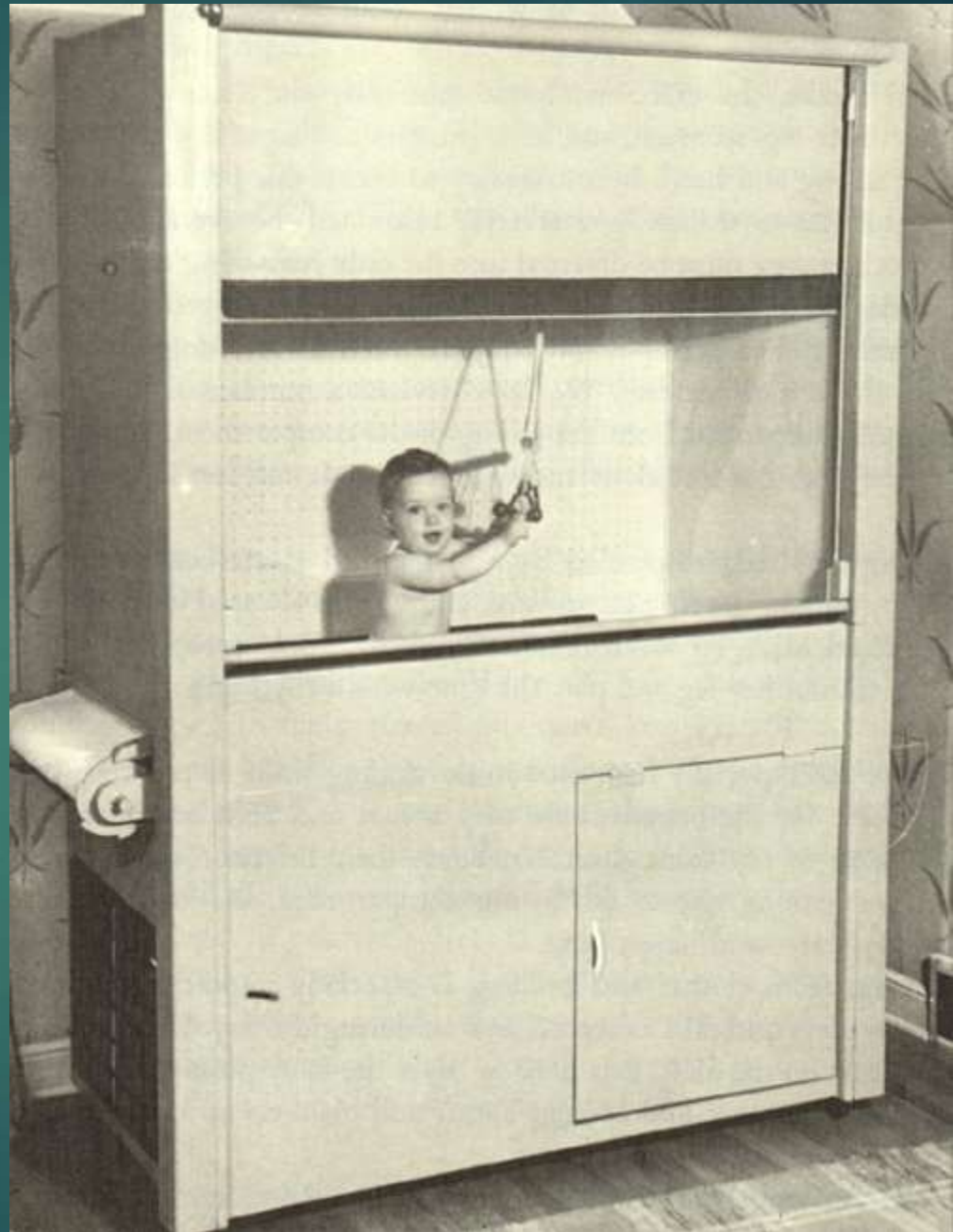
Skinner Box >> Pigeon

During World War II (1944) there were no missile guidance systems so Skinner decided to try and discover one. Skinner got funding for a top secret project to train pigeons to guide bombs. He trained pigeons to keep pecking a target that would hold a missile onto a target. The pigeons pecked reliably, even when falling rapidly and working with warlike noise all around them. (Learned behavior)

Skinner trained the pigeon's to peck at a particular colored disk



baby tender



Consequences



Reinforcement

- The process in which a behavior is strengthened, and thus, more likely to happen again.



Punishment

- The process in which behavior is weakened, and thus, less likely to happen again



Reinforcement

Positive Reinforcement

Making a behavior stronger by following the behavior with a pleasant stimulus

Negative Reinforcement

Making a behavior stronger by taking away a negative stimulus

Punishment

Negative Punishment

- Reducing a behavior by removing a pleasant stimulus

Positive Punishment

- Reducing a behavior by presenting an unpleasant stimulus

Consequences

Positive Reinforcement

A rat presses a lever and receives food

Positive Punishment

If the rat previously pressed the lever and received food and now receives a shock, the rat will learn not to press the lever

Negative Reinforcement

A rat presses a lever and turns off the electric shock

Negative Punishment

If rat now receives food consistently without pressing the lever, the rat will learn to stop pressing the lever

Operant Conditioning

**Reinforcement
+ Behavior**

**Punishment
- Behavior**

POSITIVE

ADD Appetitive
stimulus following
correct behavior

Giving a food when the dog sits.

NEGATIVE

POSITIVE

ADD Unpleasant
Stimuli following
behavior

Spanking a child for cursing

NEGATIVE

REMOVE Appetite
stimulus following
behavior

Telling a child to go to his room
for fighting her sister

Escape

Remove
unpleasant
stimuli following
correct behavior

Turning off an alarm clock the push
button to stop

Active Avoidance

Behavior avoids
noxious stimulus
following correct
behavior

Studying to avoid getting a bad grades

2. Operant Conditioning (B.F. Skinner)

- ▶ Types of reinforcements:
 - ▶ **Positive** (reinforcers increase response frequency)
 - ▶ **Negative** (strengthens behavior by removal of something which will reinforce learning as a result)
 - ▶ **Primary** (food, water, sleep)
 - ▶ **Secondary** (money, grades, stars, tokens)

2. Operant Conditioning (B.F. Skinner)

- ▶ Classifications of reinforcements:
 - ▶ **Verbal** (praise, encouragement)
 - ▶ **Physical** (touches, pats, hugs)
 - ▶ **Non-Verbal** (smiles, winks, warm looks)
 - ▶ **Activity** (being allowed to play games, listen to music)
 - ▶ **Token** (points, chips, stars)
 - ▶ **Consumable** (cookies, food)

Shaping

- ▶ Technique of reinforcement used to teach new behaviors.
- ▶ At the beginning, people/animals are reinforced for easy tasks, and then increasingly need to perform more difficult tasks in order to receive reinforcement.

Originally the rat is given a food pellet for one lever press, but we gradually increase the number of times it needs to press to receive food, the rat will increase the number of presses.

Extinction

- ▶ The elimination of the behavior by stopping reinforcement of the behavior

A rat who received food when pressing a bar, receives food no longer, will gradually decrease the amount of lever presses until the rat eventually stops lever pressing.

Generalization

- ▶ A behavior may be performed in more than one situation.

A rat who receives food by pressing one lever, may press a second lever in the cage in hopes that it will receive food. It is the tendency for a stimulus that is similar to the original conditioned stimulus to elicit a response that is similar to the new stimulus

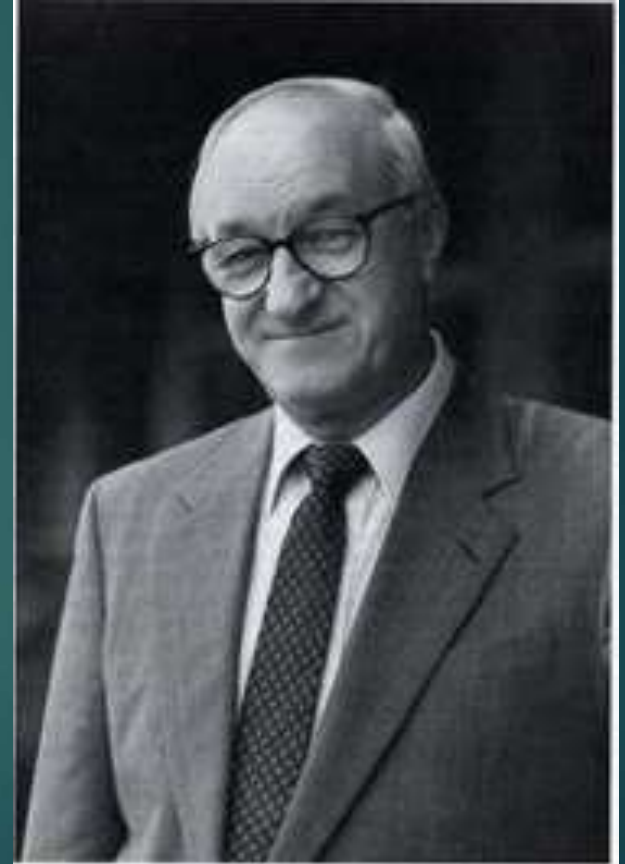


Identify what reinforcement type (positive or negative) is characterized in the examples:

- ▶ Buying toys after having good grades
- ▶ After having a relief in taking aspirin the first time, you take it when headache always occurs
- ▶ As reward, time is shortened for those who finish early in their tasks
- ▶ Appreciative nod of the teacher makes the students continue their work

SOCIAL LEARNING THEORY

ALBERT BANDURA



Social Learning Theory (Albert Bandura)

- ▶ Also called **observational learning theory**
- ▶ States that learning takes place when one person **observes and then imitates the behavior of others**
- ▶ Process of learning through vicarious conditioning and observational learning

BOBO DOLL EXPERIMENT



Social Learning Theory (Albert Bandura)

- ▶ Stresses the importance of **models** because the observer-learner may:
 - ▶ Acquire new responses
 - ▶ Strengthen or weaken every existing response
 - ▶ Cause the reappearance of responses that have been apparently forgotten

Social Learning Theory (Albert Bandura)

Elements of Observational Learning:

- ▶ **Attention** (mental focus / concentration)
- ▶ **Retention** (ability to retain the information)
- ▶ **Motor Reproduction Process** (perform what was observed)
- ▶ **Motivational Process** (encouragement, force that drives one to act)

Three Forms of Reinforcement

- ▶ **Direct Reinforcement** (student receiving verbal praises or reward, or even punishment)
- ▶ **Vicarious Reinforcement** (student waits for a reward because someone else has been rewarded)
- ▶ **Self-Reinforcement** (student strives to meet personal standards and doesn't care about others)



COGNITIVISM

- ▶ Meaningful Reception
- ▶ Insight Theory
- ▶ Field Theory
- ▶ Gestalt Psychology
- ▶ Information Processing

MEANINGFUL RECEPTION LEARNING

DAVID AUSUBEL



Meaningful Reception Learning

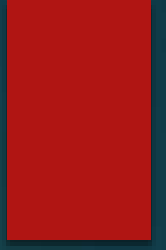
- ▶ Concerned with how students learn large amounts of **meaningful material from verbal/textual presentations** in a school setting
- ▶ Meaningful learning results when information is acquired by **linking the new information in the learner's own cognitive structure**
- ▶ A major instructional mechanism proposed by Ausubel is the use of **advanced organizers** which help **to link new learning material** with existing related ideas

INSIGHT LEARNING THEORY

WOLFGANG KOHLER



Insight Theory



- ▶ Gaining insight is a gradual process of **exploring**, **analyzing**, and **restructuring** perceptions until a solution is arrived at.
- ▶ When the learner is confronted with a situation requiring a new **adjustment** on his part and he is able to **analyze the situation**, then insight has been involved.

What is Insight Learning?

► **Insight learning** is a type of learning or problem solving that happens all-of-a-sudden through understanding the relationships of various parts of a problem rather than through trial and error.



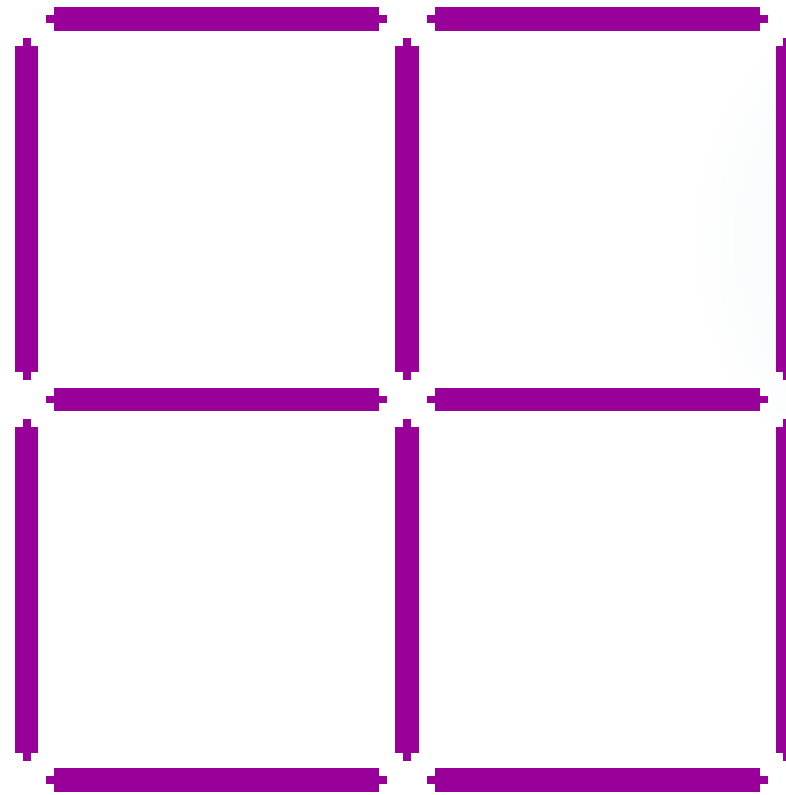
Epiphany

Another noteworthy term that describes insight learning is "epiphany." Epiphanies involve a sudden revelation or abrupt awareness bringing seemingly chaotic data into symmetry.

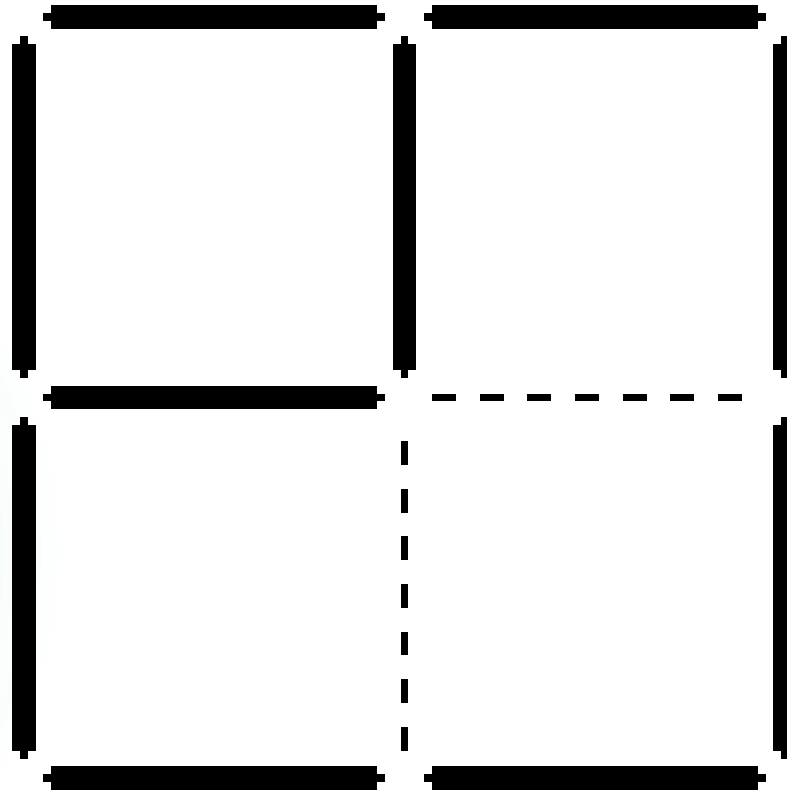
Eureka!

- ▶ Insight learning also involves the "**I have found it!**" feeling or "**eureka**".
- ▶ Insight learning is also expressed as the "**Aha moment**,"

Can you leave two complete squares only, by taking away two sticks from the pattern shown below?

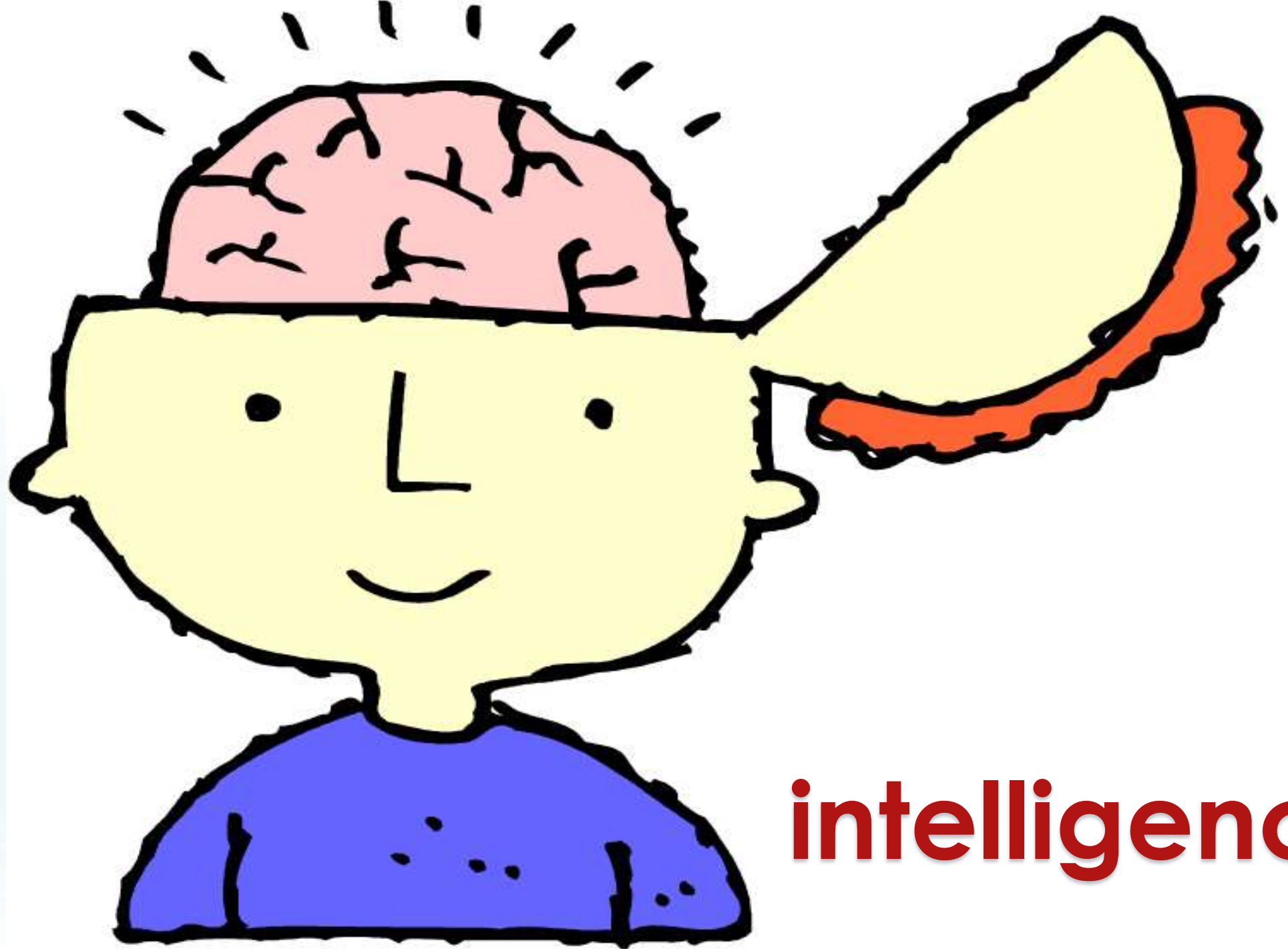


Answer is.....



Insightful learning depends on certain factors:

experience



intelligence



Learning Situation



Initial Efforts

Repetition

&

Generalization









TOPOLOGICAL & VECTOR / FIELD THEORY

KURT LEWIN





Learning is more effective when it is
an active rather than a passive
process.

— *Kurt Lewin* —

AZ QUOTES

Topological and Vector or Field Theory

- ▶ The behavior of an individual at a given moment is the result of **existing forces** operating simultaneously in his **life space** (internal and external forces).
- ▶ Derived from the terms **vector** (quantity that has magnitude and direction) and **topology** (geometric configuration)



Lewin said that an individual has inner and outer forces that affect his perception and also his learning.

▶ **Inner Forces**

include his own motivation, attitudes and feelings.

▶ **Outer Forces**

include the attitude and behavior of the teacher and classmates.

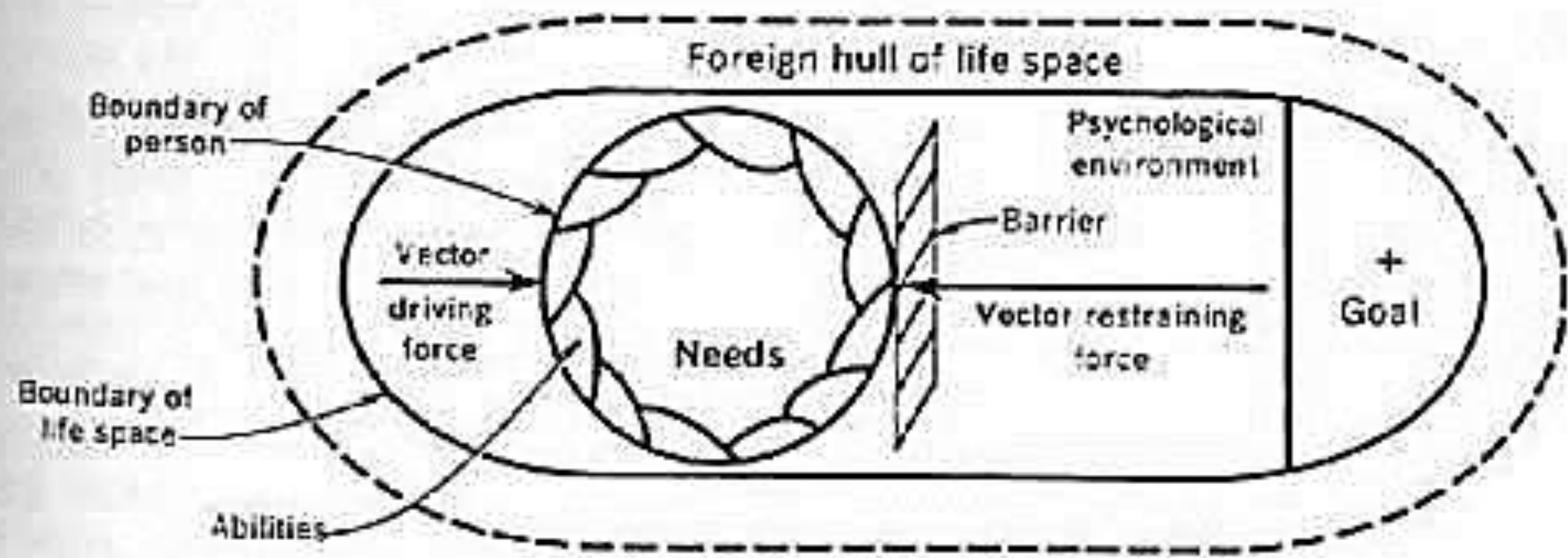
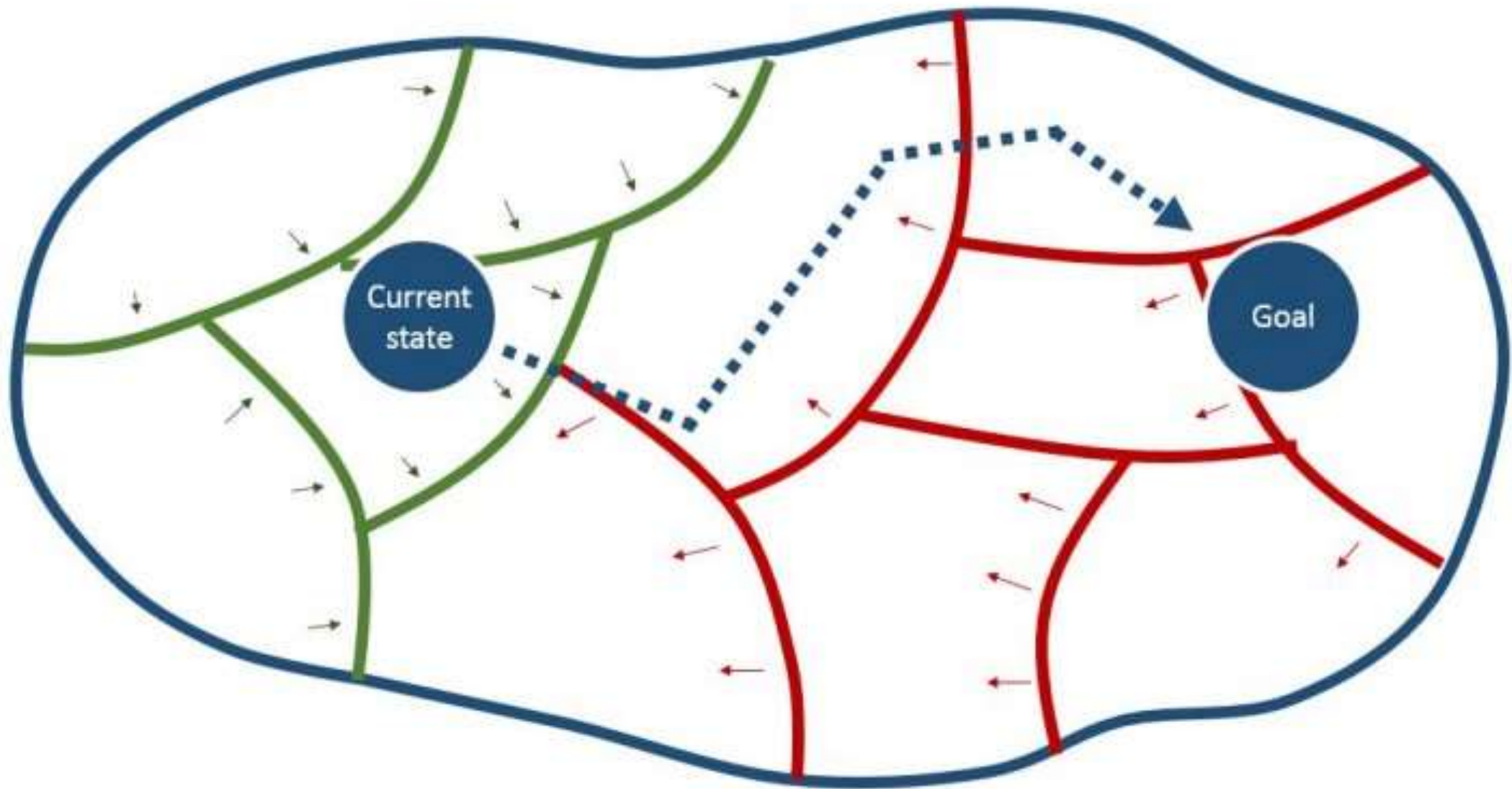


Fig. 3.5. Life space of an individual according to Lewin.

A depiction of a life space, reflecting Kurt Lewin's field theory



Adapted from Burnes & Cook (2013), Kurt Lewin's Field Theory: A Review and Re-evaluation,
International Journal of Management Reviews, Vol. 15, 408-425
www.sidewaysthoughts.com

GESTALT LEARNING THEORY

KOHLER, WERTHEIMER, KOFFKA, AND LEWIN



Gestalt Psychology

- ▶ Focused on **perception** and how people assign meanings to visual stimuli
- ▶ *"The whole is more than the sum of its parts."*
- ▶ **Gestalt** = structure, configuration, or pattern

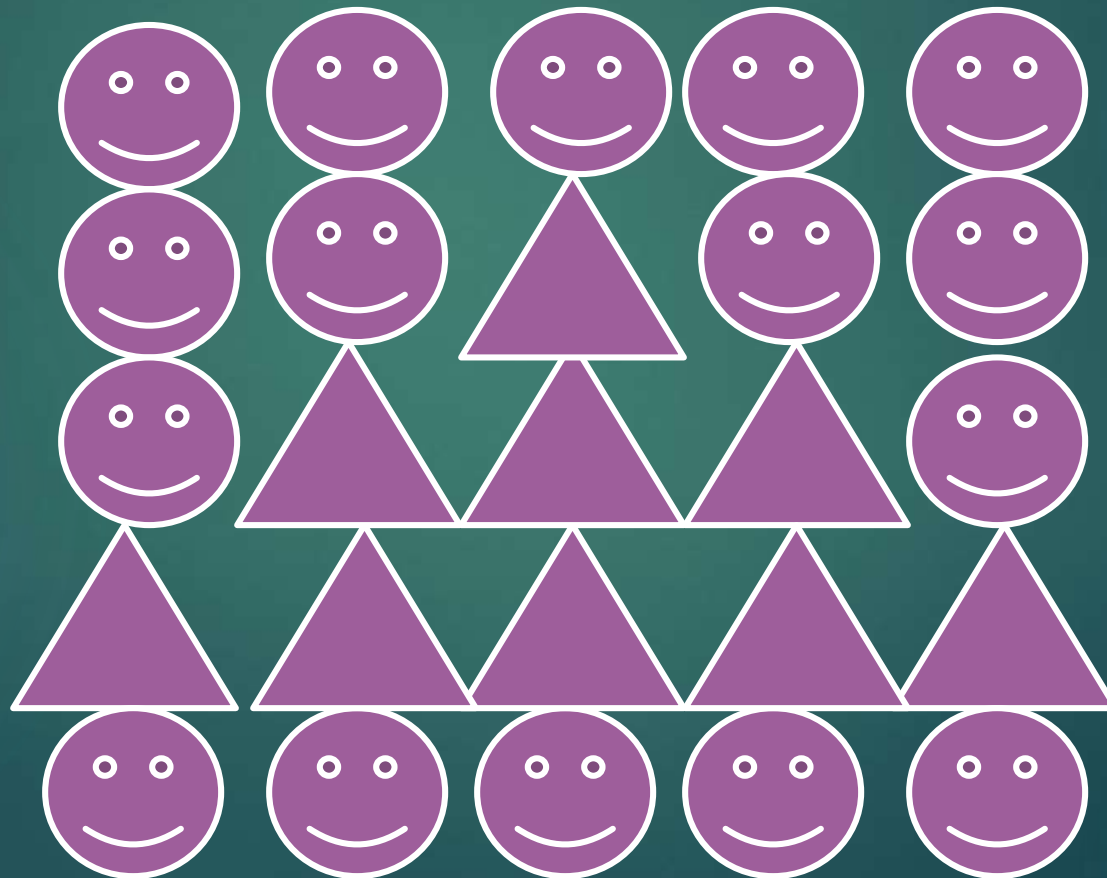
Law of Proximity

these are the objects near each other or closer together that tend to be grouped together.



Law of Similarity

- ▶ elements that look similar we perceived as part of the same form.



Law of Closure

- ▶ objects grouped together are seen as a whole.



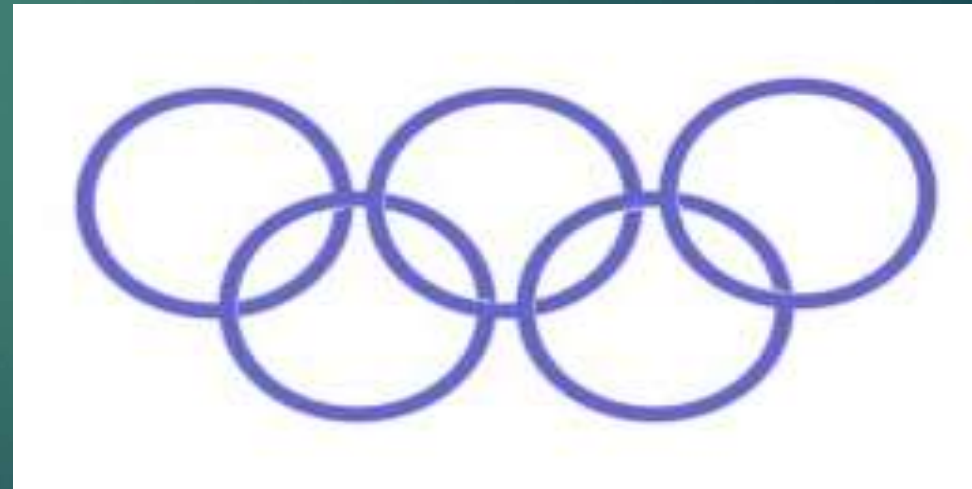
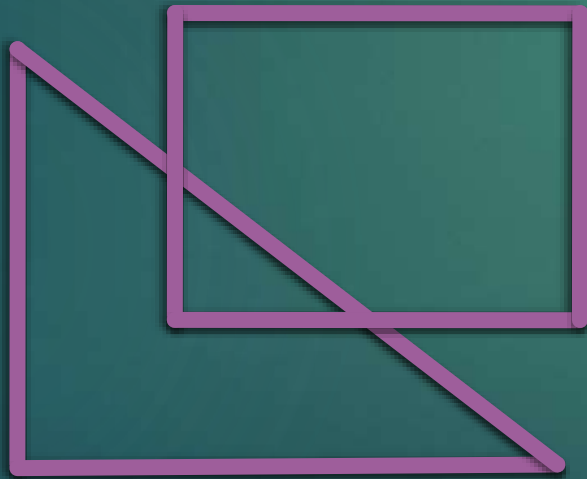
Law of Good Continuation

- ▶ holds the points that are connected by straight or curving lines are seen in a way that the smoothest path, rather than seeing separate lines and angles, lines are seen that are belonging together.



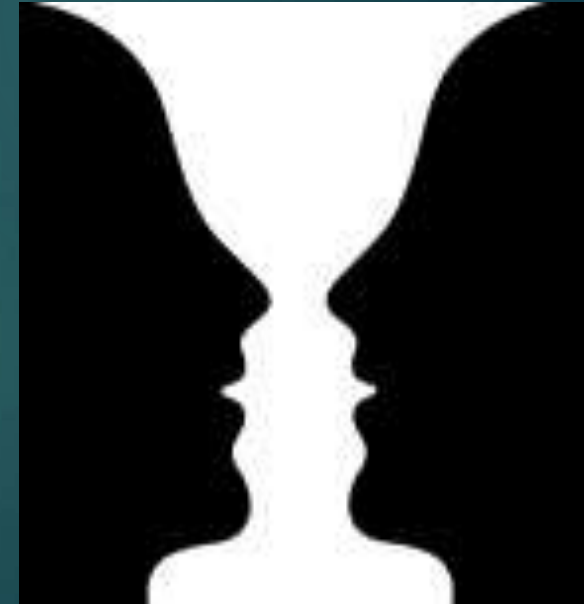
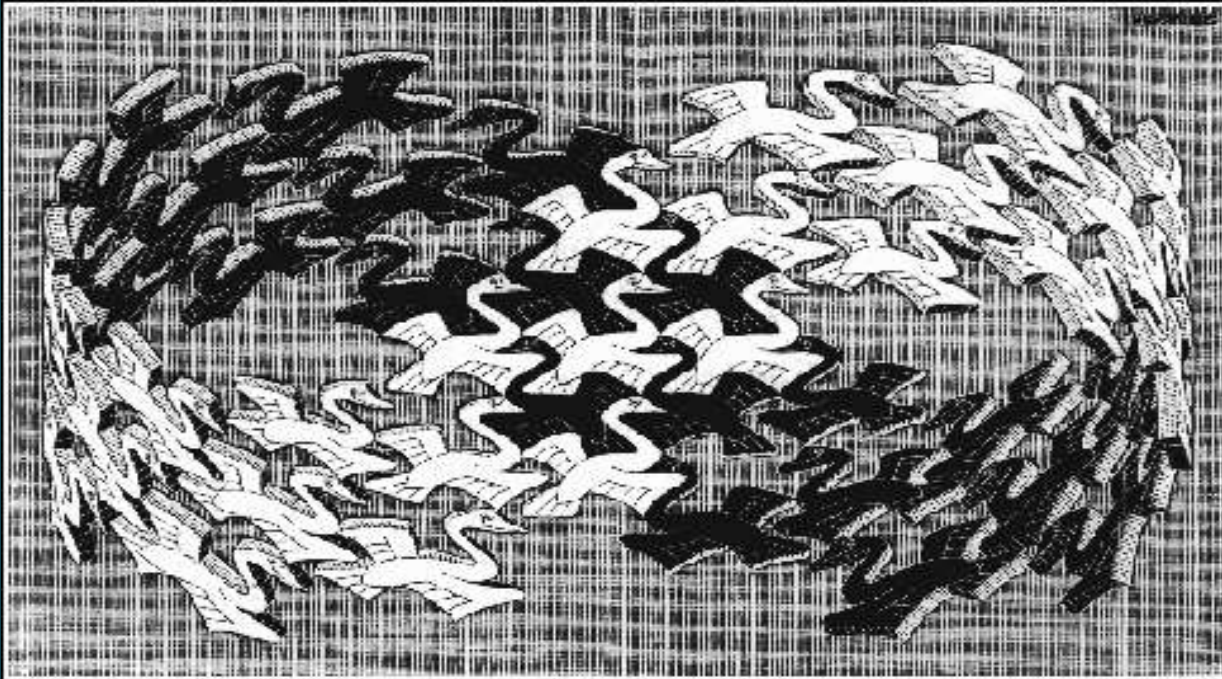
Law of Good Pragnanz

- ▶ the word *pragnanz* a German term meaning “good figure”. The law of pragnanz is sometimes referred to as law of good figure or law of simplicity. This hold that objects in the environment are seen in a way that makes them appear as simple as possible.



Law of Figure/Ground

we pay attention and perceived things in the foreground first. A stimulus will be perceived as separate from its ground.



Olny srmat poelpe can raed tihs.

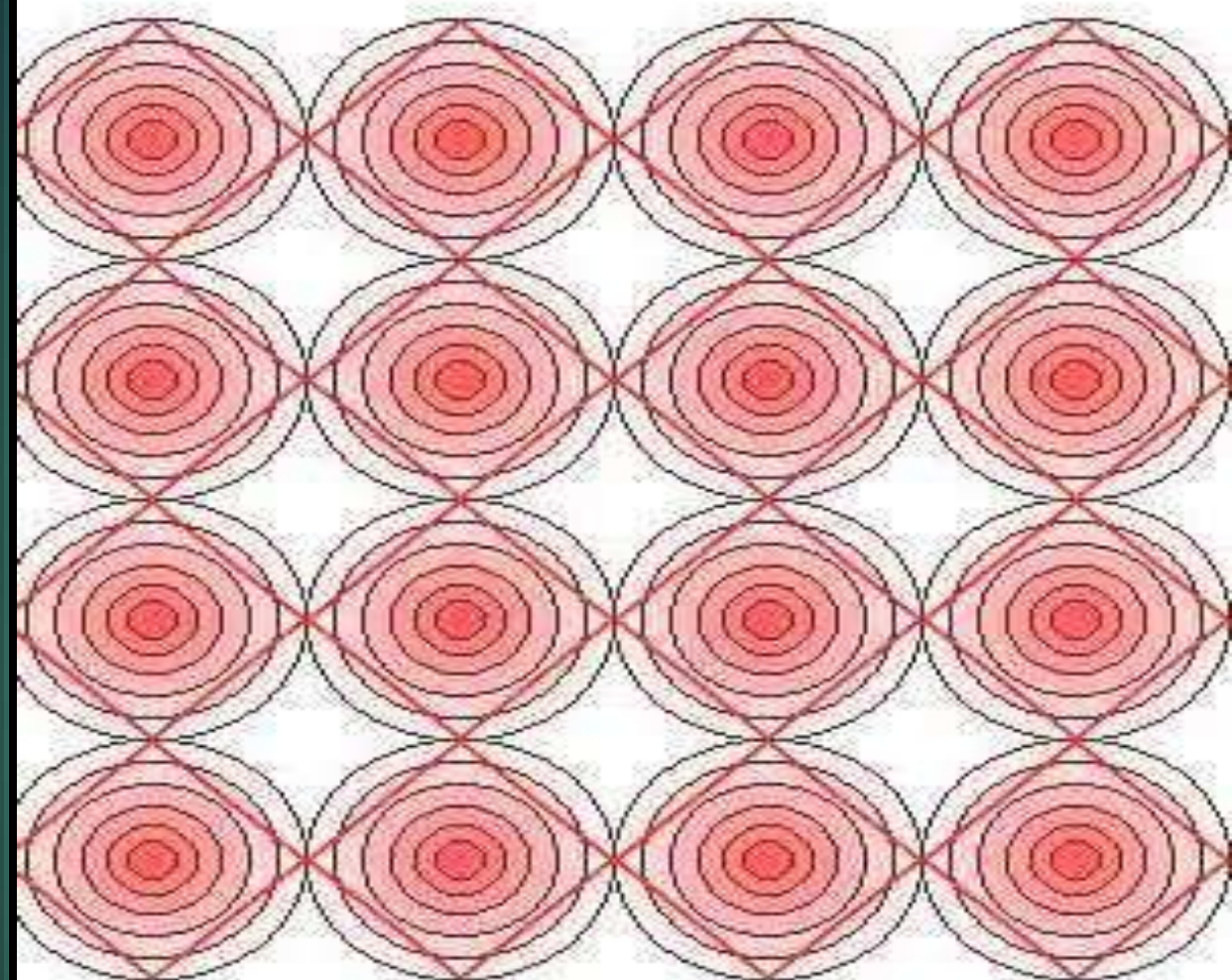
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CAN BE A TAOTL MSES AND YOU CAN SITLL RAED
IT WOUTHIT A PORBELM.*

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WLOHE. AMZANIG HUH? YAEH AND I
AWLYAS TGHUHOT SLPELING WAS IPMORANTT! IF
YOU CAN RAED TIHS PSAS IT ON!!*



Two shades of Green ?

Actually there is only
one shade of green.



Can you believe that all the
lines are perfectly straight?



Which soldier is taller? Neither one. They're all the same size!!!



INFORMATION PROCESSING



Take out a piece of paper...

- ▶ Name the seven dwarves...



Was it easy or hard?



- It depends on several things....
- If you like Disney movies?
- When was the last time you have seen the movie or have read the book?

Answers:

Snow White
and the
Seven Dwarfs
1937

Doc

Grumpy

Happy

Sleepy

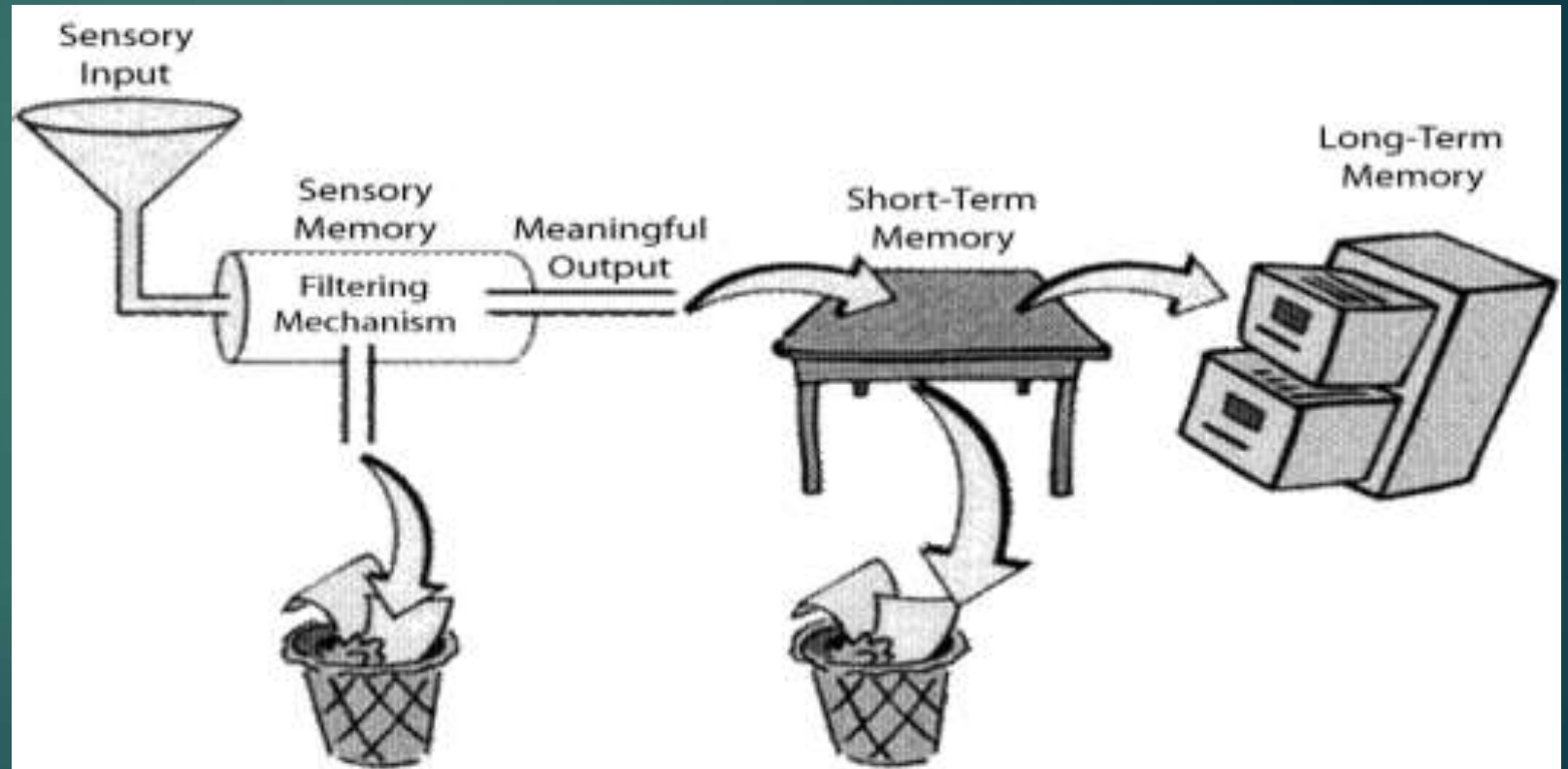
Bashful

Sneezy

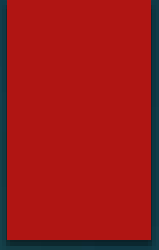
Dopey

Information Processing

- ▶ Model that facilitates understanding of how people store and retrieve information based on how computers process info



Retaining Information in Working Memory



- **Chunking:**
Organizing items into familiar, manageable units.

- **Mnemonic devices**
(knuckle mnemonic)

- **Rehearsal**

- **Maintenance Rehearsal**
involves **repeating the information** in our mind (phone number or a location on a map)
- **Elaborative Rehearsal** involves **connecting the information** you are trying to remember with something you already know, with knowledge from long-term memory. (ex. You meet someone at a party whose name is the same as your brother's—you make the **ASSOCIATION**)

Forgetting

- Information may be lost from working memory through **interference or decay**
- **DECAY** – the weakening and fading of memory with the passage of time
- **INTERFERENCE** – processing new information interferes with old information
- **Retroactive Interference:** new information blocks out old information.
- **Proactive Interference:** old information blocks out new information.

Contents of Long Term Memory

1. Declarative

Knowledge- is knowledge that can be declared, through **words and symbol systems** of all kinds

- **verbal information; facts; “knowing that”** something is the case.

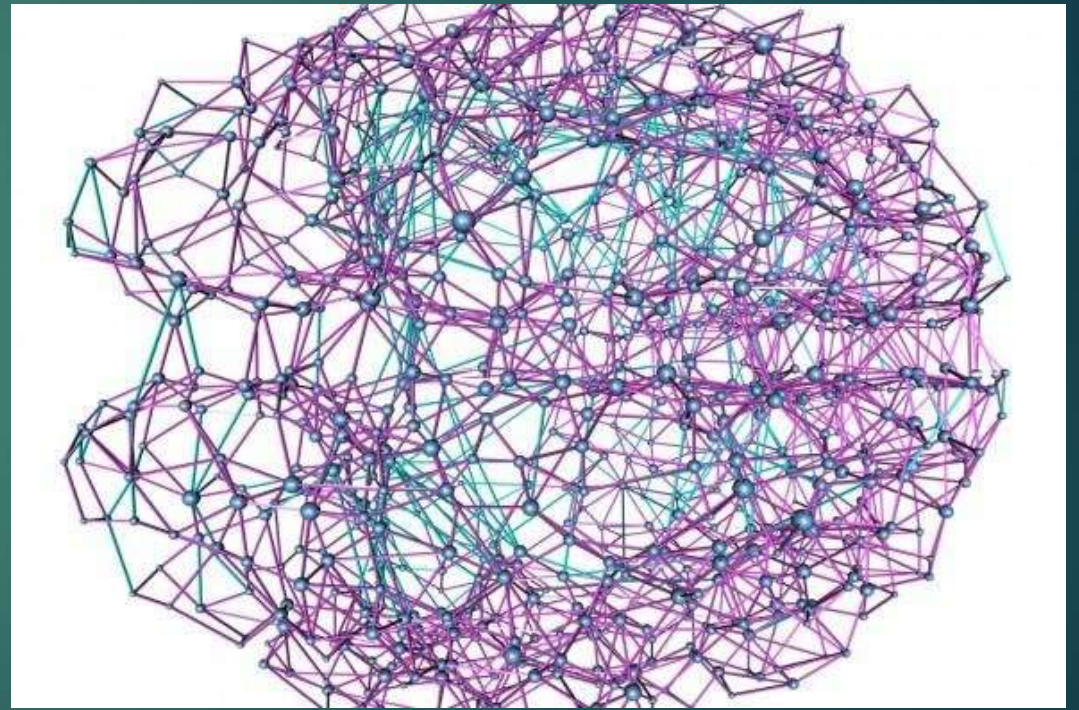
2. Procedural

Knowledge- knowledge that is demonstrated when we perform a task; **“knowing how”**

3. Conditional

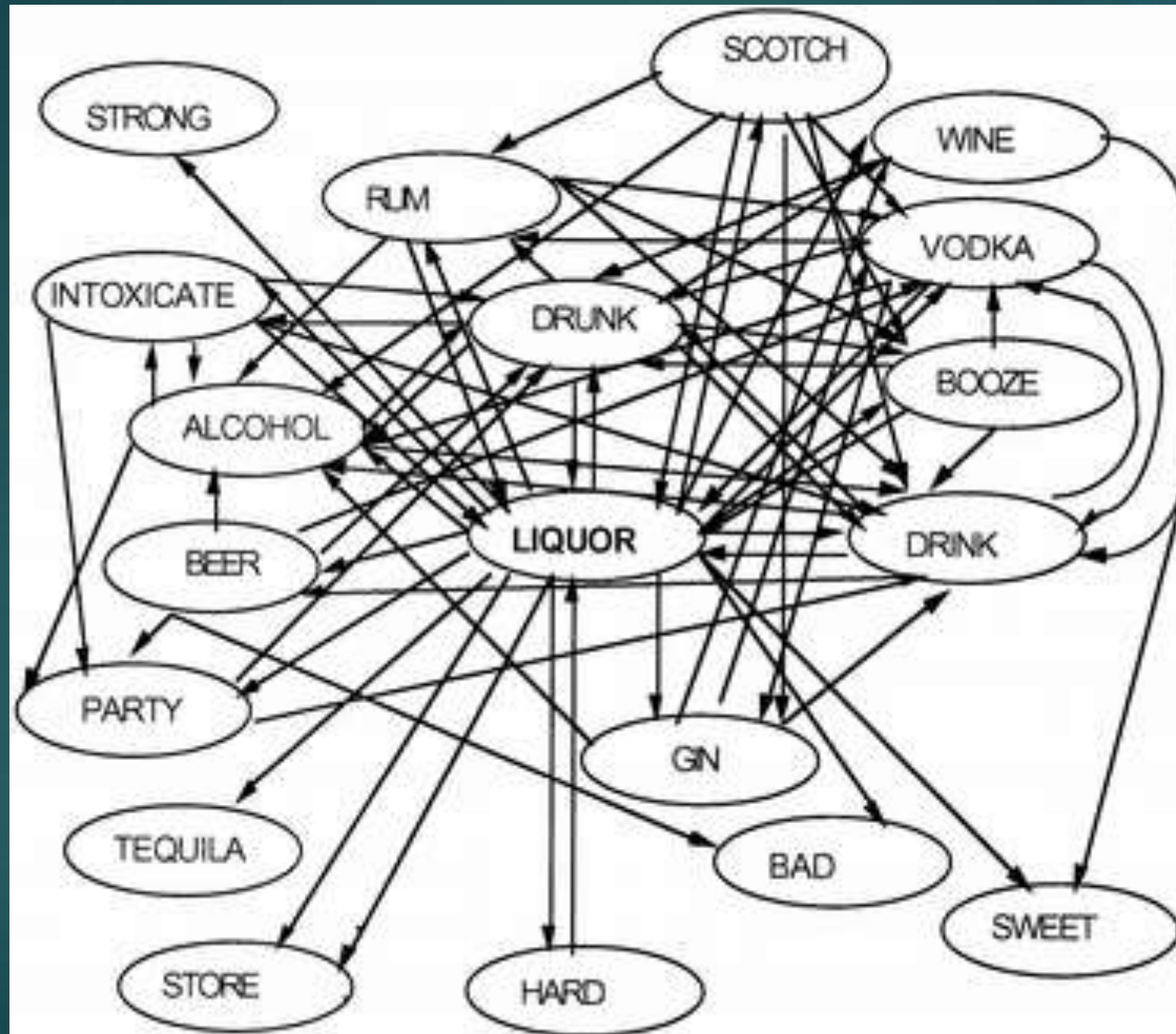
Knowledge – “knowing when and why” to use declarative and procedural knowledge

NETWORK MODEL



Network Model

- ▶ A model that attempts to mimic human memory through **associative networks**
- ▶ Resembling a net, a chain – connected in a huge network with related information connected **hierarchically**
- ▶ Semantic memory – memory of information regarding facts, concepts, and principles; information about general categories

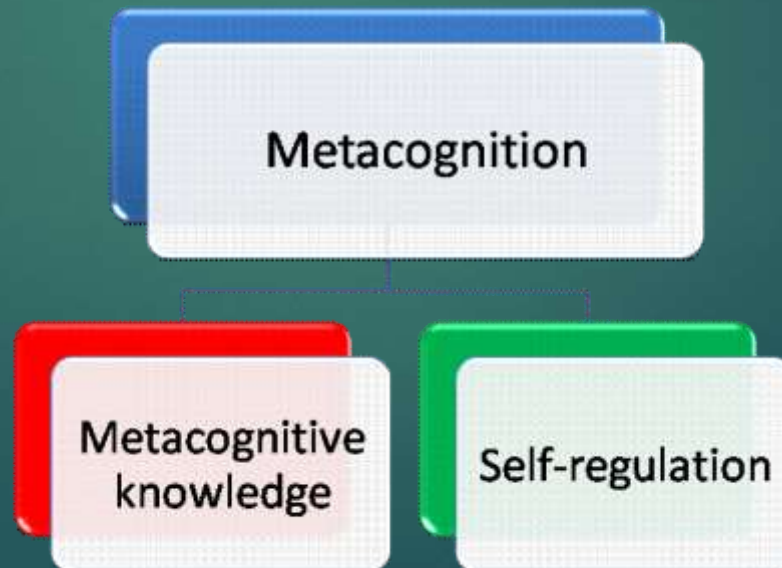


METACOGNITION



What is metacognition?

- ▶ “thinking about thinking”
- ▶ Knowledge and understanding of what we know and how we think, including the ability to regulate our thinking as we work on a task



Metacognition

```
graph TD; Metacognition[Metacognition] --> Metacognitive_knowledge[Metacognitive knowledge]; Metacognition --> Self-regulation[Self-regulation]; Metacognitive_knowledge --> Declarative_Knowledge[Declarative Knowledge]; Metacognitive_knowledge --> Procedural_Knowledge[Procedural Knowledge]; Metacognitive_knowledge --> Conditional_Knowledge[Conditional Knowledge]; Self-regulation --> Planning[Planning]; Self-regulation --> Monitoring[Monitoring]; Self-regulation --> Evaluation[Evaluation];
```

Metacognitive knowledge

Declarative Knowledge

Procedural Knowledge

Conditional Knowledge

Self-regulation

Planning

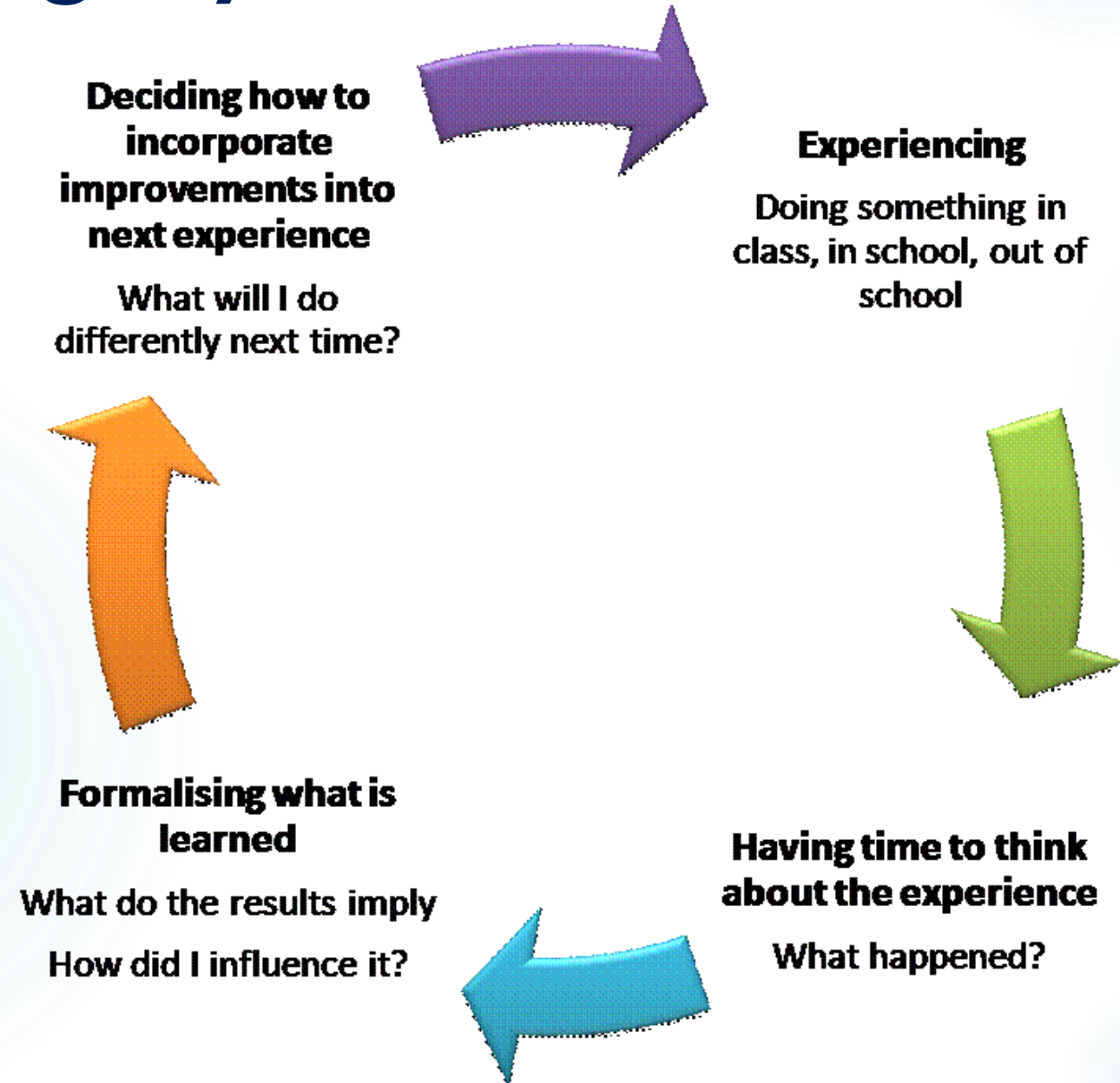
Monitoring

Evaluation

Why is metacognition important?

- ▶ ...if it happens of its own accord anyway?
- ▶ Shapes active rather than passive learners
- ▶ Gives pupils sense of control over learning
- ▶ Learning how to learn.
- ▶ Helps to promote “deep learning”

The Learning Cycle



Strategy	How to Use	When to Use	What is it for?
Skim/Survey	Search for headings, highlighted words previews, summaries	Before you read a long piece of text	Gives an overview of the key concepts, helps you to focus on the important points
Slow down	Stop, read and think about information	When information seems important. If you realise you don't understand what you have just read.	Improves your focus on important information.
Activate prior knowledge	Stop and think about what you already know about a topic.	Before you read something or do an unfamiliar task.	Makes new information easier to remember and allows you to see links between subjects. Information is less daunting if you already know something about the topic
Fit ideas together	Relate main ideas to one another. Look for themes that connect the main ideas, or a conclusion	When thinking about complex information, when deep understanding is needed.	Once you know how ideas are related they are easier to remember than learning as if they are separate facts. Also helps to understand them more deeply
Draw Diagrams	Identify main ideas, connect them, classify ideas, decide which information is most important and which is supporting	When there is a lot of factual information that is interrelated	Helps to identify main ideas and organise them into categories. Reduces memory load. May be easier to visualise

Planning

What is the nature of the task?

What is my goal?

What kind of information and strategies do I need?

How much time and resources do I need?

Monitoring

Do I have a clear understanding of what I am doing?

Does the task make sense to me?

Am I reaching my goals?

Do I need to make changes?

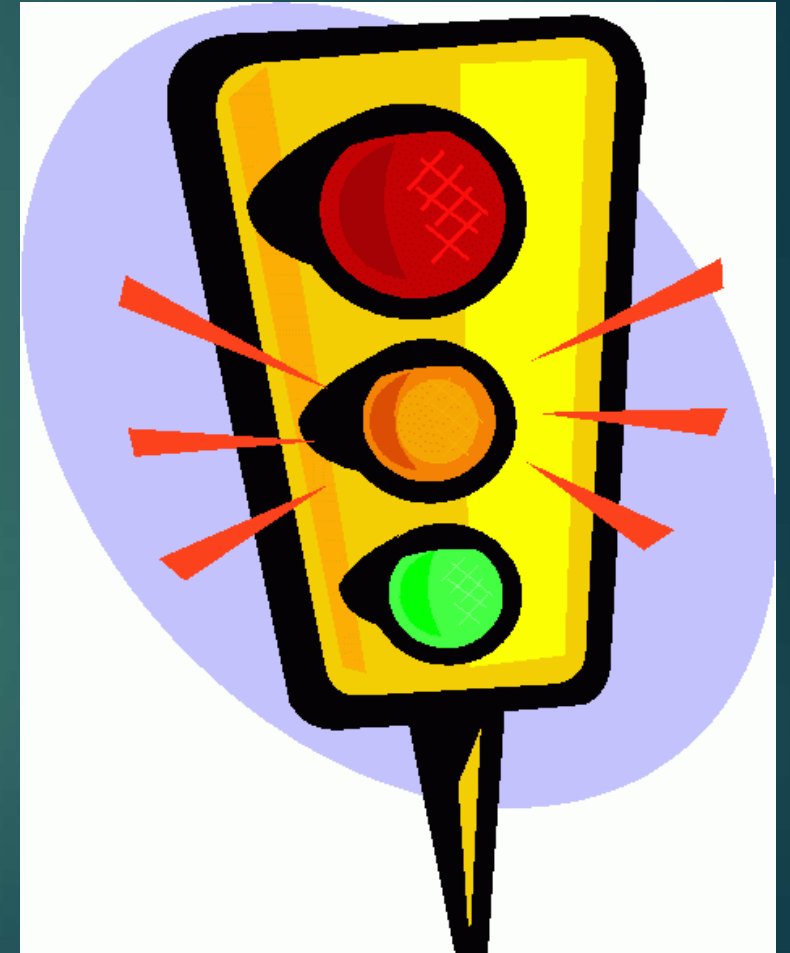
Evaluating

Have I reached my goal?

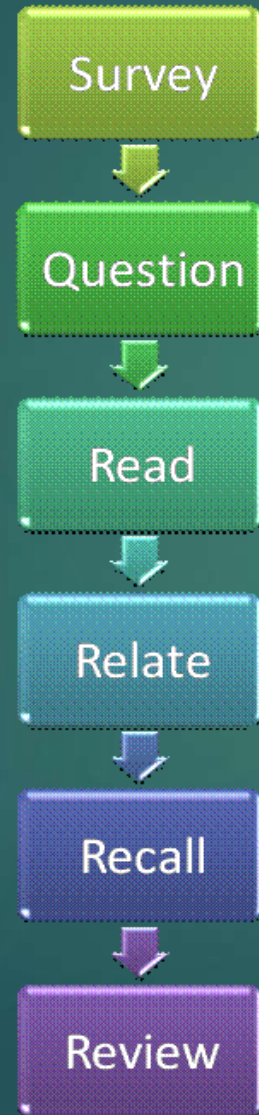
What worked?

What didn't work?

Would I do things differently the next time?



The SQ4R method for reading



KWL Grids

Ireland in World War II

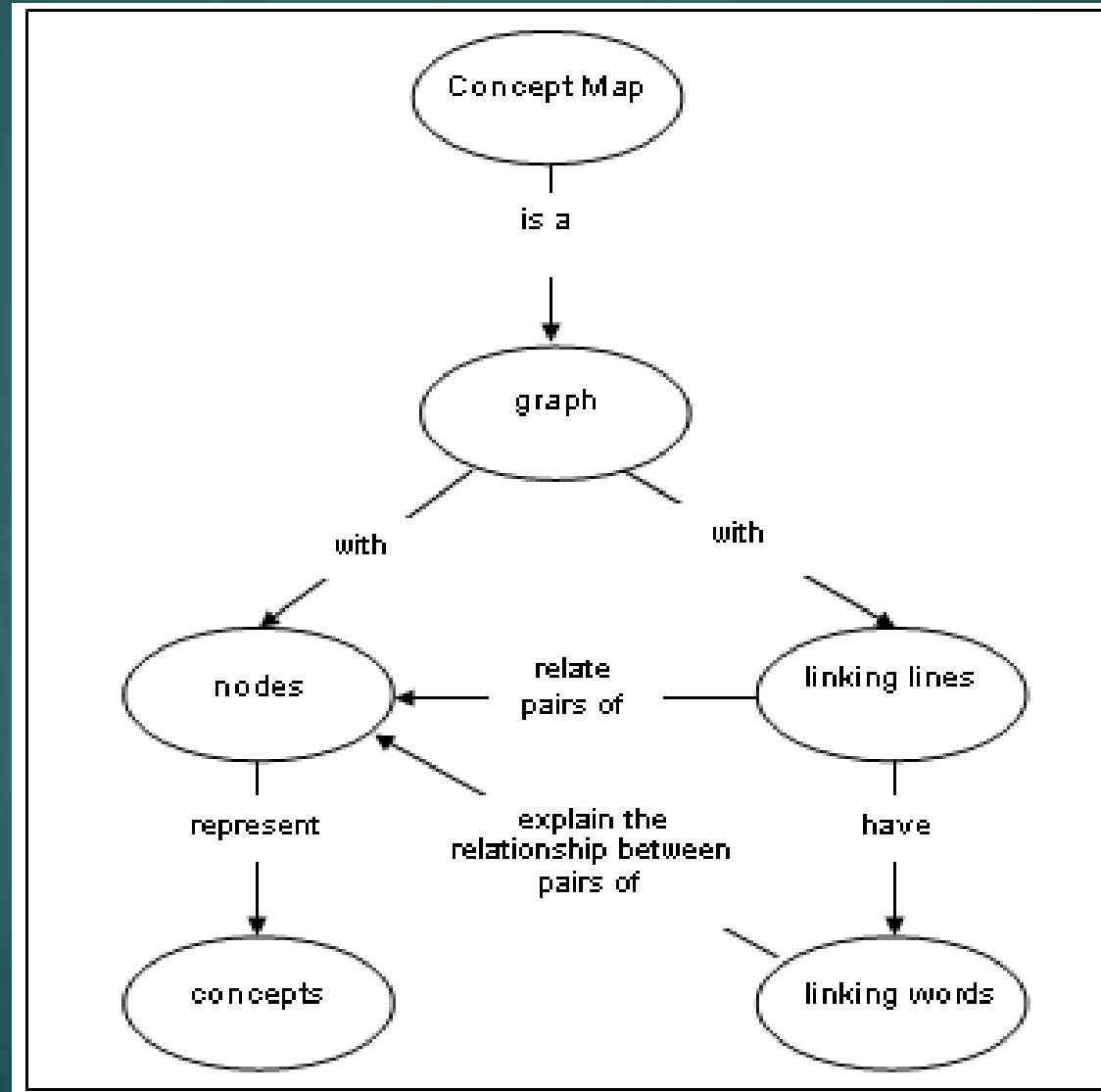
K (What I know already)	W (What I want to know)	L (What I have learned)
Dates of WW 2	Why called "The Emergency"?	
Hitler & The Holocaust	Why did Ireland decide to stay neutral?	
Role of GB & USA	Impact on life in Ireland?	
"The Emergency"	Impact on Ireland's domestic & foreign policy?	

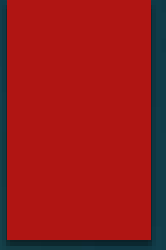
PMI : Plus Minus Interesting



- ▶ Take three minutes to think about this question; 1 min for positives, 1 min negatives and 1 min for anything interesting that strikes you
- ▶ What if the human body had a third arm?
- ▶ What if children were paid a wage to come to school?

What is a concept map?





“ He who learns but
does not think is lost ”

CHINESE PROVERB

MOTIVATION



Motivation

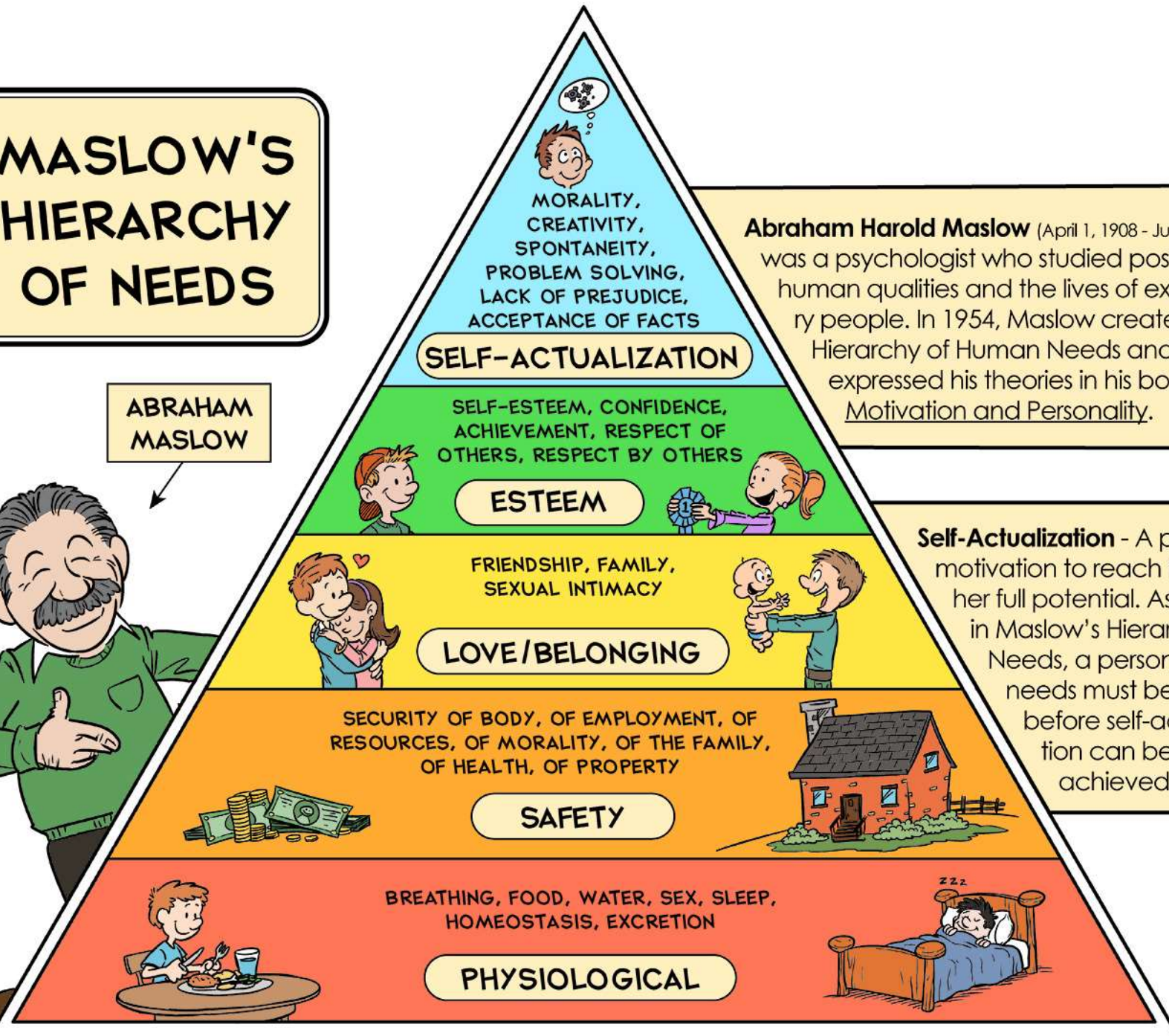
- ▶ from the Latin word “*movere*” which means **to move**.
- ▶ the processes that account for an individual’s **intensity, direction, and persistence** of effort towards attaining a goal.

Types of Motivation

- ▶ **Extrinsic** – fueled by anticipation and expectation of some kind of payoff from an external force
- ▶ **Intrinsic** – fueled by one's own goal or ambitions, self-starting, self-perpetuating and requires only an inward interest

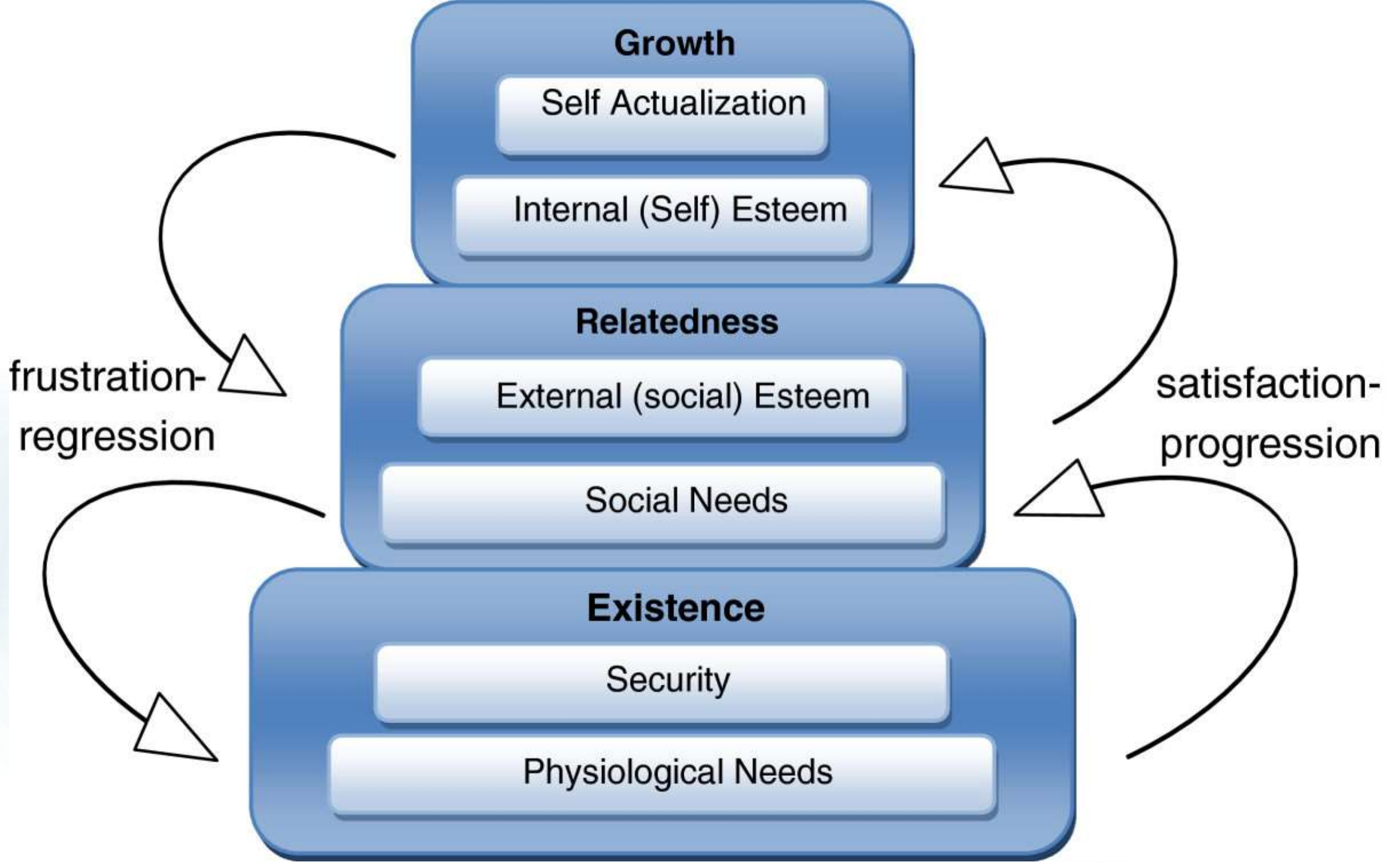
MASLOW'S HIERARCHY OF NEEDS

ABRAHAM MASLOW



Abraham Harold Maslow (April 1, 1908 - June 8, 1970) was a psychologist who studied positive human qualities and the lives of exemplary people. In 1954, Maslow created the Hierarchy of Human Needs and expressed his theories in his book, Motivation and Personality.

Self-Actualization - A person's motivation to reach his or her full potential. As shown in Maslow's Hierarchy of Needs, a person's basic needs must be met before self-actualization can be achieved.



Theory X and Theory Y (Douglas McGregor)

Theory X

The assumption that employees dislike work, are lazy, dislike responsibility, and must be coerced to perform.



Theory Y

The assumption that employees like work, are creative, seek responsibility, and can exercise self-direction.



McClelland's Motivational Needs



nACH

(Need for
Achievement)

HIGH

Must win at any cost
Must be on top and receive
credit

LOW

Fears Failure
Avoids Responsibility

nAFF

(Need for
Affiliation)

HIGH

Demands blind loyalty and
harmony
Does not tolerate disagreement

LOW

Remains aloof
Maintains social distance

nPOW

(Need for Power)

HIGH

Desires control of everyone and
everything
Exaggerates own position and
resources

LOW

Dependent/subordinate
Minimizes own position and
resources

Principles of Motivation

- ▶ Environment can be used to focus the student's situation on what needs to be learned
- ▶ Incentives motivate learning
- ▶ Internal motivation is longer lasting and more self-directive than is external motivation which must be repeatedly reinforced by praise or concrete rewards



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dubsmash.com



End of presentation.
Thank you!